

INFORMATION MEMORANDUM

in relation to the offering of RADIX Tokens

(to be achieved by the offering of Interim Tokens which will be exchangeable at the option of the Tokenholder for RADIX Tokens on a 1:1 basis as described herein)

RADIX Tokens (Jersey) Limited

A private company incorporated in Jersey with registration number 132555

22 OCTOBER 2020

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1. **RISK WARNINGS**

Prospective Tokenholders should carefully review the risk factors set forth in this section prior to purchasing any Tokens. Prospective Tokenholders should consider the purchase of any Tokens as involving a degree of financial risk and should therefore carefully consider the following risk factors (together with other matters set out elsewhere in this Information Memorandum).

Prospective Tokenholders should carefully consider the following factors, among others, in deciding whether to purchase any Tokens and should consult their own legal, tax and financial advisors as to these risks before purchasing any Tokens.

The following is a brief description of certain factors which should be considered prior to purchasing any Tokens. The following, however, does not purport to be a comprehensive summary of all the risks associated with a purchase of Tokens.

Please see the Defi White Paper, the Consensus White Paper and the Economics White Paper (together the "**White Papers**") (attached as Appendices 2, 3, and 4 to this Information Memorandum, respectively) for further information on the Token Offering.

1.1 **No Speculative Investment**

A purchaser of Tokens, represents and warrants that:

- 1.1.1 **it is not acquiring Tokens with a view to holding them as a speculative investment;**
- 1.1.2 **its motivation for acquiring the Tokens is to facilitate the development of the RADIX Network through participation in the consensus validation protocol and operations on the RADIX Network; and**
- 1.1.3 **it is acquiring the Tokens to facilitate transactions on the RADIX Ledger and/or to obtain access to the functionality that will be provided by the RADIX Ledger and/or to facilitate the purchaser's participation on the RADIX Ledger as a Node Runner.**

1.2 **General risk warnings**

- 1.2.1 If you are considering purchasing Tokens during the Token Offering or any subsequent issuances of Tokens (if any) or have already purchased Tokens, be aware of the many risks that purchase and ownership of Tokens may entail, including the total loss of the price you paid for any Tokens. The Token Offering is unregulated and may result in substantial risks for purchasers of the Tokens.
- 1.2.2 Be aware that:
 - (a) prospective Tokenholders will have no protection where the token offerings is unregulated;

- (b) token offerings are a highly speculative form of investment;
- (c) prospective Tokenholders should be prepared for the possibility of losing their investment completely; and
- (d) token offerings are not subject to existing capital markets regulations.

1.3 Many financial regulators, including ESMA, have observed a rapid growth in token offerings, which raise capital for enterprises, and are concerned that purchasers may not realise the high risks that they are taking when purchasing Tokens.

1.4 Token offerings are highly speculative. Token offerings, depending on how they are structured, may fall outside of the regulated space, in which case Tokenholders do not benefit from the protection that comes with regulated investments. Token offerings are also vulnerable to fraud or illicit activities, owing to their anonymity and their capacity to raise large amounts of money in a short timeframe.

1.5 If you are in any doubt as to the risks, or suitability of the purchase, of Tokens, you should seek advice from an appropriately qualified financial adviser.

1.6 Token offerings are extremely risky and highly speculative – we repeat here the guidance provided by ESMA with respect to general risks associated with token offerings. Prospective Tokenholders should realise that they are exposed to the following risks when purchasing Tokens:

1.6.1 **Token offerings are unregulated, and vulnerable to fraud or illicit activities:** depending on how they are structured, token offerings may not be captured by the existing rules and may fall outside of the regulated space. Some token offerings may be used for fraudulent or illicit activities, with several recent initial coin offerings ("ICOs") having been identified as frauds, while financial regulators cannot exclude that some are being used for money laundering purposes. In the case where a token offering does not fall under the scope of the laws and regulations of national or supra-national jurisdictions (such as ESMA), purchasers cannot benefit from the protection that these laws and regulations provide.

1.6.2 **Risks arising from lack of governance rights:** as Tokens confer no governance rights of any kind with respect to the Issuer, all decisions involving the Issuer (including to sell or liquidate the Issuer) will be made by the Issuer acting in its sole and absolute discretion, and all decisions involving the Issuer, including, but not limited to, decisions to create and issue more Tokens, will be made by the Issuer. These decisions could adversely affect the RADIX Network, and the value or utility of the Tokens.

1.6.3 **Jurisdiction related risks:** residents, tax residents or persons having a relevant connection with certain jurisdictions are excluded from the Token Offering. Changes in a Tokenholder's place of domicile or the applicable law may result in that Tokenholder violating legal or regulatory requirements of his applicable jurisdiction. Each Tokenholder is responsible for ensuring that the distribution, holding, use or exchange of Tokens is, and remains, lawful despite changes to applicable laws, or the Tokenholder's residence and circumstances.

- 1.6.4 **Risks arising from the market in which RADIX operates:** the crypto exchange market, the token offering market, token offerings, and, by extension, RADIX, are subject to a variety of federal, state and international laws and regulations, including those with respect to know your customer, anti-money laundering and customer due diligence procedures, privacy and data protection, consumer protection, data security, and others. These laws and regulations, and the interpretation or application of these laws and regulations, could change. In addition, new laws or regulations affecting the RADIX ecosystem could be enacted, which could impact the utility of Tokens in the RADIX ecosystem. In addition, Tokenholders are subject to or may be adversely affected by industry specific laws and regulations or licensing requirements. If any of these parties fails to comply with any of these licensing requirements or other applicable laws or regulations, or if such laws and regulations or licensing requirements become more stringent or are otherwise expanded, it could adversely impact RADIX or the Tokens.
- 1.6.5 **High risk of losing all of the capital paid:** the vast majority of token offerings are launched by businesses that are at a very early stage of development. Those businesses have an inherently high risk of failure. Many of the tokens that are being issued have no intrinsic value other than the possibility to use them to access or use a service/product that is to be developed by respective Issuers. There is no guarantee that the services/products will be successfully developed and, even assuming that a project is successful, any eventual benefit or utility may be extremely low relative to the capital paid for the Tokens.
- 1.6.6 **Lack of exit options and extreme price volatility:** Tokenholders may not be able to trade their Tokens or exchange them for fiat. The Tokens are traded on virtual currency exchanges but, like most crypto tokens, their exchange value may be extremely volatile. Many exchanges are unregulated and vulnerable to market price manipulation and fraudulent activities. Tokenholders may be exposed to the lack of exit options and a Token buyback option may not be available.
- 1.6.7 **Risk of competing platforms:** it is possible that alternative platforms could be established with code that replicates the functionality of RADIX Network. The RADIX Network may compete with these alternatives, which could negatively impact the RADIX Network and the Tokens.
- 1.6.8 **Inadequate information:** the information that is made available to prospective token purchasers, e.g. in so-called "white papers", is in many cases unaudited, incomplete, unbalanced or even misleading. It typically puts the emphasis on the potential benefits but not the risks. Information is often technical and not easily comprehensible by laypersons. Prospective Tokenholders may therefore not understand, or be capable of understanding, the risks that they are taking and make decisions as to purchasing the products that are not appropriate to their circumstances.

- 1.6.9 **Flaws in the technology:** the distributed ledger or blockchain technology that underpins the Tokens is still largely untested. There may be flaws in the code or programs that are used to create, transfer or store the Tokens. Tokenholders may not be able to access or control their Tokens and Tokens may be transferred out of a Tokenholder's possession by negligent, fraudulent, dishonest or erroneous actions (including the Tokenholders' own such actions). More generally, the technology may not function efficiently, e.g. during peaks of activity, or securely.

Purchasing Tokens involves a high degree of risk. You should consider carefully the risks described below, together with all of the other information contained in this Information Memorandum, before making a decision to purchase any Tokens. The following risks entail circumstances under which, our business, financial condition, results of operations and prospects could suffer.

1.7 **Risks associated with purchasing the Tokens.**

- 1.7.1 There may not be a successfully developed market for the Tokens. It will require the expertise, time and effort of the Issuer's management to develop the Tokens. It is possible that the Tokens may not and cannot meet a Tokenholder's expectations at the time of purchase or beyond. Furthermore, despite good faith and efforts to maintain the Tokens, it is still possible that the Tokens will experience malfunctions or otherwise fail to be adequately developed or maintained, which may negatively impact the Tokens.
- 1.7.2 While the Issuer has sought to retain and continue to competitively recruit experts, there is a general scarcity of management, technical, scientific, research and marketing personnel with appropriate experience to build, maintain and improve the Tokens.

Investing in new crypto projects involves a high degree of risk. Participating in token offerings may involve an even higher degree of risk.

- 1.7.3 Financial and operating risks confronting new crypto projects are significant: the Issuer is not immune to these. The market in which the Issuer operates is highly competitive and the percentage of companies that survive and prosper is small. New crypto projects often experience unexpected problems in the areas of product development, marketing, financing, and general management, amongst others, which frequently cannot be solved.

Risk of losing access to tokens due to loss of private key(s), custodial error or Tokenholder error.

- 1.7.4 Tokens can only be accessed by using an Ethereum wallet and a Tokenholder's Tokens may only be transferred by the person in possession of the respective private key(s). Tokenholders acknowledge, understand and accept that if their private key(s) becomes lost or stolen, the Tokens associated with such private key(s) will likely be unrecoverable and permanently lost. In addition, any third party that gains access to a Tokenholder's private key(s), including by gaining access to the login credentials relating to the wallet, may be able to misappropriate the Tokenholder's Tokens. Any errors or malfunctions caused by or

otherwise related to the digital wallets or vaults to which Tokenholders choose to receive the Tokens or in which Tokenholders choose to store the Tokens, including a Tokenholder's own failure to properly maintain, update, backup or use such digital wallet or vault, may also result in the loss of a Tokenholder's Tokens.

Risk of incompatible wallet service.

- 1.7.5 The wallet or wallet service provider used to receive Tokens must conform to the relevant Ethereum token standard in order to be technically compatible with Tokens. The failure to ensure such conformity may have the result that that Tokenholder will not gain access to his Tokens or may lose access to his Tokens.

Risk of inadequate resources.

- 1.7.6 The Token sale and RADIX will require intensive computing resources. The demand for these resources may exceed the Issuer's estimates. Ultimately, the Issuer's resources may prove inadequate to support the Token sale or to develop the Tokens, which may affect the distribution or utility of the Tokens.

The Issuer may be forced to cease operations or take actions that result in the Issuer's dissolution.

- 1.7.7 It is possible that, due to any number of reasons, including, but not limited to, an unfavourable fluctuation in the value of cryptographic and fiat currencies, the inability by the Issuer to establish the Tokens' utility, the failure of commercial relationships, or intellectual property ownership challenges, the Issuer may no longer be viable to operate and the Issuer may dissolve or take actions that result in its dissolution.

The tax treatment of the Tokens, the purchase rights contained therein and the Tokens' distribution is uncertain and there may be adverse tax consequences for Tokenholders upon certain future events.

- 1.7.8 The tax characterisation of the Tokens is uncertain, and each Tokenholder must seek its own tax advice in connection with a purchase of Tokens which may result in adverse tax consequences to Tokenholders, including withholding taxes, income taxes and tax reporting requirements. Each Tokenholder should consult with and must rely upon the advice of its own professional tax advisors with respect to the treatment of any purchase of the Tokens and the rights contained therein.

1.8 Risks associated with the Tokens.

The Tokens may not be widely adopted and may have limited users.

- 1.8.1 It is possible that the Tokens will not be used by a large number of individuals, companies and other entities. Such a lack of use or interest could negatively impact the development of the Tokens and therefore the potential utility of the Tokens.

Risk of hard-fork.

- 1.8.2 RADIX will need to go through substantial development works as part of which it may become the subject of significant conceptual, technical and commercial changes before release. As part of the development, an upgrade to Tokens may be required (a hard-fork of the Tokens) and that, if Tokenholders decide not to participate in such upgrade, Tokenholders may no longer be able to use their Tokens and any non-upgraded or otherwise deprecated Tokens may lose their utility in full.

Risk of uninsured losses.

- 1.8.3 Unlike bank accounts or accounts at some other financial institutions, Tokens are uninsured unless Tokenholders specifically obtain private insurance to insure them. Thus, in the event of loss or loss of utility value, there is no public insurer or private insurance arranged by the Issuer, to offer recourse to Tokenholders.

Alternative tokens may be established that compete with or are more widely used than the Tokens.

- 1.8.4 It is possible that alternative tokens could be established that utilise similar code and/or protocol to the Tokens. The Tokens may be in direct competition with these alternative tokens, which could negatively impact the Tokens.

Risks associated with markets for Tokens.

- 1.8.5 The Issuer may choose not to enable or otherwise facilitate any secondary speculative trading or any such external valuation of Tokens. This may restrict the contemplated avenues for using Tokens to the token utility described in the Whitepapers and could therefore create illiquidity risk with respect to any Tokens held. Even if secondary trading of Tokens is facilitated by third-party exchanges, such exchanges may be relatively new and subject to little or no regulatory oversight, making them more susceptible to fraud or manipulation. Furthermore, to the extent that any third party ascribes an external exchange value to Tokens (e.g. as denominated in a crypto or fiat currency), such value may be extremely volatile and diminish to zero. Tokenholders accept that Tokens are not offered by the Issuer or its affiliates on an investment basis. Each Tokenholder further acknowledges that any payment made for Tokens will not be protected, guaranteed or reimbursed by any governmental, regulatory or other entity, and is unlikely to be protected by any jurisdiction.
- 1.8.6 The open-source nature of underlying elements of the Tokens means that the Tokens may be susceptible to developments by users or contributors that could damage the Tokens and the Issuer's reputation and could affect the utilisation of the Tokens.
- 1.8.7 The Tokens will operate and exist on the Ethereum blockchain, both of which are maintained and developed by community contributors. The open-source nature of the Ethereum protocol means that it may be difficult to maintain and neither the contributors nor the Issuer may have adequate resources to address emerging issues or malicious programs that

develop within Ethereum adequately or in a timely manner. Third parties not affiliated with the Issuer may introduce weaknesses or bugs into the core infrastructure elements of Ethereum and open-source code which may negatively impact the Tokens. Such events may result in a loss of trust in the security and operation of the Tokens and a decline in user activity and could negatively impact the market price of the Tokens.

1.8.8 The Tokens may be the target of malicious cyberattacks or may contain exploitable flaws in its underlying code, which may result in security breaches and the loss or theft of the Tokens. If the Tokens' security is compromised or if the Tokens are subjected to attacks that frustrate or thwart the ability to use or transfer Tokens, users may cut back on or stop using the Tokens altogether, which could seriously curtail the utilisation of the Tokens and cause a decline in the market price of the Tokens.

1.8.9 The Tokens' structural foundation, the open-source protocol, the software application and other interfaces or applications built upon the Tokens are still in an early development stage and are unproven, and there can be no assurances that the Tokens and the creating, transfer or storage of the Tokens will be uninterrupted or fully secure which may result in a complete loss of users' Tokens or an unwillingness of users to use Tokens. Further, the Tokens or their suppliers may also be the target of malicious attacks seeking to identify and exploit weaknesses in the software or the Tokens which may result in the loss or theft of the Tokens. For example, if the Tokens are subject to unknown and known security attacks, this may materially and adversely affect the value of the Tokens.

Lack of Initial Demand.

1.8.10 If there is insufficient initial demand there is a risk that the Tokens fail to gain support and grow as a currency. This risk is unknown until the Token Offering has commenced and demand has been identified.

Unanticipated risks.

1.8.11 Cryptographic tokens such as the Tokens are a new and untested technology. In addition to the risks set out in this section, there are other risks associated with acquisition, storage, transfer and use of Tokens, including those that the Issuer may not be able to anticipate. Such risks may further materialise as unanticipated variations or combinations of the risks set out in this section.

1.9 Risks related to blockchain technologies and digital assets.

The regulatory regime governing the blockchain technologies, cryptocurrencies, Tokens and coin offerings such as the Tokens is uncertain, and new regulations or policies may materially adversely affect the development and the utility of the Tokens.

1.9.1 Regulation of coins (including the Tokens) and coin offerings such as the Token Offering, cryptocurrencies, blockchain technologies, and cryptocurrency exchanges currently is

undeveloped and likely to rapidly evolve, varies significantly amongst international, state and local jurisdictions and is subject to significant uncertainty. Various legislative and executive bodies may in the future, adopt laws, regulations, guidance, or take other actions, which may severely impact the development and growth of the Tokens and the adoption and use of the Tokens. Failure by the Issuer or certain users of the Tokens to comply with any laws, rules and regulations, some of which may not exist yet or are subject to interpretation and may be subject to change, could result in a variety of adverse consequences, including civil penalties and fines.

- 1.9.2 As blockchain networks and blockchain assets have grown in popularity and in market size, state, national and supra-national agencies have begun to take interest in, and in some cases regulate, their use and operation.
- 1.9.3 The regulation of non-currency use of blockchain assets is also uncertain. To the extent that a domestic government or quasi-governmental agency exerts regulatory authority over a blockchain network or asset, the Tokens may be materially and adversely affected.
- 1.9.4 Various jurisdictions may, in the near future, adopt laws, regulations or directives that affect the Tokens. Such laws, regulations or directives may directly and negatively impact our business. The effect of any future regulatory change is impossible to predict, but such change could be substantial and materially adverse to the development and growth of the Tokens and the adoption and widespread use of the Tokens.
- 1.9.5 New or changing laws and regulations or interpretations of existing laws and regulations may materially and adversely impact the value of the currency in which the Tokens may be exchanged, the liquidity of the Tokens, the ability to access marketplaces or exchanges on which to trade the Tokens, and the structure, rights and transferability of the Tokens.

Risk of software weaknesses.

- 1.9.6 As Tokens, the smart contract system and the Tokens are based on the Ethereum protocol, any malfunction, breakdown or abandonment of the Ethereum protocol may have a material adverse effect on Tokens, the smart contract system and/or the RADIX ecosystem. Moreover, advances in cryptography, or technical advances such as the development of quantum computing, could present risks to RADIX (including the utility of Tokens for obtaining services), the smart contract system and/or RADIX, by rendering ineffective the cryptographic consensus mechanism that underpins the Ethereum protocol. The smart contract system concept, the underlying software application and software platform (i.e. the Ethereum blockchain) is still in an early development stage and unproven. There is no warranty or assurance that the process for creating Tokens will be uninterrupted or error-free and there is an inherent risk that the software could contain defects, weaknesses, vulnerabilities, viruses or bugs causing, amongst other things, the complete loss of contributions and/or Tokens.

Risk of necessary alteration to protocol or blockchain.

- 1.9.7 In order to stay up to date with technology, the RADIX ecosystem may be required to update or change the smart contract system, the protocol or even the blockchain that the Tokens operate on. In the event that such an alteration is required, the Issuer will make a good faith effort to credit every user's account in such a way that no Tokens are lost. However, the Issuer does not, and cannot, promise that every user will be accommodated.
- 1.9.8 The Tokenholders will have no control and the Issuer may only have limited control following the issuance of the Tokens.
- 1.9.9 The Tokens depend on a network of computers to run certain software programs to process transactions. Because of this decentralised model, the Issuer has limited control over the recording of the Tokens once issued. In addition, purchasers of Tokens are not and will not be entitled, to vote or receive dividends or be deemed the holder of capital stock of the Issuer for any purpose, nor will anything be construed to confer on the holders any of the rights of a stockholder of the Issuer or any right to vote for the election of directors or upon any matter submitted to Shareholders at any meeting thereof, or to give or withhold consent to any corporate action or to receive notice of meetings, or otherwise.
- 1.9.10 There may be occasions when certain individuals involved in the development and launch of the Tokens may encounter potential conflicts of interest in connection with this Token Offerings, such that said party may avoid a loss, or even realise a gain, when other purchasers of the Tokens are suffering losses. Purchasers of the Tokens may also have conflicting purchase, tax, and other interests with respect to the Tokens, the Tokens' code, the timing of the network launch or other Token pre-sales, or other factors. Decisions made by the key employees of the Issuer and/or its affiliates on such matters may be more beneficial for some Tokenholders than for others.

Holders may lack information for monitoring the Tokens.

- 1.9.11 Tokenholders may not be able to obtain all information they would wish to have access to in respect of the Tokens on a timely basis or at all. It is possible that a holder may not be aware on a timely basis of material adverse changes that have occurred with respect to the Tokens.

The Issuer and the Tokens have no history.

- 1.9.12 A Token will be a newly formed token and neither it nor the Issuer has any operating history. Each prospective Tokenholder should evaluate such potential purchase on the basis that the Tokens, the Issuer or any third party's assessment of the prospects of the Tokens may not prove accurate. Past performance of the Issuer or any similar token is not predictive of future results.

If the Tokens are unable to satisfy data protection, security, privacy, and other government and industry-specific requirements, their growth could be harmed.

1.9.13 There are a number of data protection, security, privacy and other government- and industry-specific requirements, including those that require companies to notify individuals of data security incidents involving certain types of personal data. Security compromises could harm the Token's reputation, erode user confidence in the effectiveness of its security measures, negatively impact its ability to attract further people to use the Tokens, or cause existing holders to sell their Tokens.

The further development and acceptance of blockchain networks, which are part of a new and rapidly changing industry, are subject to a variety of factors that are difficult to evaluate. The slowing or stopping of the development or acceptance of blockchain networks and blockchain assets would have an adverse material effect on the successful development and adoption of the Tokens.

1.9.14 The growth of the blockchain industry in general, as well as the blockchain networks with which the Tokens will rely and interact, is subject to a high degree of uncertainty. The factors affecting the further development of the cryptocurrency industry, as well as blockchain networks, include, without limitation:

- (a) worldwide growth in the adoption and use of Bitcoin, Ethereum and other blockchain technologies;
- (b) government and quasi-government regulation of Bitcoin, Ethereum and other blockchain assets and their use, or restrictions on or regulation of access to and operation of blockchain networks or similar systems;
- (c) the maintenance and development of the open-source software protocol of the Ethereum network;
- (d) changes in consumer demographics and public tastes and preferences;
- (e) the availability and popularity of other forms or methods of buying and selling goods and services, or trading assets including new means of using fiat currencies or existing networks;
- (f) general economic conditions and the regulatory environment relating to cryptocurrencies; or
- (g) a decline in the popularity or acceptance of Bitcoin, Ether or other blockchain-based coins or tokens would adversely affect our results of operations.

- 1.9.15 The slowing or stopping of the development, general acceptance and adoption and usage of blockchain networks and blockchain assets may deter or delay the acceptance and adoption of the Tokens.

Risks associated with third party contractors.

- 1.9.16 Development of the Tokens and of RADIX, and the operation of the Token Offering, will require third-party contractors with particular expertise in Ethereum and blockchain technology. The availability of such contractors is limited. There may not be a sufficient number of contractors available on terms deemed acceptable by the Issuer. The costs associated with any such contractors may be significantly greater than currently estimated. Furthermore, the quality, reliability and timely delivery of services by such contractors may vary significantly.

The prices of blockchain assets are extremely volatile. Fluctuations in the price of digital assets could materially and adversely affect our business, and the Tokens may also be subject to significant price volatility.

- 1.9.17 The prices of blockchain assets such as Bitcoin have historically been subject to dramatic fluctuations and are highly volatile, and the market price of the Tokens may also be highly volatile. Several factors may influence the market price of the Tokens, including, but not limited to:
- (a) global blockchain asset supply;
 - (b) global blockchain asset demand, which can be influenced by the growth of retail merchants' and commercial businesses' acceptance of blockchain assets like cryptocurrencies as payment for goods and services, the security of online blockchain asset exchanges and digital wallets that hold blockchain assets, the perception that the use and holding of blockchain assets is safe and secure, and the regulatory restrictions on their use;
 - (c) Tokenholders' expectations with respect to the rate of inflation;
 - (d) changes in the software, software requirements or hardware requirements underlying the Tokens;
 - (e) changes in the rights, obligations, incentives, or rewards for the various participants in the Tokens;
 - (f) interest rates;
 - (g) currency exchange rates, including the rates at which digital assets may be exchanged for fiat currencies;

- (h) fiat currency withdrawal and deposit policies of blockchain asset exchanges on which the Tokens may be traded and liquidity on such exchanges;
- (i) interruptions in service from or failures of major blockchain asset exchanges on which the Tokens may be traded;
- (j) purchase and trading activities of large purchasers that may directly or indirectly purchase Tokens or other blockchain assets;
- (k) monetary policies of governments, trade restrictions, currency devaluations and revaluations;
- (l) regulatory measures, if any, that affect the use of blockchain assets such as the Tokens;
- (m) the maintenance and development of the open-source software protocol of the Tokens;
- (n) global or regional political, economic or financial events and situations; or
- (o) expectations amongst the Issuer or other blockchain assets participants that the value of the Tokens or other blockchain assets will soon change.

1.9.18 A decrease in the price of a single blockchain asset may cause volatility in the entire blockchain asset industry and may affect other blockchain assets including the Tokens since crypto historically appears to be correlated. For example, a security breach that affects user confidence in Bitcoin may affect the industry as a whole and may also cause the price of the Tokens and other blockchain assets to fluctuate.

Bank misappropriation or insolvency

1.9.19 The Issuer may allocate some the proceeds from the sale of the Tokens into deposit accounts held with various banks. There is a risk that the banks could become insolvent and/or misappropriate the funds.

General hacking: risk of theft and vulnerabilities.

1.9.20 The smart contract system concept, the underlying software application and software platform (i.e. the Ethereum blockchain) may be exposed to attacks by hackers or other individuals including, but not limited to, malware attacks, denial of service attacks, consensus-based attacks, Sybil attacks, smurfing and spoofing. Any such successful attacks could result in theft or loss of contributions or Tokens, adversely impacting the ability to develop the Tokens and derive any usage or functionality from the Tokens. Tokenholders must take appropriate steps to satisfy themselves of the integrity and veracity of relevant websites, systems and communications. Furthermore, because the RADIX Network is based

on open-source software, there is a risk that a third party or a member of the Issuer's team may intentionally or unintentionally introduce weaknesses or defects into the core infrastructure of the RADIX Network, which could negatively affect the RADIX Network and the Tokens.

Risk of Ethereum mining attacks.

1.9.21 As with other cryptocurrencies, the blockchain used for the smart contract system is susceptible to mining attacks, including but not limited to double-spend attacks, majority mining power attacks, "selfish-mining" attacks, and rare condition attacks. Any successful attacks present a risk to the smart contract system, expected proper execution and sequencing of token transactions, and expected proper execution and sequencing of contract computations. Tokenholders understand and accept that the use of miners will ultimately be in control of the distribution of Tokens via the smart contract system, and that a majority of miners could agree at any point to make changes, updates, modifications to, or effect a deletion or destruction of the smart contract system, and that such a scenario could lead to the Tokens losing intrinsic value and/or functionality.

Fraud

1.9.22 Until fully distributed the Tokens are not fully decentralised, and there is a risk that an employee of the Issuer, the Administrator or one of its suppliers conducts fraudulent activities which could undermine the value of the Tokens.

IN VIEW OF THE FOREGOING CONSIDERATIONS, AMONGST OTHERS, A PURCHASE OF ANY TOKEN(S) IS SUITABLE ONLY FOR PURCHASERS WHO ARE CAPABLE OF BEARING THE RELEVANT RISKS AND WHO CAN AFFORD TO PAY THE MINIMUM PURCHASE AMOUNT.

THE FOREGOING RISK FACTORS DO NOT PURPORT TO BE A COMPLETE EXPLANATION OF THE RISKS INVOLVED IN THIS OFFERING. PROSPECTIVE TOKENHOLDERS MUST READ THE ENTIRE INFORMATION MEMORANDUM INCLUDING THE RELEVANT SUPPLEMENTS AND ALL ATTACHMENTS AND MUST CONSULT THEIR OWN PROFESSIONAL ADVISERS, BEFORE DECIDING WHETHER TO PURCHASE ANY TOKENS.

2. **IMPORTANT INFORMATION AND JERSEY REGULATORY WARNINGS**

2.1 **Important Information**

2.1.1 This Information Memorandum does not constitute an offer or solicitation by any person in any jurisdiction in which the offer or solicitation is unlawful or in which the person making the offer or solicitation is not qualified to do so or to persons to whom it is unlawful to make the offer or solicitation. Persons into whose possession this Information Memorandum comes are required to inform themselves about, and to observe, the laws and regulations applicable to them in the relevant jurisdiction.

2.1.2 Prospective Tokenholders should not construe the contents of this Information Memorandum as legal, tax or financial advice. Each prospective Tokenholder should consult its own professional advisors as to:

- (a) the legal requirements within the country of its residence for the purchase, holding or disposal of the Tokens;
- (b) any foreign exchange restrictions that may be relevant to it and the income and other tax consequences that may be relevant to the purchase, holding or disposal of the Tokens; and
- (c) the suitability of a purchase of the Tokens to its personal circumstances and risk appetite.

2.1.3 A purchase of Tokens is only suitable for financially sophisticated purchasers who are capable of evaluating the merits and risks of such a purchase and who have sufficient resources to be able to bear any losses which may result from such purchase. Prospective Tokenholders are strongly advised to conduct their own due diligence including, without limitation, the legal, tax, and financial consequences of purchasing any Tokens.

2.1.4 This Information Memorandum includes forward-looking statements relating to, amongst other things, the future financial performance and objectives of the Issuer, the Tokens and the Token Offering, plans and expectations for the operation of the Issuer, the Tokens and the Token Offering, and estimates or expectations for fees, costs and expenses. These forward-looking statements are typically identified by terminology such as "may", "will", "aim", "target", "should", "expect", "anticipates", "plans", "intends", "believes", "estimates", "projects", "predicts", "seeks", "potential", "continue" or other similar terminology. Forward-looking statements are inherently unreliable, and prospective Tokenholders should not rely on them. The forward-looking statements are based on the Issuer's current expectations, assumptions, estimates and projections about future events. Actual results are subject to numerous risks, conditions, and uncertainties that could cause actual results to differ materially from those expressed in a forward-looking statement as a result of factors such as those described in "Risk Factors" at section 1 of this Information Memorandum and

elsewhere in this Information Memorandum. The Issuer does not have any obligation to update or otherwise revise any forward-looking statements after the date of this Information Memorandum or to reflect the occurrence of unanticipated events. The Auditor has neither reviewed nor audited any of the calculations, figures, or forward looking statements contained in this Information Memorandum.

- 2.1.5 No person has been authorised to make any representations or to give any warranties or to give any information with respect to the Tokens, except the information contained in this Information Memorandum. Neither the delivery of this Information Memorandum at any time nor any sale made pursuant hereto shall imply that information contained herein is correct as of any time subsequent to the date set forth on the cover of this Information Memorandum. Any reproduction or distribution of this Information Memorandum or retransmission of their contents in whole or in part to any person other than a prospective Tokenholder's professional advisers, without the consent of the Issuer, is prohibited.
- 2.1.6 The Issuer reserves the right to refuse to accept the offer of any prospective Tokenholder to purchase any Tokens for any reason or no reason.
- 2.1.7 The advisers of the Issuer named in this Information Memorandum act only for the Issuer in connection with the Token Offering described in this Information Memorandum and will not be responsible for providing the protections offered to their clients or for advising any other person in connection with the Token Offering.
- 2.1.8 This Information Memorandum may, from time to time, be supplemented with supplements to this Information Memorandum or other offering documents. In that event, the disclosures contained in such supplements to this Information Memorandum or other offering documents shall supersede this Information Memorandum to the extent thereof.
- 2.1.9 The Directors of the Issuer have taken all reasonable care to ensure that the facts stated in this document are true and accurate in all material respects, and that there are no other facts the omission of which would make misleading any statement in the document, whether of facts or of opinion. All the Directors accept responsibility accordingly.
- 2.1.10 Purchasing any Tokens involves special risks, and should be considered only by persons who can bear the economic risk of their purchase for an indefinite period and who can afford a total loss of their purchase. Please see section 1 (Risk Factors) of this Information Memorandum for further information on the risks involved in this Token Offering.
- 2.1.11 The Directors reserve the right to modify, withdraw or cancel any offering made pursuant to this Information Memorandum at any time prior to consummation of the offering and to reject any purchase of the Tokens, in whole or in part, in their sole discretion (subject to notifying, or obtaining the consent of, the Commission, as the case may be).

- 2.1.12 This Information Memorandum is intended solely for use on a confidential basis by those persons to whom it is transmitted by the Issuer in connection with the contemplated Initial Token Offering. Recipients, by their acceptance and retention of this Information Memorandum, acknowledge and agree to preserve the confidentiality of the contents of this Information Memorandum and all accompanying documents and to return this Information Memorandum and all such documents to the Issuer or the Administrator if the recipient does not purchase any Tokens at the Token Offering. Neither this Information Memorandum nor any of the accompanying documents may be reproduced in whole or in part, nor may they be used for any purpose other than that for which they have been submitted, without the prior written consent of the Issuer.
- 2.1.13 Neither the Directors nor the Issuer are making any representation to any prospective Tokenholder regarding the legality of purchases of any Tokens by such prospective Tokenholders under applicable laws.
- 2.1.14 The distribution of this Information Memorandum and the Token Offering in certain jurisdictions may be restricted by law. Prospective Tokenholders should inform themselves as to the legal requirements and tax consequences within the countries of their citizenship, residence, domicile and place of business with respect to the acquisition, holding or disposal of the Tokens, and any foreign exchange restrictions that may be relevant thereto.

2.2 Jersey Regulatory Warnings

- 2.2.1 A copy of this document has been delivered to the Registrar of Companies in accordance with Article 5 of the Companies (General Provisions) (Jersey) Order 2002, and he has given, and has not withdrawn, his consent to its circulation.
- 2.2.2 The Commission has given, and has not withdrawn, its consent under Article 2 of the Control of Borrowing (Jersey) Order 1958 to the issue of shares in the Issuer.
- 2.2.3 The Commission is protected by the Control of Borrowing (Jersey) Law 1947, as amended, against liability arising from the discharge of its functions under that Law.
- 2.2.4 It must be distinctly understood that, in giving these consents, neither the Registrar of Companies nor the Commission takes any responsibility for the financial soundness of the Issuer or for the correctness of any statements made, or opinions expressed, with regard to it.
- 2.2.5 If you are in any doubt about the contents of this document you should consult your stockbroker, bank manager, solicitor, accountant or other financial adviser.
- 2.2.6 The Directors of the Issuer have taken all reasonable care to ensure that the facts stated in this document are true and accurate in all material respects, and that there are no other facts the omission of which would make misleading any statement in the document, whether of facts or of opinion. All the Directors accept responsibility accordingly.

3. **SELLING RESTRICTIONS**

This Information Memorandum does not constitute an offer to sell, or the solicitation of an offer to acquire, Tokens in any jurisdiction where such an offer or solicitation is unlawful or would impose any unfulfilled registration, qualification, publication or approval requirements on the Issuer.

3.1 **NOTICE TO US PERSONS OR PERSONS RESIDENT IN THE USA**

The Tokens may not be offered, sold, pledged or otherwise transferred into or within the United States as to or for the account of any U.S. Person.

3.2 **OFFER TO ACCREDITED INVESTORS ONLY IN CERTAIN JURISDICTIONS**

In each of the following jurisdictions, only "accredited investors" (as defined by the law/regulation of each such jurisdiction) may purchase Tokens:

3.2.1 Australia;

3.2.2 Japan;

3.2.3 Canada;

3.2.4 People's Republic of China; and

3.2.5 South Korea.

4. DIRECTORY

The Issuer

RADIX Tokens (Jersey) Limited
First Floor,
La Chasse Chambers,
Ten La Chasse,
St Helier,
Jersey,
JE2 4UE

Technology Provider to the Issuer

Altcoinomy S.A.
Place des Florentins 1
1204 Geneva
Switzerland

Administrator to the Issuer

Cavendish Fiduciary (Jersey) Limited
First Floor,
La Chasse Chambers,
Ten La Chasse,
St Helier,
Jersey,
JE2 4UE

Adviser as to Jersey Law

Carey Olsen Jersey LLP,
47 Esplanade,
St Helier,
Jersey,
JE1 0BD

Directors of the Issuer

Andy Jarrett
Piers Ridyard
James Cunningham-Davis

Auditor to the Issuer

Baker Tilly (Channel Islands) Limited
46/50 Kensington Place
St Helier
Jersey
JE\$ 0ZE

Adviser as to English Law

Keystone Law
48 Chancery Lane
London
WC2A 1JF

5. OVERVIEW OF THE TOKEN OFFERING

The information set out in this summary should be read in conjunction with the full text of this Information Memorandum. Please see the White Papers (each attached as an Appendix to this Information Memorandum) for further information on the Token Offering.

RADIX Foundation and RADIX Ledger

- 5.1 Radix Foundation is a company limited by guarantee incorporated in the United Kingdom. It is a not-for-profit organization whose purpose is to conduct research and promote the education and use of Radix Ledger technology and the Radix Public Network (the “**Network**”).
- 5.2 The Radix Ledger utilizes “Cerberus,” a new, scalable, distributed ledger technology platform whose protocol employs sharding to parallelize consensus agreement to increase transaction and processing throughput (“**Cerberus**”). The Radix Ledger technology will be deployed with the Network, which is a scalable, decentralized, open source, public network with open participation that can support and connect billions of users and transactions.

The RADIX Network and decentralised finance

- 5.3 Elements of the Network, including the Radix Ledger technology, have been in development since 2013. Initial development of the Radix project was conducted exclusively by Mr. Daniel Hughes (Mr Hughes' CV can be found at Section 10.1.4(d) of this Information Memorandum), and was supported, in part, by donations of fiat and cryptocurrency from certain community contributors to the project (the “**Contributors**”) between 2013 and 2017. Mr. Hughes founded the UK company, RADIX DLT Limited (“**DTL**”) in 2017 to further the development of the Radix project and raised additional funds through sales of its equity. Mr. Hughes and DLT have contributed all their interests in the Radix project, including the Radix Ledger technology, Cerberus and the Network to RADIX Foundation.
- 5.4 The Radix project is a decentralized finance protocol whose goal is to provide frictionless access, liquidity and programmability of any digital asset in the world. The Network is a permissionless, open source, distributed ledger. The Network is designed to serve as backbone infrastructure for the growth of decentralized finance and address shortcomings with blockchain technology that prevent scalability, or make scalability prohibitively expensive. The initial version of the Radix Ledger technology, known as “Tempo,” has been tested to exceed over one million transactions per second.

The Network's consensus mechanism – Proof of Stake

- 5.5 RADIX Foundation has a commitment and a mandate to deploy the Radix Ledger via a decentralized public network. Because the Network's security will be provided through proof of stake and the novel consensus mechanism of the Cerberus technology, this decentralization threshold requires that the public control over 67% of the available RADIX Token staking supply on the Network. RADIX Foundation's deployment plan for the Network, including the initial allocation of RADIX Tokens, is programmed to reach 67% public control on day 133 of the Network's life.

Node participation in the Network and RADIX Foundation Charter

- 5.6 The Network will be operated by a network of nodes, which consists of computers running the Network client software ("**Nodes**"). Those that choose to run Nodes are called "Node Runners" ("**Node Runners**"). Anyone may become a Node Runner by downloading and running the Node Runner client software. The Network is secured against attack by a proof of stake sybil prevention mechanism. Holders of RADIX Tokens can choose to stake a portion of their RADIX Tokens. Staking is designed to cause the cost of attacking the Network to rise in proportion to the total value of RADIX Tokens staked. A Node Runner that stakes RADIX Tokens, thereby locking those RADIX Tokens, becomes a "Staking Node" ("**Staking Nodes**").
- 5.7 Although anyone may run the Network client, only Staking Nodes are able to directly participate in consensus protocols on the Network. The Radix Foundation Charter (the "**Charter**") will be released by RADIX Foundation prior to the public Launch of the Network. The Charter will describe a structure and method for achieving Radix community consensus for the adoption of changes to the Network by the Radix community. The charter will also deal with the interaction of RADIX Foundation with the proposed community-led governance of the Network, and development and refinement of the charter itself.
- 5.8 Staking Nodes are compensated daily for participating in the Network consensus through the creation and distribution of new RADIX Tokens (the "**Network Emission**"). The Network Emission is equal to an annual rate of RADIX Token inflation of 2.5%. Network Emission RADIX tokens are earned by Staking Nodes in proportion to the amount of RADIX Tokens staked by the Node compared to the total amount of RADIX Tokens staked in all Nodes. Any holder of RADIX Tokens that does not wish to run a Staking Node, or has insufficient balance to do so itself, will be able to delegate its RADIX Tokens to an existing Staking Node, thereby becoming a "Delegate" ("**Delegate**"). Delegates are eligible to receive a portion of the Network Emissions earned by their affiliated Staking Node.
- 5.9 At Launch, the Network will be unsharded and have an initial maximum of 100 Staking Nodes. Subsequent sharded versions of the Network are planned to substantially extend this initial limit. Holders of RADIX Tokens will be able to delegate their RADIX Tokens to stake with any of these initial 100 Staking Nodes. In the initial implementation of the Network, RADIX Foundation will run numerous Staking Nodes and stake all 600 million RADIX Tokens from the Network Security Allocation (please see Economics White Paper at Appendix 4 for further details). RADIX Foundation anticipates that at Launch, the minimum staking amount will be less than 20 RADIX Tokens, so that virtually anyone may participate in the Network's security and earn from the Network Emission should they wish to do so.
- 5.10 The Network is designed to encourage active participation in the Network by holders of RADIX Tokens. The decentralized nature of the operation of the Network means that the greater the level of participation in the Network by holders of RADIX Tokens, the better the Network will perform in terms of stability, security and robustness. Active participation in this context involves a holder of RADIX Tokens performing at least one of the following roles within the Network: Staking Node, delegating RADIX Tokens to a Staking Node, or engaging in operations on the Network.

6. PURPOSE OF THE RADIX TOKENS

The RADIX Tokens have the following functions:

Payment of Transaction Fees on Network

- 6.1 All Network operations (e.g. transactions, messaging, conditional state creation) will require a change to the state of the Radix Ledger and must be submitted with a transaction fee (“**Transaction Fees**”). RADIX Tokens will be the only way to pay Transaction Fees for conducting transactions involving the Radix Ledger. Once the requested operation has been validated by the Network, the Transaction Fees will be burned.

Participation in Proof of Stake consensus mechanism/dilution by no

- 6.2 Holders of RADIX Tokens will also be able to stake RADIX Tokens by becoming a Staking Node or by delegating RADIX Tokens to an existing Staking Node as a Delegate, thereby supporting the security of the Network through the Network’s proof of stake mechanism and Cerberus consensus protocol. Staking Nodes will be paid RADIX Tokens for staking services from the Network Emission. In addition, Staking Nodes that exceed performance benchmarks set by RADIX Foundation will receive RADIX Token rewards from the Network Subsidy Allocation (please see Economics White Paper at Appendix 4 to this Memorandum for further details).

Dilution of Token holdings by non-participants in staking

- 6.3 Passive holders of RADIX Tokens (ie those who do not perform roles as Staking Nodes or Delegates within the Network or otherwise build on or use the Network) may see their percentage holdings of RADIX Tokens dilute over time as more RADIX Tokens come into existence as a result of the planned Network Emission, Network Subsidy payments, and future RADIX Tokens sales. **Over time, this may result in the dilution of a passive participant’s holdings of RADIX Tokens.**

Limited other rights of RADIX Token holders

- 6.4 Holders of RADIX Tokens will have none of the rights commonly associated with an equity interest or debt instrument. Holders of RADIX Tokens are not shareholders, owners, members or equity holders of RADIX Foundation or the Issuer. Holders of RADIX Tokens cannot participate in the management of RADIX Foundation’s business through voting or contractual rights. Holders of RADIX Tokens do not have redemption or repurchase rights.
- 6.5 Tokens holders are required to acknowledge certain risk factors in the Offering Memorandum inherent in the issuance of digital tokens, including that the Network might not function properly or at all.

7. OFFERING OF INTERIM TOKENS – TARGET FUND RAISING AND TOKEN ALLOCATION

- 7.1 The Issuer, a subsidiary of RADIX Foundation, will issue 12 billion RADIX Tokens at the time the Network mainnet is published, which is currently expected to occur in the second quarter of 2021 (the “**Launch**”).
- 7.2 Prior to Launch, the Issuer intends to sell up to 720 million ERC20 tokens (“**Interim Tokens**”) via the Token Issuance that will be exchanged for or converted into an equal number of RADIX Tokens when the mainnet has been published.
- 7.3 A detailed breakdown of the allocation of Tokens is set out in the Economics White Paper at Appendix 4.

The target fundraising and Minimum Raise

- 7.4 Purchasers will pay for their tokens either:
- 7.4.1 in a "stablecoin" pegged 1:1 to the US\$ known as "USDC", which is administered by Coinbase; and/or
 - 7.4.2 in Ethereum.
- 7.5 Each Ethereum payment will be time-stamped (for example: <https://etherscan.io/tx/0xa1f6bdc74e902c98e1be8a6cf2bf975794846925a44524a1a414e7cd9a3eb1e0>). On the basis of a time stamp, a price feed application programming interface (known as "API") (for example CoinGecko: <https://www.coingecko.com/en/api>) will be utilised to establish the relevant Ethereum to US\$ to conversion rate, and the corresponding amount of Interim Tokens purchased.
- 7.6 The Issuer is seeking to raise up to US\$25 million in the Token Issuance.
- 7.7 If the Token offering fails to attract US\$3,000,000 (the "**Minimum Raise**") the Token Issuance will not take place and Crypto Payments will be returned to purchasers.

Minimum purchase amount/price per Token

- 7.8 The minimum purchase amount is US\$5,000.
- 7.9 The purchase price per Token is US\$0.039 (in USDC and/or Ethereum equivalent).

8. TOKEN SALE MECHANICS

- 8.1 RADIX wish to pre-sell their Interim Tokens to take advantage of current market conditions.
- 8.2 As at the date of this Information Memorandum, a potential purchaser has already registered their interest in purchasing Interim Tokens with Altcoinomy SA, a Swiss regulated crypto service provider and financial intermediary ("**Altcoinomy**") via Altcoinomy's online onboarding platform, by providing personal KYC data, to include ID, utility bill, electronically signed beneficial owner form in addition to

other required information based on their tier classification (please refer to the Altcoinomy online onboarding document).

- 8.3 As part of the registration process, a purchaser has electronically signed the Radix Token Purchase Agreement which has been uploaded on the Altcoinomy web portal (the "TPA") as part of the Altcoinomy KYC and AML checks on the purchaser. Altcoinomy has acted as the pre-incorporation agent for the Issuer pursuant to Article 21 of the Companies Law. **Please note that at no time is Altcoinomy acting as the issuer of the Interim Tokens.**
- 8.4 The TPA stipulated the following:
- 8.4.1 the purchaser will provide satisfactory KYC and AML information to Altcoinomy;
 - 8.4.2 the purchaser has sent USDC (US\$ pegged Coinbase/Circle stablecoin) and/or Ethereum equivalent to pay for the purchase of his Interim Tokens (the "**Crypto Payments**"). There is no payment in fiat currency;
 - 8.4.3 The Issuer will be responsible for the initiation of the Token Issuance;
 - 8.4.4 the TPA is contingent on the Commission's approval of the incorporation of the Issuer. As the Commission has approved the Issuer's Incorporation this contingency falls away.
 - 8.4.5 Each purchaser is required to re-affirm the TPA and confirm (by clicking on a button on the Altcoinomy onboarding platform) that he has read and understood this Information Memorandum.
 - 8.4.6 The Issuer will then draw down the purchaser's Crypto Payments and issue the Interim Tokens to him.
- 8.5 Prior to the Token Issuance, Altcoinomy is custodying Crypto Payments via a 2/3 USDC / Ethereum ERC20 multi-signature wallet which is smart contract based, where Altcoinomy and Radix each hold a key and transfers can only be done with the validation of both keys. Under this scenario, there is no limitation on duration of custody. In the past Altcoinomy used Gnosis MultiSig Wallet (<https://wallet.gnosis.pm>), it has been now superseded by Gnosis Safe (<https://gnosis-safe.io/>).
- 8.6 When the mainnet is launched, the Issuer will exchange the Interim Tokens for mainnet RADIX Tokens on a 1:1 basis.

Transferability in secondary market post Token Issuance

- 8.7 As an ERC20 Token, the Interim Token will be automatically be accepted onto many exchanges and be capable of being traded on the secondary market.
- 8.8 Following exchange of the Interim Token for the RADIX Token, the RADIX team hope that the token will be accepted for listing (and therefore trading) on one or more of the large crypto exchanges.

JERSEY TREATMENT OF ISSUER

- 9.1 There are currently no existing capital markets regulations in Jersey specifically governing token offerings. However, the Commission is aware of the desire of token promoters to use a Jersey incorporated issuer because of Jersey's reputation as a well-regulated and reputable jurisdiction.

- 9.2 Accordingly, the Commission has established certain conditions that any issuer of cryptographic token registered in Jersey is required to satisfy. These are implemented through a consent (a "**COBO Consent**") granted under the Control of Borrowing (Jersey) Order 1958, which any Jersey entity wishing to issue token offering must obtain.
- 9.3 The conditions require the Issuer to take certain measures to manage, amongst other things, financial crime and investor risks. The conditions reflect the guiding principles pursuant to which the Commission discharges its functions as the Island's financial services regulator (the "**Guiding Principles**") which are to have regard to:
- 9.3.1 the reduction of the risk to the public of financial loss due to dishonesty, incompetence, malpractice or the financial unsoundness of financial service providers;
 - 9.3.2 the protection and enhancement of Jersey's reputation and integrity in commercial and financial matters;
 - 9.3.3 the best economic interests of Jersey; and
 - 9.3.4 the need to counter financial crime both in Jersey and elsewhere.
- 9.4 However, whilst the Commission has established certain conditions that issuers of tokens are required to satisfy, it does not regulate or supervise the token offerings or the issuers.
- 9.5 The COBO Consent imposes on the issuer certain requirements which reflect the Guiding Principles, including to:
- 9.5.1 acknowledge that token offerings are a "sensitive activity" falling within the Commission's Sound Business Practice Policy. Accordingly, the issuer must maintain and adopt systems, controls, policies and procedures for the customer take-on, profiling and transaction monitoring at enhanced levels ensuring reporting of suspicions of money-laundering and financing of terrorism activity;
 - 9.5.2 apply relevant AML/CFT requirements to persons that either purchase tokens from, or sell tokens back to, the issuer of those tokens;
 - 9.5.3 appoint and maintain a registered Jersey administrator that is registered by the Commission for "trust company business" under the Financial Services (Jersey) Law 1998 (the "**TCSP**");
 - 9.5.4 appoint and maintain a Jersey resident director on the board of the issuer; where the Jersey resident director is a natural person and a principal person of TCSP appointed by the issuer;
 - 9.5.5 obtain the Commission's prior approval to any change to the TCSP appointed by the issuer , the Jersey resident director of the issuer or additional specified counterparties of the issuer as set out in the COBO consent;
 - 9.5.6 prepare and file annual audited accounts with the Jersey Companies Registry;

- 9.5.7 have procedures and processes in place to (i) mitigate and manage the risk of retail investors investing inappropriately in the token offering, and (ii) to ensure retail investors understand the risks involved;
- 9.5.8 prepare and submit to the Commission an Information Memorandum (which may be in the form of a White Paper) which complies with certain content requirements required of a prospectus issued by a company under the Companies (Jersey) Law 1991; and
- 9.5.9 ensure that any marketing material (including the Information Memorandum) is clear, fair and not misleading.

10. THE ISSUER AND ITS SERVICE PROVIDERS

10.1 The Issuer

- 10.1.1 The Issuer is a private limited company established in Jersey with registered number 132555 and whose registered office address is at First Floor, La Chasse Chambers, Ten La Chasse, St Helier, Jersey, JE2 4UE. 100% of the Issuer's issued share capital (comprising 2 ordinary shares of £1.00 each) are owned by RADIX Foundation.
- 10.1.2 The Directors of the Issuer have control and authority over and responsibility for the operations and management of the Issuer. The Directors of the Issuer may appoint additional persons to act as Directors from time to time. The Directors will be responsible for ensuring compliance with the conditions imposed on the Issuer by the Commission's ICO Guidance Note.
- 10.1.3 The business address of each of the Directors is First Floor, La Chasse Chambers, Ten La Chasse, St Helier, Jersey, JE2 4UE.
- 10.1.4 The Directors of the Issuer are:

- (a) **Piers Ridyard**

A serial entrepreneur, Piers started his professional life in investment banking and law, getting his CFA level 1 before graduating from Manchester University in Chinese and Business, and then earning his Bachelors of Law, as part of his training contract with Linklaters. Piers never actually practiced law as his first startup, Nifty, became the UK most overfunded Kickstarter project in 2012, subsequently allowing him to build a consumer electronic company with products that were distributed through Apple's retail stores globally, and sold through retail and online stores in over 78 countries.

Piers first got into the crypto space in 2015, joining the Ethereum community prior to the first mainnet launch, and mining the Ethereum native currency, ETH. In 2017 Piers launched his first commercial crypto company, Surematics, which used decentralised

ledger technology to create the world's first decentralised dealroom software for clearing and settling large syndicated insurance deals. Surematics was accepted into the world's most prestigious accelerator program, YCombinator, moving to Silicon Valley for the Summer 2017 batch, before selling Surematics to Radix DLT in a 100% equity deal.

In October of 2017, Piers became CEO of Radix DLT, building out a team of international talent and raising two rounds of equity funding to continue the development of the core Radix technology as well as building the Radix community. Piers is an experienced board member and has served for over 8 years as both CEO and executive director at each of his three VC backed companies.

Piers is also a noted public speaker on blockchain and DLT, speaking at both private seminars and as keynote at public crypto conferences.

(b) Andy Jarrett

Andy is an experienced, energetic and Institute of Directors' Chartered director used to working in highly competitive markets with a strong, entrepreneurial approach to acquiring new business. Andy was approached by Digital Jersey in 2014 to help establish Jersey as one of the leading technology centres in Europe. During the next two years, Andy developed strategic plans for industry verticals such as Fintech, eGaming, crypto currency, blockchain and data analytics. Andy led the strategic thinking, consultancy and working group to enable the delivery of the crypto currency regulation. Andy in Jersey during this time, Andy established the founding team of Brainnwave, which specialises in the commercial application of data science and geographical information. The business intelligence platform deploys bespoke tools, proprietary algorithms and technology to help businesses understand their data.

Andy joined Twelve40 as CEO to turn the boards vision into reality, raising finance, developing the board and executive teams to deliver a global client base. Part of this growth included attaining approval for gambling licenses and operating both B2B and B2C platforms. Andy re-engineered the platform to cater for 18 decimal places required for handling crypto currency and introduced elements of blockchain reporting and payments into the product. Andy also, consulted to a number of businesses on the use of blockchain and crypto currency within the gambling sector and the regulatory landscape.

More recently Andy developed two businesses that, whilst commercial, are philanthropic at heart and support the UN Sustainable Development Goals. Andy designed the commercial model and roll out strategy which was scrutinised by Raine Capital in New York as part of the Series A process; one business is a global prize winning application in which Andy designed two regulatory models that enable participation in most countries without regulation including the US; The second

business uses market crossover from the first business, being a donations platform that simultaneously creates a market for fractional ownership of art. Andy designed the technical architecture based on blockchain technology and secured strategic partnerships with exchanges delivering liquidity.

(c) **James Cunningham-Davis**

With over 27 years' experience in the international legal and financial services sector, James is an English qualified Solicitor having qualified with Appleby (formerly, Appleby Spurling & Kempe) in the Isle of Man. In addition James is a Board member of a Regulated entity in Jersey and has responsibility for a significant portfolio of global clients with varied and diverse business interests, ranging from Aviation, online gaming, Employee Benefit Trusts and also a number of multinational trading enterprises and many other entities dealing with asset classes as diverse as Residential and Commercial Real Estate and cryptocurrencies. Many of these relationships are client facing and 'hands on'.

James has held various Directorships and Company Secretarial roles of companies listed on both NEX Exchange (formerly ISDX/ PLUS market) and AIM quoted companies and has had specific involvement, from the initial seeding of the companies and progressing through IPO and thereafter maintaining registration on the respective markets. He has also dealt with de-listing from PLUS to re-list on AIM and in various reverse takeover situations and including involvement with a flotation on NEX Exchange (formerly ISDX).

Additionally, in the past 10+ years James has been increasingly involved in digital enterprises ranging from e characters/ emoji, mobile telephony including prepaid cards and more recently mobile games services. James is also a founding director of a UK licensed, online gaming company and of a digital platform providing next generation platforms for gaming companies, and those engaging with Blockchain for various industries ranging from provenance in food, logistics in aviation cargo and determining carbon emissions for offset to meet with corporations' environmental, social and corporate governance objectives.

(d) **Dan Hughes**

As stated in section 5 of this Information Memorandum, Dan was instrumental in developing the Network and RADIX Ledger technology.

Dan has a long history of breaking new ground in technology and was one of the first to develop applications for NFC technology and contactless payment services such as mobile wallets. He has worked closely with a number of mobile OEM's and operators such as Nokia, Sony Ericsson, T-Mobile, Samsung and others via his previous

successful start-up, KDB Technology where he held the positions of CEO/CTO and chief architect.

After exiting KDB, Dan discovered Bitcoin and the digital currency movement. Bitcoin was great first step, but is not without flaws. After a period of investigation he decided to attempt to develop a more complete solution. This started a 7-year research project into ways that you could build a trustless decentralised ledger that would be capable of supporting 7 billion people and 24 billion devices simultaneously.

Once Dan had finally cracked the formula in 2017 Radix DLT Ltd was formed. Since then, it has grown from a team of 2 people to 20 people. With engineers coming from all over the world to work with Dan in Stoke on Trent in England.

10.2 The Administrator

10.2.1 The Administrator is Cavendish Fiduciary Jersey Limited, a limited company incorporated in Jersey on 23 July 1992 with registration number 53151 and whose registered office address is at First Floor, La Chasse Chambers, Ten La Chasse, St Helier, Jersey, JE2 4UE.

10.2.2 The Issuer has delegated the administration of the Issuer to the Administrator on the terms of an Administration Agreement. Such administration services include the Administrator acting as the company secretary of the Issuer.

10.2.3 The Administrator is the holder of a licence to conduct trust company business and is regulated by the Commission.

10.3 RADIX DLT

10.4 RADIX DLT Limited ("**DLT**") is a UK company, whose share capital is divided into:

10.4.1 ordinary shares; and

10.4.2 "Seed Preferred" Shares.

10.5 Piers Ridyard and Daniel Hughes are the majority ordinary shareholders (and the only shareholders with over 10% of DLT's ordinary share capital). There are a number of smaller ordinary shareholders.

10.6 DLT has also conducted some external fundraising pursuant to the issue of Seed Preferred Shares – here is a link to DLT's latest fundraising <https://www.finextra.com/pressarticle/83252/radix-dlt-raises-41m-for-public-release-of-defi-protocol>.

10.7 **Altcoinomy S.A.** -. Altcoinomy SA is a Swiss financial intermediary and digital asset service provider supervised by the Financial Services Standards Association (VQF), a financial regulatory authority specialized in monitoring companies active in the fintech industry. Altcoinomy provides KYC/AML services to corporate issuers as well as OTC brokerage services to digital asset owners. As part of its services, Altcoinomy complies with stringent AML procedures and is bound to the Swiss banking act that ensures the highest level of confidentiality. Altcoinomy's affiliation with VQF can be found by consulting the official Swiss Financial Market Supervisory Authority (FINMA) website:

<https://www.finma.ch/en/authorisation/self-regulatory-organisations-sros/sro-member-search/#query=altcoinomy&Order=1> (please enter Altcoinomy SA in the search bar)

10.8 Altcoinomy and the Issuer have entered into an "**AMC – Customer Due Diligence Services Agreement**" pursuant to which Altcoinomy will undertake certain Token purchasers on-boarding services for the Issuer, including conducting AML/KYC checks on potential purchasers of the Tokens.

10.9 **Copper** will provide custody services to the Issuer for the Interim Token once the Issuer has been incorporated and the Altcoinomy has transferred the Interim Tokens to the Issuer. "Copper" is the trading name of Copper Technologies Limited, a UK company. Here is a link to Copper's website <https://copper.co>.

10.10 **The Auditor**

10.10.1 The Auditor of the Issuer is Baker Tilly (Channel Islands) Limited, a company incorporated in Jersey on 12 July 2001 with company number 80451 and whose registered office address is PO Box 437, 1st Floor Kensington Chambers, 46/50 Kensington Place, St Helier, Jersey JE4 0ZE.

10.10.2 The Auditor's principal responsibilities are to audit and express an opinion on the financial statements of the Issuer in accordance with applicable law and accounting standards.

10.11 **Additional Service Providers**

10.11.1 Additional service providers may be appointed from time to time according to the requirements of the Issuer.

11. **RIGHTS ATTACHING TO A TOKEN**

11.1 **Economic Rights attaching to a Token**

11.1.1 As Tokenholders are not shareholders of the Issuer, they have no rights under the Articles of Association.

11.1.2 Tokenholders are not entitled to:

- (a) vote at any meetings of the Issuer;
- (b) vote for the election of Directors or upon any matter submitted to Shareholders at any meeting thereof;
- (c) give or withhold consent to any corporate action or to receive notice of meetings, or otherwise;
- (d) receive dividends from the Issuer; or
- (e) be deemed the holder of capital stock of the Issuer for any purpose.

11.2 Tokenholders will be entitled to information on developments in the RADIX project via multiple sources, including:¹

11.2.1 the Issuer's website; and

11.2.2 social media feeds such as Facebook and Twitter.

12. **TAXATION**

12.1 **Jersey Taxation**

12.1.1 It is the responsibility of all prospective Tokenholders to inform themselves as to any tax consequences arising from their purchase of any Tokens and the Issuer's operations or management, as well as any foreign exchange or other fiscal or legal restrictions, which are relevant to their particular circumstances in connection with the acquisition, holding or disposition of the Tokens. Prospective Tokenholders should therefore seek their own separate tax advice in relation to their holding of the Tokens and accordingly the Issuer does not accept any responsibility for the taxation consequences of any purchase of Tokens by any Tokenholder.

12.1.2 The Issuer is treated as a separate 'zero rated company' and will fall to be assessed under Article 123C of the Income Tax (Jersey) Law 1961, as amended (the "**1961 Law**"), as a Jersey resident company which is neither a "utility company" nor a "financial services company" and as such will be charged to Jersey income tax at a rate of zero per cent (0%) on its income (other than on receipts chargeable to tax under Schedule A of the 1961 Law - which relates broadly to income or profits derived from the ownership, disposal or development of land in Jersey).

12.1.3 No death duties, capital gains tax, gift, inheritance or capital transfer taxes are levied in Jersey. No stamp duty is levied in Jersey on the issue, transfer or buyback of Tokens held, but probate stamp fees may be payable at the rate of up to 0.75% of the value of the Jersey estate in the event of the death of the Tokenholder.

12.1.4 The attention of Jersey residents is drawn to the provisions of Article 134A of the 1961 Law which may in certain circumstances render such a resident liable to income tax on any undistributed income or profits of the Issuer.

12.1.5 A Jersey goods and services tax ("**GST**") applies at a standard rate of five per cent (5%) on the majority of goods and services supplied in Jersey for local use or benefit. The Issuer will apply for International Services Entity status under the Goods and Services Tax (Jersey) Law 2007. In connection with their International Services Entity status the Issuer will pay an annual fee to the Comptroller of Income Tax in Jersey. As an International Services Entity,

¹ RADIX to confirm/expand

the Issuer will not be required to charge GST and in most situations will not be subject to a GST charge on goods and services provided to them.

- 12.2 **This Information Memorandum does not address legal, regulatory or taxation issues outside of Jersey. Accordingly, prospective Tokenholders should consult their professional advisers on the potential tax, exchange control and other consequences of purchasing, holding or selling Tokens under the laws of their country of citizenship, domicile or residence.**

13. **CONFLICTS OF INTEREST**

- 13.1 The Directors, the Issuer and the Administrator or companies with which any of them are associated may from time to time act in relation to, or be otherwise involved in, other companies which have similar objectives to those of the Issuer. It is therefore possible that any of them may, in the course of business, have potential conflicts of interest with the Issuer. Each will, at all times, have regard in such event to its obligations to the Issuer and will endeavour to ensure that such conflicts are resolved fairly. In addition, any of the foregoing may deal as principal or agent with the Issuer, provided that such dealings are carried out as if effected on normal commercial terms negotiated on an arm's length basis. Neither the Issuer nor any of its affiliates nor any person connected with it is under any obligation to offer any token offering opportunities of which any of them becomes aware to the Issuer or to account to the Issuer in respect of (or share with the Issuer or inform the Issuer of) any such transaction or any benefit received by any of them from any such transaction. In determining the net asset value of the Issuer, the Directors may rely on valuations provided or attributed to any asset or liability by the Administrator and/or the Issuer.
- 13.2 Members of the Issuer's group, their affiliates or any person connected with them advise, sponsor or manage other companies, vehicles or accounts in which only the Issuer's investors, entities and partners, employees, affiliates or other persons connected with the Issuer's group may invest. Such companies, vehicles or accounts may pursue the same or a similar investment objective and use the same or a similar investment approach as the Issuer, or may employ investment approaches that are more or less leveraged or risky. The partners, employees or affiliates of members of, or other persons connected with, the Issuer's group, or other investment professionals, involved in advising, sponsoring or managing such companies, vehicles or accounts may, or may not, provide similar services to, or fulfil similar roles in respect of, the Issuer. Accordingly, such proprietary companies, vehicles or accounts may produce investment results that are substantially different from those of the Issuer. To the extent that the Issuer invests in similar markets and investments at or about the same time, such other companies, vehicles or accounts may compete with the Issuer with respect to such investments. The potential fees payable to the Issuer or another member of the Issuer's Group by another Issuer entity might in certain circumstances exceed the potential fees payable by the Issuer. Members of the Issuer's Group will allocate resources as they in their sole discretion consider appropriate in managing the assets of the Issuer and any other proprietary and/or non-proprietary companies, vehicles or accounts in accordance with their respective investment objectives and approaches.

- 13.3 Any Director of the Issuer who has, directly or indirectly, an interest in a transaction entered into or proposed to be entered into by the Issuer or by a subsidiary of the Issuer which to a material extent conflicts or may conflict with the interests of the Issuer, and of which the Director is aware, will be required to disclose to the Issuer the nature and extent of their interest. Notwithstanding such interests, each of the Directors will be entitled to be counted in the quorum and vote on the matters to be discussed at the Issuer's board meetings.
- 13.4 The Directors will seek to ensure that any conflict of interest of which they are aware is resolved fairly.
- 13.5 By acquiring or continuing to hold Tokens, each Tokenholder will be deemed to have acknowledged the existence of the actual or potential conflicts of interests described above and to have waived, to the fullest extent permitted by applicable law, any claim with respect to the existence of any such conflicts.
- 13.6 The foregoing does not purport to be a complete list of all potential conflicts of interest involved in a purchase of any Tokens.

14. **STATUTORY AND GENERAL INFORMATION**

This section contains statutory and general information in relation to the Token Offering. Please see the White Papers (attached as Appendices to this Information Memorandum) for further information on the Token Offering.

14.1 **The Token Offering**

Having fully considered the White Papers, applicants are invited, subject to the terms of the White Papers, and this Information Memorandum, to purchase Tokens issued by the Issuer.

14.2 **Opening and Closing dates of the Token Offering**

From the date of this Information Memorandum, purchasers have 90 calendar days (the "**Offer Period**") to re-affirm the TPA and confirm (by clicking on the Altcoinomy's on-boarding platform) that they still wish to purchase Tokens, in which event the Issuer will draw down the purchaser's Crypto Payments and issue Interim Tokens.

14.3 **The Maximum Number of Tokens pursuant to the Token Offering**

14.3.1 A maximum of 720 million Tokens will be issued by the Issuer pursuant to this Token offering.

14.3.2 Please see the Economics White Paper for further details.

14.4 **Token Issue Price**

14.4.1 The Issuer will issue Tokens at US\$0.039 (in USDC and/or Ethereum equivalent) during the Offer Period.

14.5 **Minimum Amount Required to be Raised by the Token Offering**

The minimum amount required to be raised by the Token Offering is US\$3,000,000 (the "**Minimum Raise**").

14.6 **Proceeds of the Token Offering**

Please see Sections 5 and 6 of this information Memorandum for details of the use of proceeds of the Token Offering.

14.7 **Allocation**

The Issuer reserves the right to decline in whole or in part any attempt to purchase Tokens. Accordingly, prospective Tokenholders may, in certain circumstances, not be allotted and issued the number of Tokens for which they have applied.

14.8 **Changes to the Information Memorandum**

In the event that there are any significant changes affecting any of the matters described in this Information Memorandum or where any significant new matters have arisen after the publication of this Information Memorandum and prior to the Token Offering, the Issuer may publish a supplementary document. Any such supplementary document will give details of the significant change(s) or the significant new matter(s).

14.9 **Payment**

14.9.1 Payment for the Tokens should be made in accordance with the TPA.

14.9.2 To the extent that any attempt to purchase Tokens is rejected in whole or in part (whether by scaling back or otherwise), monies received will be returned without interest at the risk of the applicant as soon as practicable thereafter.

14.9.3 The Issuer will not be held liable for any costs incurred in returning funds, or the movements in exchange rates if any funds have been converted from one currency to another during the purchase process.

14.10 **Costs and expenses of the Token Offering**

14.10.1 From the proceeds of the Token Offering, the minimum proceeds allocated to the costs and expenses of the Token Offering and the subsequent development of the Network (together the "**Total Costs**") will be US\$500,000, provided that if the Token Offering raises more than US\$5,000,000, Total Costs will be capped at 10% of such proceeds.

14.10.2 In the event the Token Offering is abandoned for whatever reason, DLT will cover all expenses incurred in connection with the Token Offering.

14.11 Anti-money laundering / due diligence

14.11.1 Each of the Issuer and its agents, reserves the right to request such information as is necessary:

- (a) to satisfy applicable due diligence requirements; and
- (b) to verify the identity of a Tokenholder or prospective Tokenholder and (if any) the underlying beneficial owner or prospective beneficial owner of a Tokenholder's or prospective Tokenholder's Tokens.

14.11.2 In the event of delay or failure by the Tokenholder or prospective Tokenholder to produce any information required for due diligence and / or verification purposes, the Directors, in consultation with any of the Issuer's agents, will refuse to accept an attempt to purchase Tokens. Tokens will only be issued once KYC has been completed.

14.12 Purchase and transfer restrictions

This Information Memorandum does not constitute an offer to sell, or the solicitation of an offer to acquire, Tokens in any jurisdiction where such an offer or solicitation is unlawful or would impose any unfulfilled registration, qualification, publication or approval requirements on the Issuer.

14.13 Legal implications of purchasing Tokens in the Token Offering

14.13.1 A prospective Tokenholder irrevocably offers to purchase Tokens, which offer is capable of acceptance or rejection by the Directors either in whole or in part. If the offer is accepted by the Directors either in whole or in part this Information Memorandum forms a binding contract between the Issuer and the Tokenholders.

14.13.2 None of the agreements appointing the Administrator, Auditor, legal counsel or any other of the Issuer's service providers provide for any third party rights in favour of the Tokenholders.

14.14 Rights Attaching to a Token

Please see section 6.4 of this Information Memorandum for more information on the rights attaching to a Token.

14.15 Financial Information and Reports

14.15.1 The Issuer's year-end will be 31 December. The first financial year of the Issuer will end on 31 December 2021.

14.15.2 The audited annual accounts of the Issuer will be available on the Issuer's website within 6 months from the end of the financial year to which they relate. The Issuer may also issue management updates from time to time.

14.16 **Distribution Policy**

Directors currently intend to use any proceeds from the Token Offering in accordance with Section 5 of this Information Memorandum and do not intend to make any distributions, dividends or ex gratia payments of any kind to Tokenholders.

14.17 **Incorporation and Share Capital**

14.17.1 The Issuer was incorporated in Jersey on 16 October 2020 with registered number 132555 and with its registered office and principal operating establishment at First Floor, La Chasse Chambers, Ten La Chasse, St Helier, Jersey, JE2 4UE.

14.17.2 As at the date of this Information Memorandum:

- (a) the Issuer's authorised share capital is £10,000 made up of shares of par value £1.00 each; and
- (b) the Issuer's issued share capital comprises 2 ordinary shares (fully paid) issued to RADIX Foundation.

14.17.3 The Directors retain the right to issue an unlimited number of shares.

14.17.4 No other subscriptions, allotments or options in the Issuer have been given or are already existing in respect of any securities of the Issuer.

14.18 **Registered Office and Register of Members**

14.18.1 The registered office of the Issuer is First Floor, La Chasse Chambers, Ten La Chasse, St Helier, Jersey, JE2 4UE.

14.18.2 The Issuer's register of members will be held at First Floor, La Chasse Chambers, Ten La Chasse, St Helier, Jersey, JE2 4UE.

14.18.3 There will not be a register of members in relation to the Tokenholders, as ownership of the Tokens is evidenced on a blockchain.

14.19 **Directors of the Issuer**

14.19.1 The Directors of the Issuer are as follows:

- (a) Piers Ridyard
- (b) Andy Jarrett
- (c) James Cunningham-Davis

14.19.2 For further details on the Directors please view the biographies at Section 10 (Management and Administration) of this Information Memorandum.

14.19.3 Piers Ridyard is a Director of the Issuer and both a guarantor member of RADIX Foundation and a shareholder in DLT.

14.19.4 James Cunningham-Davis is a Director of the Issuer and a director of the Administrator.

14.19.5 Please see section 13 (Conflicts of Interest) of this Information Memorandum for more information on conflicts of interest.

14.19.6 Note that the use of the term 'founder' in any Token Offering documentation (including the White Papers) is not a term of art and such persons, to the extent that they are not specifically listed as Directors or shareholders of the Issuer, are to be considered 'middle management'.

14.20 Litigation

The Issuer has not since its incorporation been in nor is it engaged in any legal or arbitration proceedings and no legal or arbitration proceedings are pending or threatened against the Issuer which may have or have had a significant effect on the financial position of the Issuer.

14.21 Documents available for inspection

Copies of the following documents may be inspected during usual business hours on any Business Day at the registered office of the Issuer in Jersey at the address stated in Section 4 (Directory) of this Information Memorandum:

14.21.1 this Information Memorandum; and

14.21.2 the latest annual report and accounts of the Issuer.

14.22 Material Contracts

The Issuer in the ordinary course of its business has, or will have, entered into agreements for the procurement of necessary facilities and services, including the Administration Agreement.

14.23 The Issuer has not entered into any contracts which are not in the ordinary course of the business of the Issuer, and since the Issuer was incorporated on 16 October 2020, it cannot have entered into any contracts before that date.

14.24 Data Protection

14.24.1 The Privacy Notice confirming how personal information is collected, processed and disclosed by the Issuer, together with a Tokenholder's rights under (i) the General Data Protection Regulation (2016/679) and any national law issued under that regulation; and (ii)

the Data Protection (Jersey) Law 2018 and any other legislation in Jersey concerning data protection, as amended from time to time is included at Appendix 1 to this Information Memorandum.

14.24.2 It is a condition of issuing Tokens to a Tokenholder that every Tokenholder agrees that:

- (a) any personal data provided to the Issuer and/or its affiliates is accurate and complete and that it may be lawfully processed by the Issuer and/or its affiliates for the purposes set out in the Privacy Notice;
- (b) where the consent of any data subject is required, it has all necessary authority to provide the personal information on behalf of any relevant individual; and
- (c) it will make the Privacy Notice available to each Relevant Individual (as defined in Appendix 1) and draw their attention to it.

14.24.3 The data protection registration of the Issuer can be found on the website of the Jersey Information Commissioner: <https://oicjersey.org/>.

15. DEFINITIONS

The following definitions will apply throughout this Information Memorandum:

"Administrator"	has the meaning as set out at Section 10.2 of this Information Memorandum;
"Administration Agreement"	means the agreement entered into between the Administrator and the Issuer, as amended, substituted or supplemented from time to time;
"Articles of Association"	means the memorandum and articles of association of the Issuer, as amended, substituted or supplemented from time to time;
"Auditor"	has the meaning as set out at Section 10.10 of this Information Memorandum;
"Business Day"	means any day on which banks in Jersey and London are open for normal banking business (excluding Saturdays and Sundays);
"Cerberus"	has the meaning set out in Section 5.2 of this Information Memorandum;
"Charter"	has the meaning set out in Section 5.7 of this Information Memorandum;
"COBO Consent"	has the meaning set out in Section 9.2 of this Information Memorandum;
"Consensus White Paper"	means the "consensus white paper" published by RADIX Foundation and available on the RADIX Website, a copy of which is attached at Appendix 3 to this Information Memorandum;
"Commission"	means the Jersey Financial Services Commission;
"Companies Law"	means the Companies (Jersey) Law 1991 (as amended);
"Crypto Payments"	has the meaning set out in Section 8.4.2 of this Information Memorandum;
"DLT"	means RADIX DLT Limited;

"Defi White Paper"	Means the "defi white paper" published by RADIX Foundation and available on the RADIX Website, a copy of which is attached at Appendix 2 to this Information Memorandum;
"Delegate"	has the meaning set out in Section 5.8 of this Information Memorandum;
"Directors"	means the directors of the Issuer for the time being, or as the case may be, the directors assembled as a board or as a committee thereof and "Board of Directors" shall have a corresponding meaning;
"Economics White Paper"	means the "economic white paper published by RADIX Foundation and available on the RADIX website, a copy of which is attached at Appendix 4 to this Information Memorandum;
"ESMA"	means the European Securities and Markets Authority;
"Information Memorandum"	means this information memorandum as amended, substituted or supplemented from time to time;
"Interim Token"	means an ERC 20 token issued in accordance with Section 7.2 of this Information Memorandum;
"Issuer"	has the meaning as set out at section 10.1 of this Information Memorandum;
"Launch"	has the meaning set out in Section 7.1 of this Information Memorandum;
"Minimum Raise"	has the meaning at as set out at Section 14.5 of this Information Memorandum;
"Network" or "RADIX Network"	has the meaning set out in Section 5.1 of this Information Memorandum;
"Network Emission"	has the meaning set out in Section 5.8 of this Information Memorandum;
"Nodes"	has the meaning set out in Section 5.6 of this Information Memorandum;
"Node Runners"	has the meaning set out in Section 5.6 of this Information Memorandum;

"Offer Period"	has the meaning as set out at Section 14.2 of this Information Memorandum;
"Privacy Notice"	means the privacy notice published by the Issuer, a copy of which is attached as Appendix 1 to this Information Memorandum, which sets out how personal information is collected, processed and disclosed by the Issuer;
"RADIX Token"	means a token as described in Sections 5 and 6 of this Information Memorandum;
"RADIX Website"	means the website www.radixdlt.com ;
"Staking Nodes"	has the meaning set out in Section 5.6 of this Information Memorandum;
"Token"	means in relation to the Initial Offering, an Interim Token and, upon exchange for a RADIX Token in accordance with Section 8 of this Information Memorandum, means a RADIX Token;
"Tokenholder"	means a holder of any amount of Tokens;
"Token Offering"	means the initial offering of the Tokens;
"Transaction Fees"	has the meaning set out in Section 6.1 of this Information Memorandum;
"TPA"	has the meaning set out in Section 8.3 of this Information Memorandum;
"US\$"	means the lawful currency of the United States of America;
"USDC"	has the meaning set out in Section 7.4 of this Information Memorandum;
"U.S. Person"	has the meaning of "U.S. person" in Rule 902 of Regulation S under the U.S. Securities Act;
"White Papers"	means the Defi White Paper, the Consensus White Paper and the Economics White Paper;
"£"	means the lawful currency of the United Kingdom.

APPENDIX 1 – PRIVACY NOTICE

1. INTRODUCTION

- 1.1 This Privacy Notice sets out how personal information is collected, processed and disclosed in connection with the issue of Tokens.
- 1.2 As a result of a proposed or actual purchase of Tokens (the "**Tokens**") issued by RADIX Tokens (Jersey) Limited (the "**Issuer**") either made by you or a firm or entity with which you have a connection (the "**Applicant**"), your personal information and/or the personal information of other relevant individuals of the Applicant (such as directors, officers, employees or beneficial owners or persons connected to the foregoing) may be provided to the Issuer, Cavendish Fiduciary Limited (the "**Administrator**") and/or any of their affiliates (where such relevant individuals and the Applicant shall together be referred to as the "**Relevant Individual**").
- 1.3 The Issuer will act as data controller in respect of its use of personal information provided by a Relevant Individual. In this Privacy Notice, "**we**" or "**us**" refers to the Issuer and/or its affiliates (as applicable) and "**you**" refers to the Relevant Individual.
- 1.4 Service providers appointed by the Issuer (including the Administrator, Altcoinomy, the Auditor, and any third party AML provider) may also process personal information relating to Relevant Individuals when conducting administrative and other activities relating to the issue of Tokens. Where they are required to do so in order to comply with their own legal and regulatory obligations, they will do so as data controllers in their own right.
- 1.5 This Privacy Notice sets out how the personal information of Relevant Individuals is collected, processed and disclosed in connection with the Applicant's purchase of Tokens from the Issuer. Unless otherwise provided for in this Privacy Notice, terms and expressions defined elsewhere in this Information Memorandum shall have the same meaning where used in this Privacy Notice.

2. THE PERSONAL INFORMATION WE PROCESS

- 2.1 We may collect various types of personal data about you, including:
 - 2.1.1 identification information (which may include your name, ID card and passport numbers, nationality, place and date of birth, gender, photograph and/or IP address and personal data relating to claims, court cases and convictions, politically exposed person (PEP) status, personal data available in the public domain and such other information as may be necessary for the Issuer and its service providers to perform their functions and comply with their CDD processes and discharge AML/CFT obligations);
 - 2.1.2 tax status information (which may include your tax residency, tax ID and/or your tax status);
 - 2.1.3 contact information (which may include postal address and e-mail address and your home and mobile telephone numbers);

- 2.1.4 about your family relationships (which may include your marital status, the identity of your spouse and the number of children that you have);
 - 2.1.5 professional and employment information (which may include your level of education and professional qualifications, your employment, employer's name and details of directorships and other offices which you may hold); and
 - 2.1.6 financial information, sources of wealth and your assets (which may include details of your, shareholdings and your beneficial interest in assets, your bank details and your credit history).
- 2.2 We may also collect and process personal data regarding people connected to you, either by way of professional (or other) association or by way of family relationship.
3. **WHERE WE OBTAIN YOUR PERSONAL INFORMATION:**
- 3.1 We collect personal information from the following sources:
- 3.1.1 personal information which data subjects provide to us or which is given to us by the Applicant, including but not limited to:
 - 3.1.2 information collected via the Issuer's website;
 - 3.1.3 such other forms and documents as we may request that are completed in relation to the administration/management of any purchase of Tokens from the Issuer;
 - 3.1.4 information gathered through client due diligence carried out as part of our compliance with regulatory requirements; or
 - 3.1.5 any personal information provided by way of correspondence with us by phone, e-mail or otherwise;
 - 3.1.6 personal information we receive from third party sources, such as:
 - 3.1.7 entities in which you or someone connected to you has an interest;
 - 3.1.8 your legal and/or financial advisors;
 - 3.1.9 other financial institutions who hold and process your personal information; and
 - 3.1.10 credit reference agencies and financial crime databases for the purposes of complying with our regulatory requirements; and
 - 3.1.11 personal information received in the course of dealing with advisors, regulators, official authorities and service providers by whom you are employed or engaged or for whom you act.

4. **WHY WE COLLECT YOUR PERSONAL INFORMATION:**

Lawful grounds for processing:

4.1 We may hold and process your personal information on the following lawful grounds, namely where:

- 4.1.1 the processing is necessary for our legitimate interests, provided your interests and fundamental rights do not override those interests;
- 4.1.2 the processing is necessary to comply with our respective contractual duties to the Applicant under the terms of the RADIX website, this Information Memorandum and any supplemental agreements thereto;
- 4.1.3 the processing is necessary to comply with our legal and regulatory obligations;
- 4.1.4 (on rare occasions) where we have obtained consent to processing personal information for a specific purpose; and
- 4.1.5 on rare occasions, where it is needed in the public interest.

Purposes of processing

4.2 Pursuant to paragraph 4.1 above, personal information may be processed for the purposes set out below ("**Purposes**"). The Purposes based on our legitimate interests are set out in paragraphs 4.2.1 to 4.2.2 inclusive):

- 4.2.1 facilitating the administration of the Issuer and its service providers;
- 4.2.2 communicating with data subjects as necessary in connection with the Applicant's purchase of Tokens from the Issuer;
- 4.2.3 monitoring and recording telephone and electronic communications and transactions:
 - (a) for quality, business analysis, training and related purposes in order to improve service delivery; and
 - (b) for investigation and fraud prevention purposes, for crime detection, prevention, investigation and prosecution of any unlawful act (or omission to act);
- 4.2.4 to comply with the listing, legal, regulatory, reporting and/or financial obligations of the Issuer or any legal or regulatory obligations of any service provider or functionary (or his/her employer) of the Issuer;
- 4.2.5 disclosing your personal information (including your identity and interest in the Issuer) to any bank, financial institution or other third party lender providing any form of facility, loan, finance or other form of credit or guarantee to the Issuer;

- 4.2.6 to enforce or defend the rights of the Issuer, or those of third parties who provide services to the Issuer;
- 4.2.7 collecting, processing, transferring and storing customer due diligence, source of funds information and verification data under applicable anti-money laundering and terrorist financing laws and regulations; and
- 4.2.8 liaising with or reporting to any regulatory authority (including tax authorities) with whom we are either required to cooperate or report to, or with whom we decide or deem it is appropriate to cooperate in relation to an investment, and which has jurisdiction over the Issuer or its investments.

5. **SHARING PERSONAL INFORMATION**

- 5.1 We may share your personal information with our group companies and third party service providers, including the Administrator, any third party AML provider, banks, financial institutions or other third party lenders, IT service providers, auditors and legal professionals to facilitate the running of the issuance of the Tokens. The group companies and third party service providers to whom your information may be transferred are based in Jersey, the UK and Ireland.
- 5.2 Where we share your information with a third party, we require the recipients of that personal information to put in place adequate measures to protect it.
- 5.3 Where we transfer your personal information outside the European Economic Area, we will ensure that it is protected and transferred in a manner consistent with legal requirements applicable to the information. This can be done in a number of different ways, for instance:
 - 5.3.1 the country to which we send the personal information may be approved by the European Commission as providing adequate protection for personal data;
 - 5.3.2 by utilising a contract based on "model contractual clauses" approved by the European Commission; or
 - 5.3.3 where the recipient is in the US, it may belong to the EU-US Privacy Shield scheme.
- 5.4 In other circumstances, the law may permit us to otherwise transfer your personal information outside the EEA.
- 5.5 If you would like further information about the safeguards we have in place to protect your personal information, please contact the Administrator.

6. RETENTION OF PERSONAL INFORMATION

6.1 Your personal information will be retained for as long as required:

- 6.1.1 for the Purposes for which the personal information was collected;
- 6.1.2 to establish or defend legal rights or obligations or to satisfy any reporting or accounting obligations; and/or
- 6.1.3 as required by data protection laws and any other applicable laws or regulatory requirements.

7. ACCESS TO AND CONTROL OF PERSONAL INFORMATION

7.1 You have the following rights (which are subject to certain conditions and exceptions) in respect of the personal information about you that we process:

- 7.1.1 the right to access and port personal information;
- 7.1.2 the right to rectify personal information;
- 7.1.3 the right to restrict the use of personal information;
- 7.1.4 the right to request that personal information is erased; and
- 7.1.5 the right to object to processing of personal information.

7.2 You also have the right to lodge a complaint about the processing of your personal information either with us, with the Office of the Data Protection Commissioner in Jersey (<https://oicjersey.org/>) or the data protection authority in the EU member state of your usual residence or place of work.

7.3 Where we have relied on consent to process your personal information, you have the right to withdraw consent at any time.

7.4 If you wish to exercise any of the rights set out in this paragraph 7, please contact the Administrator.

8. INACCURATE OR AMENDED INFORMATION

Please let us know as soon as possible if any of your personal information changes (including your correspondence details). Failure to provide accurate information or to update information when it changes may have a detrimental impact upon the Applicant's purchase of Tokens from the Issuer.

Failure to provide information where it is required for anti-money laundering or other legal requirements means that we may not be able to accept the Applicant as a Tokenholder.

9. **QUESTIONS**

- 9.1 If you have any questions about this Privacy Notice or how we handle your personal information (e.g. our retention procedures or the security measures we have in place), or if you would like to make a complaint, please contact the Administrator.
- 9.2 This Privacy Notice is up-to-date as of the date of this Information Memorandum. If it is updated, we will provide the Applicant with the updated version and require them to bring it to your attention.

APPENDIX 2 – DEFI WHITE PAPER

Radix DeFi White paper

Delivering Security, Interoperability,
Incentives and Scale for DeFi

🔒 13 August 2020 v1.0



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Introduction

Friction in today's financial system costs the world [0.05% of global GDP](#) (approximately \$71Bn per annum), and delivers financial products and services that are often inflexible and benefit primarily those at the top of the food chain. Radix believes this is the most important problem that the right permissionless DLT network can solve.

The concept of Decentralised Finance (DeFi) offers an alternative, where innovative new financial applications can be quickly developed and easily accessed by all. The right permissionless DLT network can enable this new era of financial innovation and democratization, providing an open infrastructure of programmable assets that can substantially replace the closed, walled gardens of traditional banking infrastructure today. Delivering on the promise of DeFi requires a platform that is equal to the task of becoming the new financial layer for the connected world. DeFi as it exists today (as a ~\$4Bn market) covers only a narrow set of financial services for a narrow set of users who are already excited about crypto. But the growth of this early DeFi market is already constrained by today's technology. Apps are reaching the limits of performance of available DLT platforms, and frequent exploits keep the majority of mainstream users away. These technology limitations will only become more urgent to solve as DeFi grows into an anticipated \$64Bn market over the next two years.

Radix intends to remove the technology barriers limiting the expansion of DeFi by building a layer-1 protocol that directly addresses the technology needs of DeFi, both for today's applications and for the future of general financial services for the world. This is a full stack approach, re-engineering consensus, distributed virtual machines, executable on-ledger code, DeFi component building, DeFi application building and developer incentives.

Re-shaping finance will be a 40 year sprint. To do so, the foundations have to be right.

Summary

This paper describes how the Radix platform provides an integrated technology solution to multiple problems that create significant barriers to DeFi's potential.

The growth of DeFi rests on the shoulders of developers who are building the new decentralized applications that will replace traditional closed systems. The barriers holding back DeFi are precisely those that hold back those developers. We believe there are four problems that DeFi developers face today that a DLT platform must solve before DeFi can truly go mainstream:

1. **Reduce smart contract app hacks, exploits, and failures**
2. **Build interoperable DeFi dApps faster**
3. **Incentivize a decentralized development community**
4. **Scale public ledger dApps without breaking DeFi composability**

While these challenges also apply to the majority of use cases for permissionless DLT platforms, DeFi needs a purpose-built solution all the more urgently because users' personal wealth is at stake in every transaction.

These problems are each associated with four crucial layers of a complete DeFi platform stack; Radix addresses them layer by layer.

Starting at the bottom of the stack, developers need a new **consensus and network** design to make DeFi scalable without breaking composability; something that "scalable" networks such as Ethereum 2.0, Near, Polkadot, Cosmos and Avalanche all fail at. This problem is the focus of our **Cerberus**¹ consensus protocol.

On that foundation, a new **development environment** is needed to let DeFi developers confidently create a new kind of smart contract that lowers the risks of hacks, exploits, and code failures. Then to accelerate the building of interoperable DeFi apps, an on-ledger system is needed that provides access to modular **DeFi "lego bricks"**. And finally, to rapidly grow a community around these developer tools, we need a system that creates a decentralised, self-incentivizing **developer ecosystem**, similar to blockchain's current self-incentivizing network infrastructure. These three problems are the focus of the **Radix Engine**² and two of its unique features: the **Component Catalog** and the **Developer Royalty System**.

Barriers to DeFi Growth	Platform Layers	Radix Solutions
Scale Dapps without breaking DeFi composability	Consensus/Network	← Cerberus
Avoid smart contract app hacks, exploits, and failures	Development Environment	← Radix Engine
Build interoperable DeFi dApps faster	DeFi "Lego Bricks"	← Component Catalog
Incentivize a decentralized development community	Developer Ecosystem	← Developer Royalties

We believe Radix's comprehensive, integrated bottom-to-top approach is what uniquely makes it the first layer-1 DLT protocol uniquely suited for DeFi.

¹ The initial Radix mainnet launch will include a simplified version of Cerberus consensus. The capability described in this paper pertains to the full version of Cerberus intended for a later mainnet update.

² The initial Radix mainnet launch will include Radix Engine v1. The Radix Engine features described in this paper pertains primarily to the following Radix Engine v2.

1. Reduce Smart Contract Hacks, Exploits, and Failures

“Smart contract” has become a generic term to refer to application code that a developer writes to deploy to a distributed ledger. Following Ethereum’s pioneering use of smart contracts to make blockchains more programmable, there has been an explosion of new smart contract blockchain platforms that largely replicate the specific implementation adopted by Ethereum: that is, a general purpose Turing-complete language and virtual machine running on-ledger. Tezos, EOS, NEO, Tron, Hashgraph, Hyperledger, and more generally follow this model.

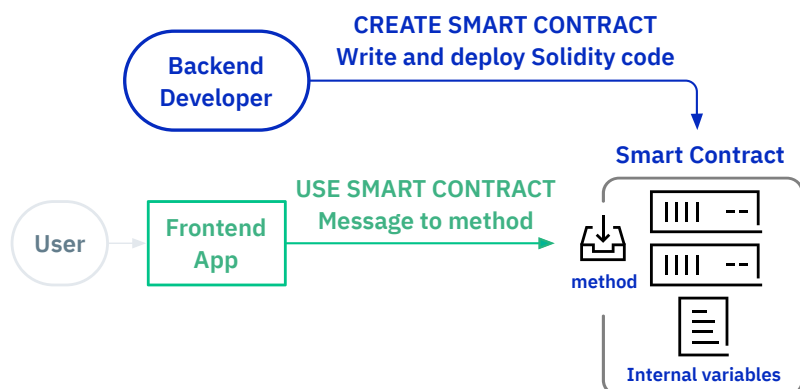
Building a reliable financial application, however, is a different class of problem than building a game, web service, or other general application. A DeFi application deployed to a DLT network is expected to run **autonomously, trustlessly, and irreversibly** while managing millions of dollars in assets. Developers building to specialized requirements like these typically use specialized development environments to make it as easy as possible to avoid bad results.

The lack of finance-appropriate smart contract development environments has led to an unfortunate sequence of high-profile losses of funds in DeFi. Simple programming errors and programming language quirks that normally would only cause annoyance in a general application have instead created [documented openings for exploits and system failures](#). Even where there are no bugs in smart contract logic, connections between multiple complex DeFi applications have made it nearly impossible to design smart contract code to cover all [edge cases that could be manipulated by an adversary](#).

The **Radix Engine** development environment is designed specifically for the creation of logic that defines predictable, correct results on-ledger in response to requests. This form of DLT programmability is based on [Finite State Machines](#) (FSMs), a class of solution that is common in mission-critical embedded systems where predictable correctness is the first priority. To make clear the difference from traditional Ethereum-style smart contracts, we give Radix Engine smart contracts a name more suggestive of their function: **Components**. Let’s compare the typical smart contract approach to Radix Engine’s Components.

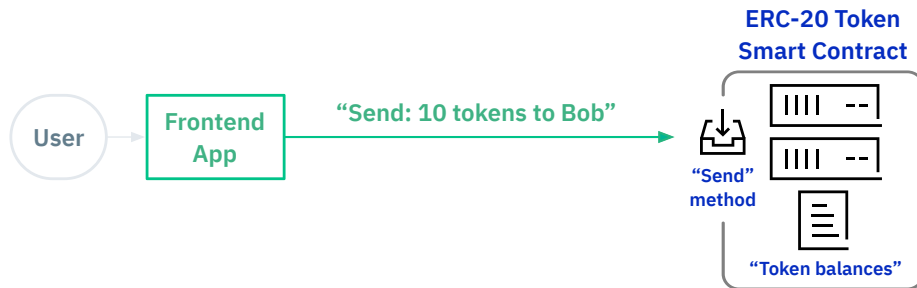
The Ethereum Smart Contract/Method Model

An Ethereum smart contract can be thought of as a black box deployed by a developer to the network. Inside that box you can imagine a little general purpose computer server running some code. To make that server-in-a-box do something, it offers “methods” that users or other apps can call by sending a signed message. The contract has its own internal variables inside the box that it can update based on the messages it receives via its methods.



*In this paper we refer to a “**backend developer**” as the creator of apps/systems, like DeFi apps, that an app created by a “**frontend developer**” may use. In many cases a backend developer may of course also develop their own frontend app.*

Those internal variables are used by the developer to represent all sorts of things. For example, an ERC-20 smart contract creates something that behaves like a supply of tokens by maintaining an internal list of balances. “Sending” a token to somebody really means using a “send” method that the contract code translates into twiddling its internal Token balance variables – ie. reducing the sender’s balance and increasing the recipient’s.

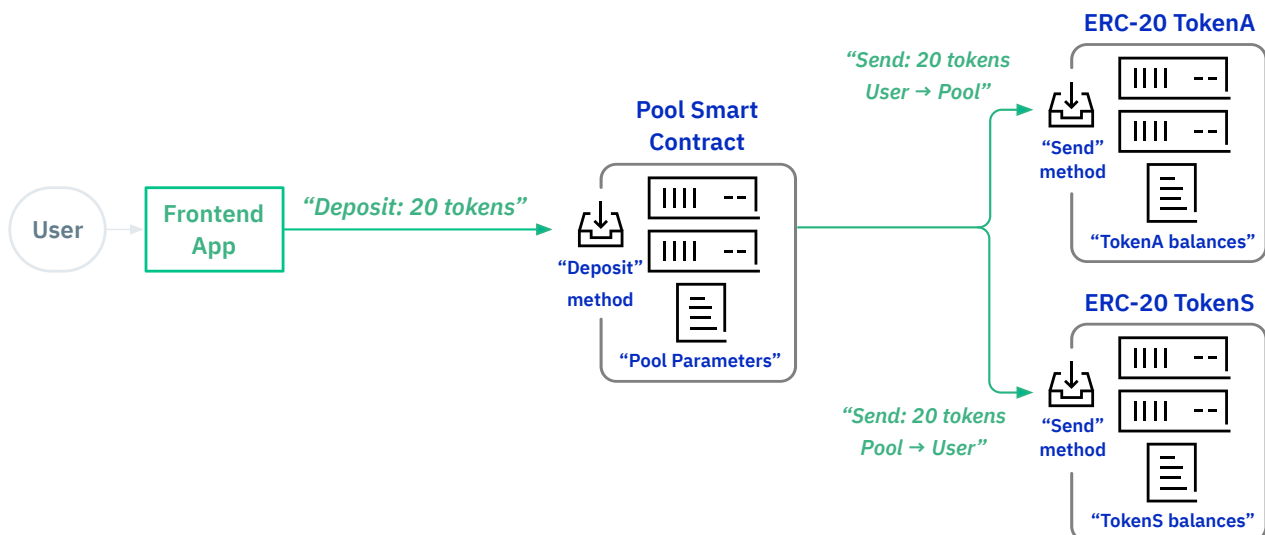


An ERC-20 smart contract represents the idea of a supply of tokens by carefully maintaining an internal list of balances – rather like balances maintained by a bank for its users.

This model is quite flexible, allowing *anything*, in theory, to run on a decentralized platform – thus the Ethereum vision of the “world computer”.

One problem however is that it puts a significant burden on the developer to ensure that their “representation” of tokens, or whatever else, within their smart contract is always correct, and that updates to the internal variables match intuitive expectations. This is doubly critical with DeFi, where potentially very expensive unexpected outcomes are immutably committed to a trustless ledger. And the situation becomes much more complex with DeFi applications where one transaction involves multiple composed smart contracts. In this case, one contract may call the methods on other contracts, with each updating their respective internal variables to produce a combined result.

For example, even a simple liquidity “pool” smart contract can quickly get complex, with results scattered across multiple contracts. If we want our pool to accept an existing (ERC-20) TokenA into the pool and mint a corresponding calculated amount of a TokenS representing a share of the pool, we end up with something like this:



For financial transactions, this doesn't behave like what we would expect for an open platform of highly composable assets. What we would prefer to think of as "tokens moving in and out of a pool" is instead expressed as negotiated balance updates across a network of black-box contracts. It starts to look a bit like the traditional world, with siloed banks communicating with each other, but each keeping privately-held books.

With greater numbers of connections between black boxes, the network of contract calls explodes and correct behavior becomes more and more difficult to reason about and predict. A wide variety of exploits of Ethereum DeFi smart contracts come down to an adversary manipulating the fact that there is nothing fundamental stopping smart contract code from doing *anything* to its internal state, often with unintuitive results that may cascade through the system.

The Radix Engine Component/Action Model

The Radix form of smart contracts, Components, are built in a way that more closely models real-world expectations for finance (and other transactional systems that ledgers are good for). Components are built from finite state machine logic, and define their behavior by **Actions** that directly translate a discrete existing input (or "before") state to an output (or "after") state.

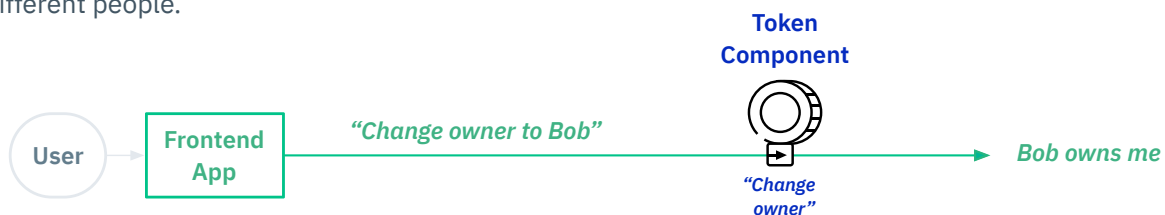
More concretely, this means that Components are defined by *what it is possible for that Component to do* via its Actions. By defining Components by their Actions in this way, we gain two important attributes when trying to avoid bad results.

First, Components can behave more intuitively like physical assets or other finance building-block "primitives", rather than as black boxes, making their behavior via Actions easier to reason about, design, and analyze. Second, *usage* of Components is similarly more intuitive and predictable – and we gain the ability for creators of Component transactions (whether a front-end app, or by reference from another Component) to directly set their own definitions of *what should be possible* for that transaction, creating explicit guard rails on what can and cannot happen in creating the final output state.

Explaining how Components and Actions work is perhaps best done through a few examples:

A User-created Token

Take the example again of a token. On Radix, developers don't need to use a monolithic ERC-20-style smart contract that keeps a list of all balances; we model each *individual indivisible token* (each "Satoshi" in Bitcoin terms) as a discrete independent Component. That token Component's primary available Action is: "*change my owner, if you have the right to do so*". So for example, the "before" input state may be *owned by Alice*, the "after" output state *owned by Bob* – if the conditions of ownership change defined in the token Component's Action are met. This means that the token Component is defined by what it is possible for it to do: to be owned by different people.

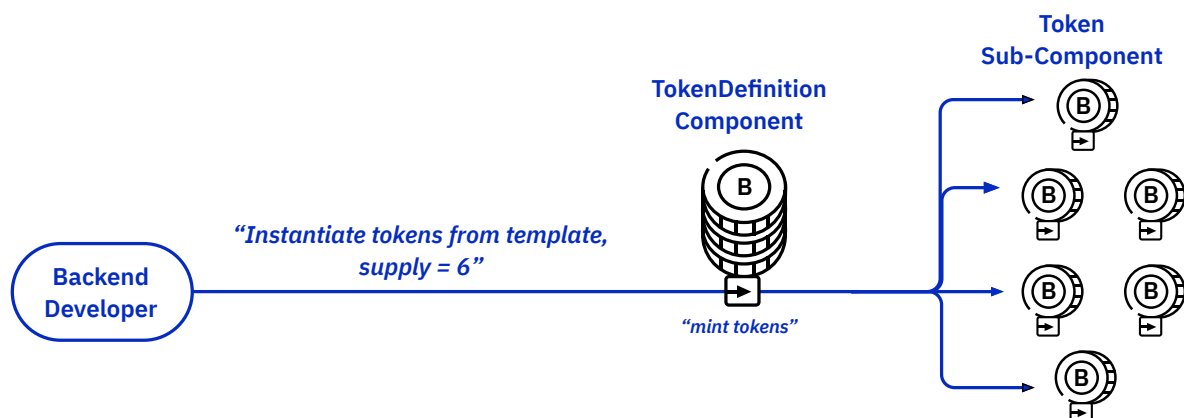


Using the FSM-based Radix Component model, each token can be its own independent Component with Actions (like "change owner") that define their behavior to match intuitive expectations.

While the Radix ledger may, behind the scenes, store a large number of these indivisible tokens collectively for efficiency, logically each token *acts intuitively like physical coins*. That is, the defining capability of each is that its ownership can be passed from one person another. There is no question of it being accidentally cloned, or a glitch in smart contract logic causing tokens to no longer be accessible.

The above picture is a little incomplete however; before we can begin sending tokens around, we need to first configure and create a supply of them. We do this using an important feature of Components: the ability to define **sub-Components**. These are Components that act independently (they have their own **Actions** and individual state), but are defined *within* a top-level “parent” Component.

A supply of user-created tokens is defined in this way. A single TokenDefinition Component³ describes the attributes of the overall token supply (name, symbol, max supply, etc.), but also includes the definition and behavior of a sub-Component: the individual Tokens.



A developer uses TokenDefinition, customizing its parameters. The TokenDefinition Component then provides the “mint tokens” Action that creates indivisible individual Token sub-Components.

These individual tokens are created with a “mint tokens” Action on the TokenDefinition, creating token copies from the sub-Component “template”. That done, the tokens can each be used as independent physical-like things just as we described above – accessed via API – while remaining associated with the parent TokenDefinition and its configuration.

A Liquidity Pool that Operates With Other Components Atomically

Not only may Components and their Actions be used individually, multiple of them may be combined within a single transaction. In this case, the mapping from input to output is *collective*, defined by all of the included Actions simultaneously. For example, let’s consider a “liquidity pool” Component. A simple version could be defined by these Actions:

Deposit Action: *“I am the owner of some reserves of TokenA and **mint** a proportional pool share TokenS to anyone who sends TokenA in.”*

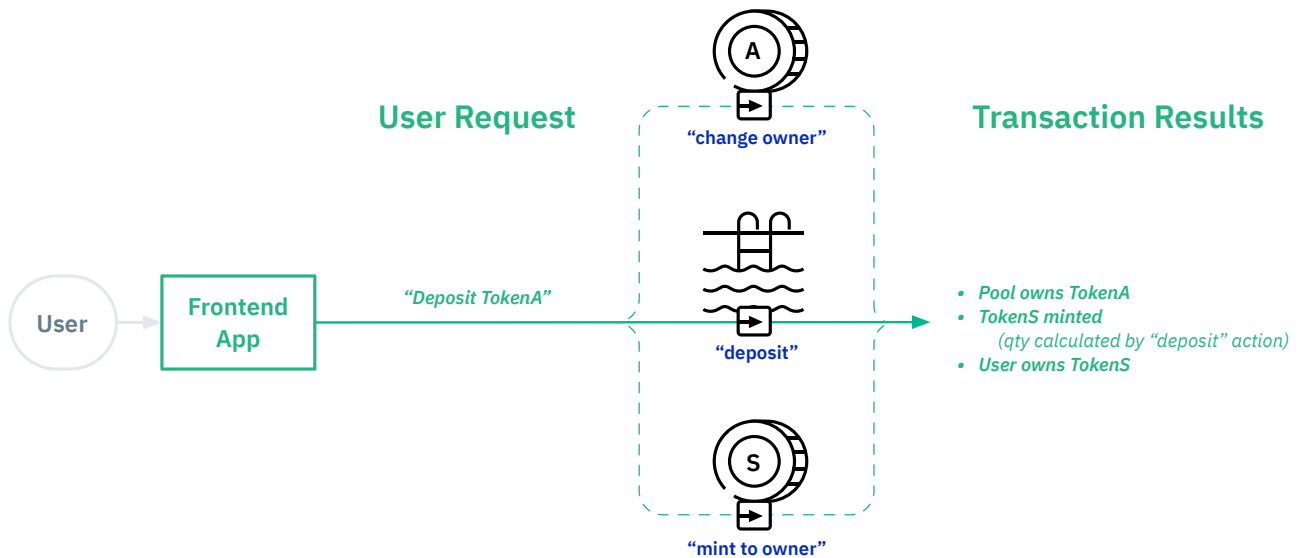
Withdraw Action: *“I am the owner of some reserves of TokenA and **send** a proportional amount of TokenA to anyone who sends me pool share TokenS (which I **burn**).”*

While these are pseudo-code descriptions of the Actions’ definitions, you can see that the words in blue indicate Actions of other existing Components (TokenA, TokenS) that need to be included in a pool transaction.

³ The TokenDefinition Component is a standard Component available on Radix. Unlike many smart contract platforms, not only many user-created tokens use this available functionality, the native RADIX platform utility token uses it as well so that both behave identically.

A nice property of FSM-driven Actions is that the input → output mappings of these multiple Actions can be combined and operated simultaneously (rather than chained together sequentially as with smart contracts). When multiple Components are required for a single transaction, *all* of the Action-driven state changes of the various Components in the transaction mesh together like gears in a gearbox. As long as all of the gears of all of the relevant Components can all successfully turn together (the input and output mappings do not conflict), the transaction is successful. If they can't, the entire transaction safely and correctly fails.

To compare with the Ethereum case above, a Radix Engine pool deposit transaction looks more like this:



A pool Component Action (deposit) includes the logic specifying that it expects to become the owner of some TokenA Components, mint a corresponding quantity of TokenS “share” Components, and make the user the owner of those.

This behaves more like an open platform of intuitively composable assets that we want!

From a front-end developer’s perspective, using the “deposit” Action of the Pool Component is no more complex than using a method on a traditional smart contract. But that Action itself can define the other Components that need to be involved (some specific TokensA and TokensS) – that is, which Components need to mesh their gears together for the transaction to be successful.

In this example, the “deposit” includes changing of ownership of a discrete set of relevant tokens using their Actions. Intuitively, the “mint” or “change owner” Actions may have their own intuitive rules about who can do those things, or who the recipient can be. If the Pool tries to do something that fails to meet those rules, the user’s request to the Pool correctly fails with clear rationale.

A Transaction Request with Consumer Safety Limits

To see how the consumer of Component Actions can explicitly prevent bad results, let’s take the example of a Token Swap Component. This Component accepts TokenA and returns an amount of TokenB equal to a market price (perhaps provided by an oracle or some market maker logic – the specifics don’t matter here). Its primary Action might look like:

Swap Action: “Any receipt of TokenA will create a *send* of TokenB by the equation: $\text{TokenA} * [\text{currently defined B/A rate}]$ ”

A user wants to perform a swap using this Action, but would prefer to set his own clear limits on the acceptable swap exchange rate rather than rely completely on whatever the rate happens to be when the transaction goes through.

The Action model allows the user (i.e. the consumer of the Token Swap “swap” Action) to also specify their own Action-style rules on the input → output mapping, as follows:

User request: *“I am **sending** TokenA to the “**swap**” Action of the Token Swap Component, **which must result in a **send** to me of TokenB of at least [desired limit]**”*

The Action-like conditions of this user request create another “gear” (for that transaction only) that must mesh with those of the relevant Components and successfully turn along with them, or else the entire transaction will correctly fail. This means that the user can place absolutely clear and direct guard rails on their usage of any Component, without having to understand the internal details of the Component’s functionality. And these safety guard rails are available not just to front-end users but to the referred usage of Components by other Components, allowing more confident composed DeFi usage that limits exposure to faults in systems built by others.

As Components become more and more complex, and more Components are composed together as is common in DeFi applications, this gives both users and creators of these apps a powerful tool to eliminate the possibility of many unexpected and expensive outcomes, even when bugs or design flaws are in play.

An On-ledger Oracle for Off-ledger Data

Another useful piece of DeFi functionality is an on-ledger data oracle. In the token swap example above, the market price data that this Component needs could be provided by an Oracle Component that offers an Action like:

Get price Action: *“When requested, I provide the current contents of a key-value pair representing the requested pair price”*

This might seem strange as the input and output states are identical; no output state change has been directly specified by this Action. But when used by the Swap Token’s Action for example, the Oracle’s “gear” meshes with that of the swap Action to determine *how the swap Action* defines an output state change (the quantity of tokens that must be sent). In this way, the oracle provides a valuable service allowing off-ledger data to drive the results of autonomous, atomic on-ledger transactions.

But how did that off-ledger data get into the Oracle’s key-value store to be atomically available? The Oracle cannot itself make an off-ledger request for new data because this would violate the atomic functioning of FSM-based Actions. In the simplest case, the Oracle would have another Action (only usable with the private key of the creator) that updates it. The creator could push updates to this data as frequently as desired, or could do so more responsively.

Creating Radix Components

Creating new Components will use a new, specialized language that we call **Scrypto**. Scrypto is a [functional language](#), providing a style of programming better suited to defining the kind of FSM-based Components described here. Functional languages are increasingly common, particularly for building reliable high-

concurrency systems. Scrypto's syntax should be familiar to developers who have worked with functional languages, and provide a set of programming primitives particularly suited to creating Component/Action logic.

Put together, we believe the Component model is a much more purpose-built solution to making the Radix network programmable for the world of finance assets and applications.

2. Build Interoperable DeFi dApps Faster

A truly useful development platform includes libraries, frameworks, and other tools that allow developers to build simple, common things quickly with a minimum learning curve. These same tools also accelerate more complex builds by providing reliable, pre-built solutions for parts of the problem that other developers have encountered and solved well already. Having good standards and off-the-shelf solutions also strongly encourages interoperability between dApps that is particularly important for a DeFi ecosystem.

In DeFi, common chunks of finance-oriented functionality recur across many applications: assets (fungible or unique), shares, accounts, multi-party control, liquidity pools, swaps, purchases, and data oracles just to list a few examples. These are prime candidates for pieces of functionality that developers would like to see pre-existing, proven, well-maintained solutions.

Traditional open source methods and community collaboration are certainly good places to start to encourage these builds. Package managers often assist in the process of discovering and using pre-existing tools. But the Radix Engine gives us an exciting new possibility: **putting community collaboration and package manager-like functionality directly on-ledger.**

An on-ledger mechanism for Components to be modularly used, leveraged, updated, versioned, extended, and combined is a powerful tool for developers. Components deployed on-ledger in this way don't just contribute to the developer ecosystem; they directly extend the effective functionality of the Radix platform. And a developer need not build a fully-functional standalone dApp Component to usefully contribute. Components that do one thing very well, and are built to be easily reused or combined with other Components, can become standards of the platform that accelerate builds and encourage interoperability for everyone.

The idea of “writing programs to do one thing well” and “writing programs to work together” was the [guiding philosophy](#) of the creators of UNIX in the 1970s. The result created the foundation of open source development and the spectacularly successful family tree of UNIX-based operating systems and applications since then. We believe that rebuilding the financial systems of the world around a decentralized platform suggests a similar philosophy, maximizing interoperability, modularity, and potential for anyone to make meaningful contributions both large and small.

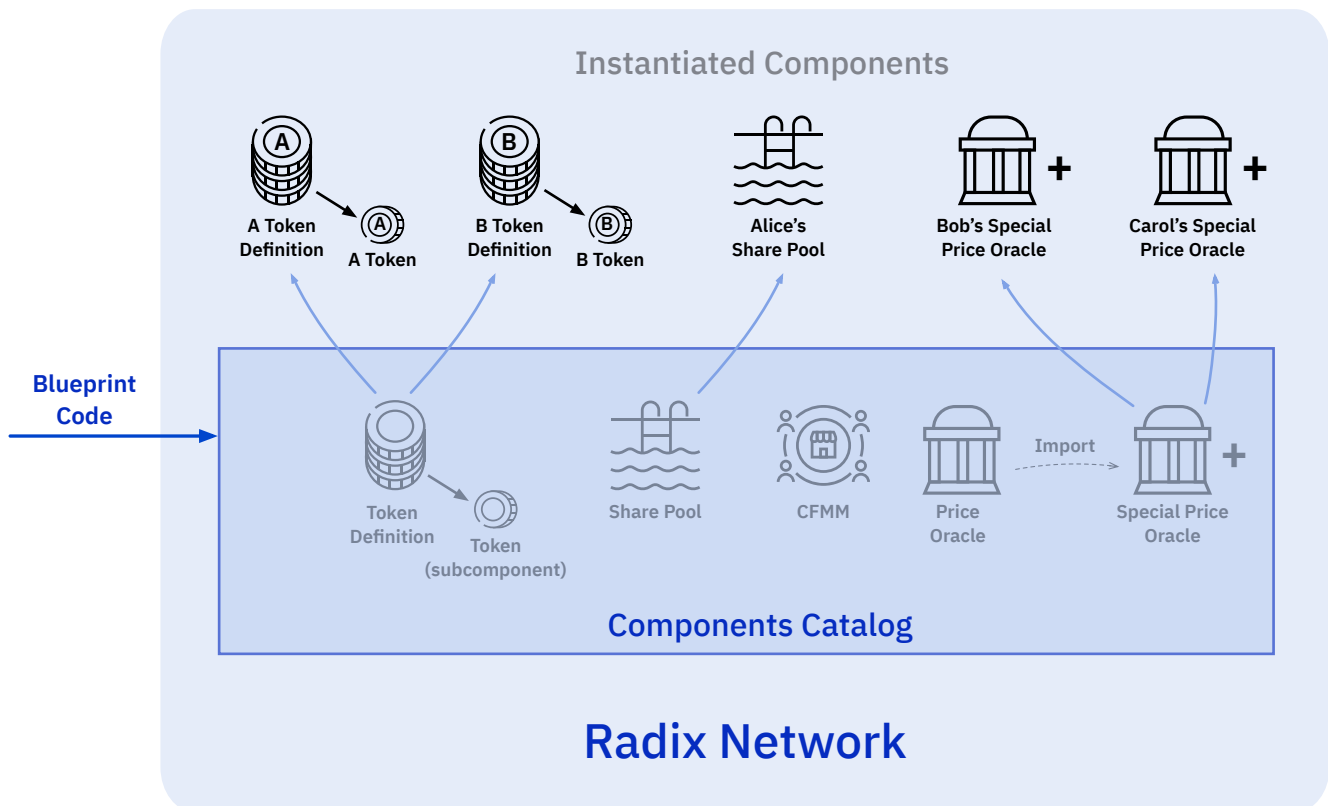
We have integrated this philosophy deeply into the way Components are deployed and used on Radix with a platform feature we call the **Component Catalog**.

The Radix Component Catalog

With typical smart contract DLTs, a developer writes some code (Solidity in the case of Ethereum) and then pushes it to the network where it becomes an active smart contract for users of the network to interact with. The Component Catalog changes this model.

When a Component's Scrypto code is pushed to the network, it is first added automatically to an on-ledger registry called the Component Catalog. Components in the Catalog are like inactive templates or blueprints that anyone may use to create multiple active Components patterned after the original in the Catalog.

To make a Component from the Catalog active for use, a developer instantiates it, creating his own **Instantiated Component** from the blueprint template. An Instantiated Component has its own unique identity on the Radix network, and its Actions become available for use by users (or other Components, as we saw previously). This means that one Component in the Catalog can be used an unlimited number of times as the template for instantiated Components that all behave in the same basic way.



New Components start life as Scripto code that is deployed into the Component Catalog on the Radix Network itself, where they can be easily configured and instantiated – or imported into other new Components.

Instantiation is simple, done via API, and requires no Scripto code. Most Components in the catalog will include configuration parameters allowing customization of the instantiated Component. For example, the “Token Definition” Component described earlier would let anyone instantiate their own Token Definition with its own unique name, symbol, maximum supply, etc. – and then begin minting their own tokens.

By instantiating Components from a universal on-ledger Catalog in this way, Radix makes it very quick, easy, and safe for any developer to issue assets and access other simple functionality created by others, without having to learn and write Scripto code.

Another way of making use of Components in the Catalog is to **import** them. A developer may want to make use of the functionality of an existing Component in the Catalog – but *add to it or customize what it can do*. This is done by creating a new Component that includes an import command for the Component that provides the core functionality desired. The additional functionality is then implemented in the developer’s own Scripto code.

Importing is *not* copying the Scripto code of the original component; it is an *on-ledger link* to the original Component (and its version). An example of this (shown above) might be a Price Oracle component that a developer wishes to customize by adding a calculation of a moving price average (and Action to access it).

The developer could create their own Special Price Oracle Component, import the original Price Oracle Component by reference to its ID in the Catalog, and add the additional Scrypto code for the price average calculation.

Both Catalog Components and Instantiated Components have their own unique Component ID, and are associated with the creator's own unique developer ID. Components are versioned, with each new deployed update requiring a revision bump. Updating a Component does not automatically force an update onto other Components making use of it; previous revisions remain immutably available on ledger and existing Components will continue to have access to the previous revision. A developer may choose to adopt a new revision by making their own update to the Component that uses it.

Standard Platform Components

The Radix Foundation is committed to filling the Catalog with useful Components that model common elements and behaviors of DeFi as standard features of the platform. This includes things like assets (fungible or non-fungible tokens) and accounts (including multi-sig control) and will expand to include functions desired by the community that could include higher-level DeFi primitives like liquidity pools, swapping systems, purchasable assets, data oracles, and more. Each of the standard Catalog Components developed by the Radix Foundation can be instantiated as-is (for example creating a supply of custom tokens via API call) or, as we saw above, they can be modularly combined in various ways to create more complex functionality.

Using these standard Catalog Components, developers can skip reimplementing common functionality themselves, accelerating development time or reducing it to zero for very simple things like issuing a token. Catalog Components created by third parties can also become standardized features of the Radix network in exactly the same way. Whoever creates them, on-ledger Catalog Components create natural composability and interoperability for the DeFi ecosystem.

Foundation-provided Components are not enough for a thriving developer ecosystem however. Neither can we simply release a platform and throw up our hands, expecting developers to fill in functionality before adoption and business demand creates a reason to do so. Our solution is to use the unique features of the Radix Engine to build in, at a protocol level, a financial incentive for developers to invest their time in contributing useful Components to Radix, leading into the next important DeFi problem Radix solves.

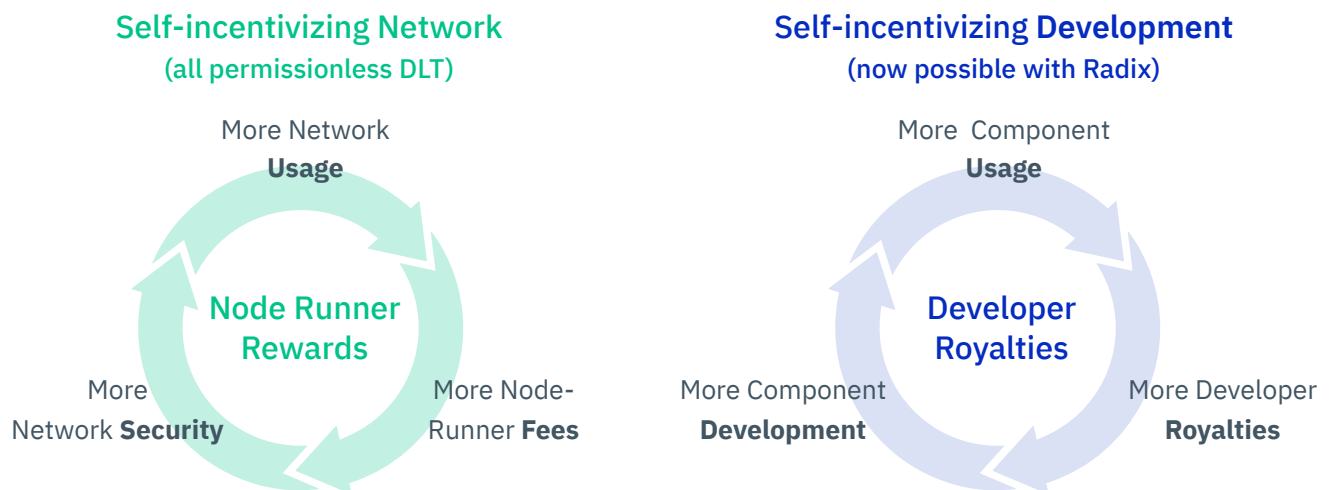
3. Incentivize a Decentralized Development Community

Perhaps the most important innovation of blockchain was the ability to create open networks that are economically self-incentivized – originally in the form of “mining”. With this innovation, a community of node-runners can be incentivized to participate from the earliest stages of the network, as well as to scale the network up (its security at least) to safely conduct transactions of billions of dollars of value.

But creating a permissionless DLT network suitable for DeFi requires more than just node-runners providing the low-level infrastructure. There must be a thriving developer ecosystem creating the kind of useful, interoperable Components that the Radix Engine and Component Catalog enable.

Developer Royalties

While traditional methods of growing a developer ecosystem can work, we believe that *decentralized developer self-incentivization* can create a breakthrough in rapid decentralised ecosystem growth, creating the same kind of market-based incentives as “mining”. The Radix Engine and the Component Catalog make this possible for the first time; we call it the **Developer Royalty System**.



The core concept is that the developer of any Component may specify a royalty in RADIX tokens (the native Radix utility token already used for transaction fees to node-runners) for *each usage of that Component in a transaction*.

Note that this isn’t an “app store” where one must pay for access to Components; it is an entirely per-transaction-use fee that must be included within the transaction itself. This means that payment of royalties is based on the real *utility* that the Component brings to the network.

The Radix protocol itself is able to calculate and charge the correct royalty fees – in the same way it handles fees for node-runners – because the Component Catalog and the various types of Component usage are all on-ledger. If a developer adds a highly useful, modular, interoperable Component to the Catalog, Radix ensures they are rewarded for the transactions enabled by their work in a completely decentralized manner. This is a unique property of the Radix Engine development environment that allows the Radix network to self-incentivize a developer community for the first time.

How Royalty-Setting Works for Developers

The fundamental innovation behind Radix Developer Royalties is the on-ledger Component Catalog and associated on-ledger mechanisms for using Components to create applications and transactions. With these, the Radix protocol directly links royalty *definition* (by the developer) to royalty *payment* (by the user), creating an open marketplace for network utility created by developers. In this marketplace, royalty payments on a per-transaction-use basis are automatic and guaranteed by exactly the same consensus mechanisms that guarantee Radix network security.

Developers must have the tools to participate in this on-ledger marketplace in whatever way they prefer. We start by allowing the developer to set a different royalty for each of the different types of Component usage that are possible once it is added to the Catalog⁴. These are:

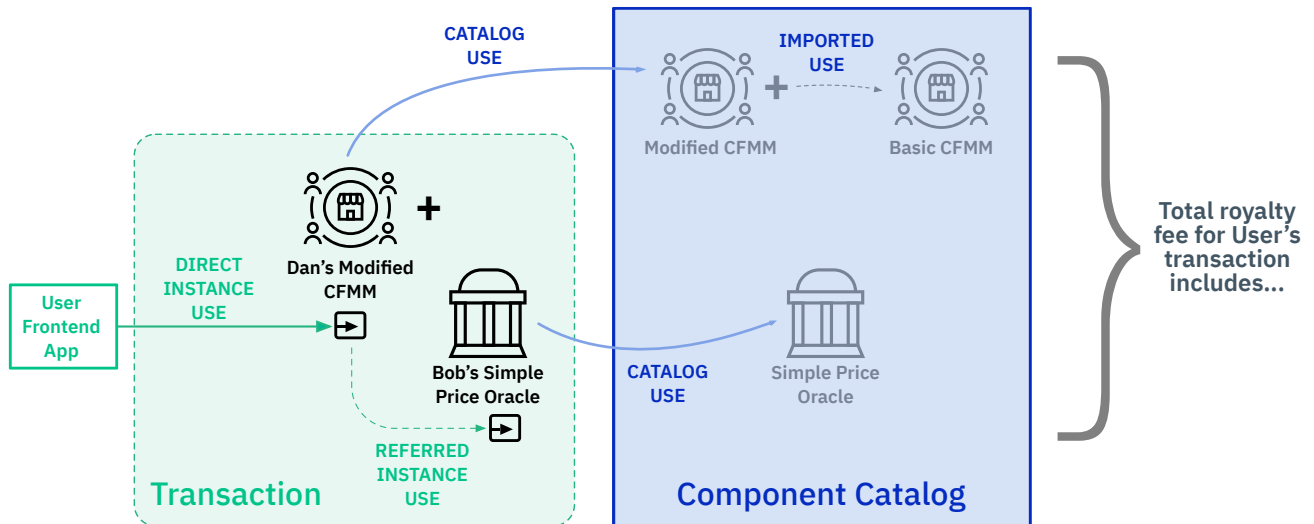
1. **DIRECT INSTANCE USE** – This is when a user transaction directly accesses an Action of an instantiated Component. For example, this might be the interface to a DeFi application like a CFMM (constant function market maker), even if this Component uses other Components “behind the scenes”.
2. **REFERRED INSTANCE USE** – When a Component is used directly, as in #1, it may specify that multiple other instantiated Components must be included in the Action for the transaction. This is “referred” instance use. For example a DeFi application might use an instantiated oracle Component to get access to its data by reference for a certain transaction.
3. **CATALOG USE** - Each instantiated Component is associated with an original template Component in the Catalog. The catalog use royalty is paid when one of these Catalog Components it is the template basis for an active, Instantiated Component that is used as in #1 and #2.
4. **IMPORTED USE** - This refers to the usage when a developer “imports” an existing Catalog Component into his own new Catalog Component. As in #3, importing itself is not paid, but rather the imported Component is considered used when another Catalog Component that has imported it is used as in #3.

Note that there are two types of usage here, indicated by the green and blue colors. A developer would specify the royalties for **CATALOG** and **IMPORTED USE** when the Component is **deployed** to the Catalog. These can be thought of as the royalties for “behind the scenes” use of Components. Royalties for **DIRECT INSTANCE** and **REFERRED INSTANCE USE** would be specified at the time that a Component is **instantiated**. These are the royalties for the use of the Actions of each active, independent “copy” of a Component from the Catalog.

For each transaction using the Action of an instantiated Component, there will be an associated set of royalties for that instance, any other instances it refers to, and the various Catalog templates that sit behind the instances. Ultimately it is this full set of Components that made the transaction possible.

The four types of usage can be seen in this example for a single transaction that a User requests to Dan’s Modified CFMM:

⁴ While fixed XRD pricing is the simplest, we anticipate developers may wish to algorithmically set pricing for each use type, perhaps to set royalties to a certain fiat price at current XRD rate using a price oracle of their choice. This is entirely feasible within the Radix protocol.



A user's transaction request to an Action of Dan's Modified CFMM (and instantiated Component) must include the summed royalty fees of the associated Components, according to type of use. These summed fees are easily determined by the Radix Engine so a User app knows what it must pay, and to ensure all royalties are paid out correctly.

Developers may tailor their royalties for each of these types of usage to suit the nature of what they build and how they expect it to be used. We believe this opens up substantial flexibility for creating decentralized on-ledger revenue streams and business models – as well as enabling royalty-free use where desired. Some use case examples are given later.

In addition, the Royalty system allows developers to set specific pricing (for any of the four types of usage) for specific other developers (via developer ID) or Components (via Component ID). For example, a developer may wish to offer discounted usage royalty pricing when accessed by a DeFi app that he expects to provide unusually high usage volume. Or the developer may simply wish to remove the royalty on his Components when accessed by other Components that he himself has built, while still applying the royalty when others use them.

Updating of royalty pricing is also possible. As with any other Component update, this would require a revision update. As described, however, previous versions (with previous royalties) continue to remain immutable and available for use. Anyone wishing to adopt a newer version of a Component would also accept the set of royalties associated with it.

With each Component defining their associated set of royalties, the Radix protocol has everything it needs to automatically calculate and assess the royalties a transaction creator must pay for that transaction. An example may make the creation of Components and the assessment of royalties in transactions more clear.

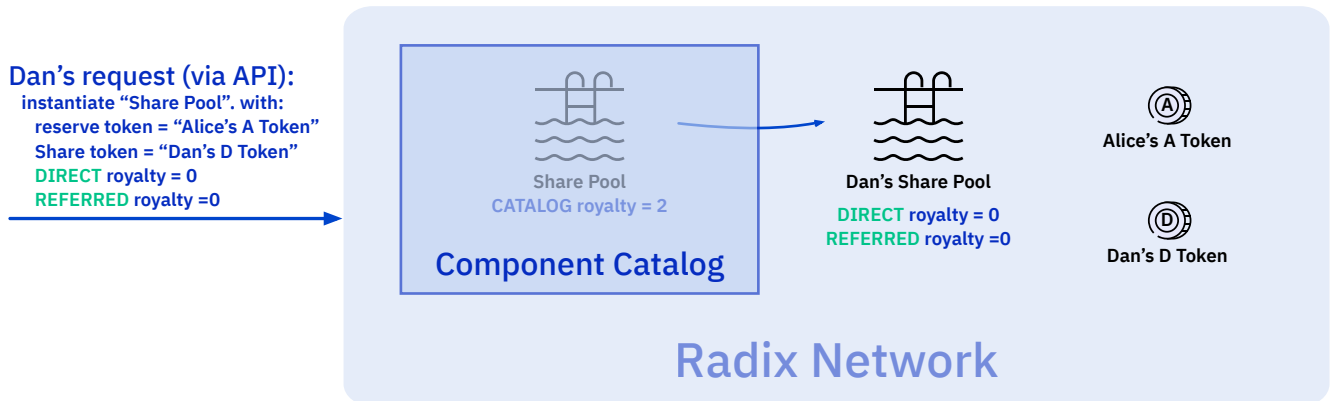
Component Creation and Royalty Example

Let's take the example of Dan: he wishes to create a DeFi service around a Constant Function Market Maker (CFMM) Component that he has some clever ideas about.

He could build it the whole thing from scratch, but first he searches the Component Catalog to see if there are existing proven solutions for any parts of the problem. One part of the CFMM is a share pool where a specified token is minted in exchange for other tokens being contributed to the pool (in proportion to the pool's

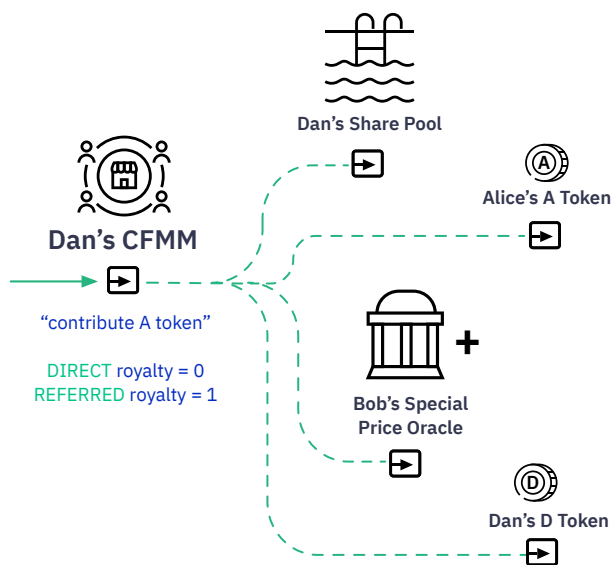
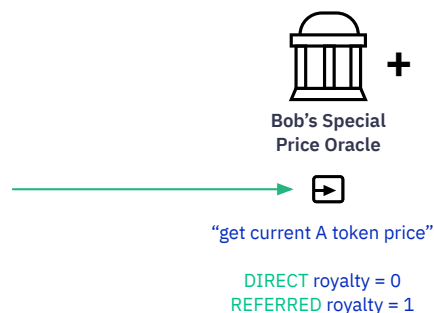
reserves). Fortunately somebody already built a “Share Pool” Component that many people seem to be using happily and safely from the Component Catalog.

Dan instantiates the share pool for himself, configuring it by setting the reserve token that can be contributed (an “A” token already instantiated by Alice) and the share token that should be minted in exchange for contributions (a “D” token that Dan already instantiated himself).



Dan sends an API request to the Radix ledger to configure and instantiate a Share Pool from the Component Catalog. His instantiated pool is then active for use alongside existing Token Components he will use.

Dan also needs a price oracle in order to get current market pricing for A Token to use as part of his CFMM's internal economics logic. Fortunately Bob has already created and instantiated a nice special price oracle Component that Bob feeds with a variety of market data for different tokens. Dan considers the REFERRED royalty that Bob charges for the oracle to be reasonable, and so he can just use the Action of this instantiated Component as-is to get the price data he needs.

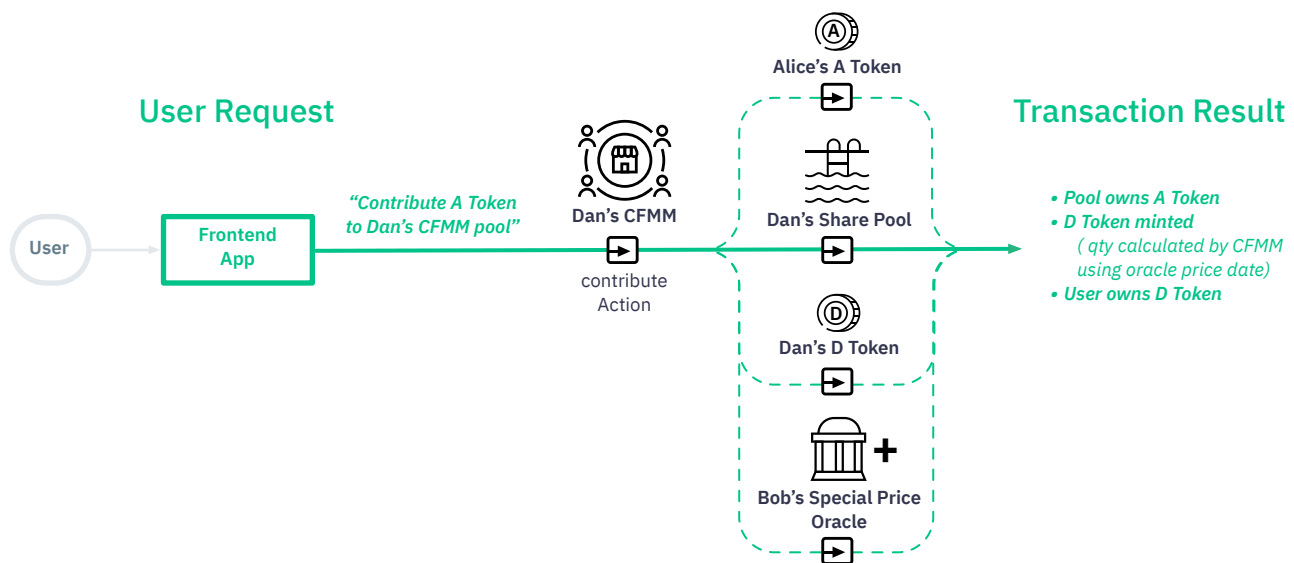


Dan then writes the Scripto code for his CFMM, defining a “contribute A token” Action (among others). This action's Scripto definition includes that it needs to use certain Actions on A token, his share pool, his D token, and Bob's special price oracle.

He doesn't need to *import* any of these Components; he can implement his own contribute Action so that transactions using it include these other Component's Actions as a referred use (thus applying their respective REFERRED royalties to that transaction).

Dan sets a royalty for his CFMM, pushes its Scripto code to the network, and instantiates it. Perhaps he creates a web UI as a frontend app to enable users to create transactions using it.

A transaction using Dan's CFMM would look something like the illustration below. The user's request from Dan's frontend app, at left, is translated by the CFMM, and the other Components it refers to, into a correct set of output result state changes at right (assuming of course that the transaction satisfies the constraints of all Actions involved).



An example of a transaction where one Component (Dan's CFMM) specifies the use of four other instantiated Components that are automatically, atomically composed to create a single output state result.

The royalty the user must pay for this transaction is calculated first by adding up the royalties for each of the Components in the Catalog that were part of the transaction. This includes the CFMM Component Dan created, the Token Definition Component of which the tokens are sub-components (provided without royalty by the Radix Foundation), and both the Special Price Oracle Component and the Price Oracle Component that it imported. (See the Components shown earlier in this document.) Then the node-runner fee is added to the royalty to determine the final fee:

Developer royalties:

Dan's CFMM

CFMM	DIRECT	5	
Alice's and Dan's Tokens	CATALOG	0	(set to zero for Dan's own use)
Token Definition x2	REFERRED	0	
Bob's Special Price Oracle	CATALOG	0	(provided by Radix Foundation)
Special Price Oracle	REFERRED	1	
Price Oracle	CATALOG	0	(again, set to zero for Bob's own use)
Dan's Share Pool	IMPORTED	0.5	
Share Pool	REFERRED	0	(set to zero for Dan's own use)
	CATALOG	2	

Node-runner fee:

Transaction fee	0.1
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Total transaction fee:

8.6 XRD

While this may seem complex when listed out this way, a front-end application can use the Radix Engine to automatically determine the full required fee for a given transaction ahead of time.

The Developer Royalty System creates a strong incentive for developers to aggressively build out the most useful functionality they can think of, as early as possible, in order to maximize the adoption and usage of their Components. A developer need not build out a full, complex, application in order to reach a point where their efforts can be rewarded; it may be even more valuable to build Components that do one thing very well, and are highly modular, in order to maximize their usability by others. These Components may become effectively new standards of the platform when widely adopted.

The Developer’s Guide to the Radix Universe

The Component Catalog and Developer Royalties provide the foundations of a fully decentralized marketplace for Component development. On one side of the market, developers of Components may freely add them to the Catalog and set their own per-transaction royalties to be enforced by the Radix protocol. On the other side, developers who wish to leverage those Components have access to immutable on-ledger data telling them what the transaction fees will be for a given Component, how broadly it is being used, its version history (fully open source), and which Components are associated with a given developer ID.

While all the right data is available on-ledger, to make it easily searchable, browsable, and visualized requires a proper front-end service. The Radix Foundation is committed to providing the first option in the form of the **Developer’s Guide**. While we believe it is important for us to provide this service right away to our community, our hope is that many such services will arise to address the needs of a diverse developer community.

The Developer’s Guide will collect the contents of the Catalog and current instantiated Components and present them in an “app store” like interface. However rather than presenting paid apps to users, the Guide will allow developers to discover Components (both Catalog and Instantiated) to accelerate their own development, and distill on-ledger data into a view of the associated reputation and history of development and usage, as well as transaction royalty pricing, in order to make good integration decisions within a highly interoperable DeFi ecosystem.

The Developer’s Guide will provide another useful service to the developer community in how it presents the work of different developers. Like any truly open marketplace, bad behavior is possible. Someone could choose to copy the Components created by others, or otherwise try to introduce exploitative Components into the Catalog. This of course cannot be stopped on a permissionless ledger. But the Developer’s Guide, as an off-ledger service, can work to detect such behavior and present relevant context in the search results it provides.

Unlike an app store, no purchases are made through the Developer’s Guide, and the Radix Foundation will take no cut of royalties (nor would it, or anyone else, have the ability to) – it is a convenient interface into a fully decentralized marketplace entirely between developers and their users.

The Radix Foundation and the Open Source Radix Project

The Radix Foundation is currently developing the Radix protocol as a fully open source code base. Long term, Radix must be a community-led movement, and we will look to the examples set by other successful open source and blockchain projects to build and support this transition to community. This project is the

infrastructure bedrock for Radix, including limitlessly scalable Cerberus consensus, and the Radix Engine that underpins the Component Catalog and Developer Royalty System.

Starting from this bedrock, the Developer Royalty System provides the right incentives at the application layer of Radix to rapidly grow the platform into a vibrant, interoperable, and open DeFi application ecosystem for developers and users.

The Radix Foundation's mission is to support developers at all levels, providing critical enabling functions where needed to avoid bottlenecks to adoption, while turning all of our work over to our community to extend. This of course includes aggressive development of core Components for developers to use and extend royalty-free – but also supporting Component developers through partner programs.

One particular area where the Radix Foundation can assist in early phase bootstrapping is to subsidize developer royalties. In the early stages of the network when there may be relatively few transactions (and associated fees), virtually all blockchains (including Radix) subsidize the rewards to node-runners via token supply inflation or simply a subsidy paid from a reserve. The Radix Foundation will explore ways in which it can offer the same to developers, multiplying the royalty rewards paid by users and encouraging developers to participate early in building Components.

Case Studies of Developer Royalties in Action

With the tools offered by Radix Components and the Developer Royalty system, we can imagine a variety of ways in which developers can contribute and customize royalty-based revenue streams based on usage. Here are some examples:

The Core Capability Developer

Cara has been playing with DeFi apps for a while and hears that Radix is an exciting new platform for DeFi. She doesn't want to create, launch, and support a full DeFi app – but she loves the idea of pooled liquidity and wants to build that capability for the many Radix dApps that she anticipates will want it.

She creates a Share Pool Component in the Catalog that can be configured to mint and burn a specified share token in response to deposit and redeem Actions for a specified reserve token, in the correct quantity to maintain the NAV (Net Asset Value) share represented by each share token. She expects others will create their own pools using her Component for various purposes. She sets both CATALOG and IMPORTED use royalties to a low 0.5 XRD to encourage broad use and extension by other developers.

Cara diligently responds to community security questions and feature requests, and so there is little reason for other developers to build this capability from scratch instead of simply accepting the small transaction royalty she asks for usage. Cara's Component becomes the trusted de facto share pool standard that others build their own Component code around, further increasing resistance to copycats or competitors. As Radix adoption grows, usage of Share Pool reaches a sustained 10,000 transactions per day, providing enough income for Cara to focus on independent development full-time.

The High-Value Service Developer

Hannah has a company with access to up-to-the-minute market data that she wishes to offer to Radix DeFi apps. She wants to provide this through a Component where this data can be used atomically, allowing other

Components to process transactions directly using current price data even when the transaction is composed across multiple apps.

Hannah's company builds, deploys, and instantiates a simple Price Oracle Component that only accepts market data update transactions from her own servers. In the Catalog she sets the Component's CATALOG and IMPORTED royalties to zero, since it is her data she wishes to monetize and she doesn't mind free usage of her open source oracle code if it helps others. When she instantiates the Component, she also sets the DIRECT royalty to zero as she doesn't mind direct access by end user applications; it only raises the awareness of the quality of data. Finally she sets the REFERRED royalty to 25 XRD, meaning that when DeFi applications want to make atomic decisions based on her data, this is where she wishes to derive her revenue. Because her data is only available through her own instantiated Component, it is the single on-ledger source for this data.

The DeFi Application Developer

Daiki wants to build a premier DeFi tokenized lending service on Radix, seeing it as a more secure and scalable platform and wanting to be among the first to build a user base there. The typical ways of monetizing this service are available – such as requiring an application-level payment of a fee in a preferred token with each use of his application's Actions. However Daiki wants to accumulate XRD tokens on the early network and sees royalties as a convenient way of generating revenue from either direct or referred (ie. composed from another DeFi app) usage – without adding the additional integration complexity of an application-level fee. He expects that his company's frontend, support, and rapid improvements are more than enough to keep copycats at bay, but nonetheless he sets the CATALOG and IMPORTED royalties for his core Lending app Component to a very high level. When instantiating it, he sets the DIRECT and REFERRED royalties to a reasonable 5 XRD level to encourage low-friction, high-volume composed use by his frontend users and other apps.

Part of Daiki's competitive advantage is access to excellent market price data from Hannah's Oracle. He went to her company early on and convinced her to create a royalty exception for Daiki's Lending app ID, allowing his app to use her data atomically at half her general royalty.

The Builder's Guild DAO

Beatriz wishes to build a completely decentralized organization in which multiple developers work together to produce useful Radix Components and share in the aggregated royalties. She first creates a Guild token (using a free Component developed by the Radix Foundation) that represents seniority within the Guild. She creates two primary Components:

- A Guild Governance Component that enables Guild token holders to vote on the policies that grant developers membership and seniority within the Guild based on their work, as well as how new Components are deployed and updated, and their royalties set
- A Revenue Sharing Component that extends basic Account functionality to allow Guild token holders to claim proportional amounts of XRD from the Guild's royalties that flow into its Account (which is used to deploy all of the Guild's Components)

She deploys both of these with zero royalty in order to contribute to the open source community and encourage the creation of other DAOs, a concept she believes strongly in.

She launches the Builder's Guild DAO, turning control over to an initial core set of developers and the Governance Component, and watches as it blossoms into a global community of contributors that develops a strong reputation for producing reliable Radix Components.

The Initial Component Offering

Itai is an independent developer that has a big idea for a DeFi app, but he needs enough capital for him to commit full-time to taking it from prototype to fully-featured platform. He is part of a network of developers that respect his skills, and so he decides to ask them for up-front funding in exchange for a portion of the royalty revenue he expects his app to produce.

Fortunately he finds Beatriz's Revenue Sharing Component. Instead of configuring it for revenue share claims based on Beatriz's Membership token, he configures it to use his own ItaiApp token as the basis for revenue claims on the royalty account. He distributes 50% of these tokens to his backers, keeping the rest for himself.

The Freemium Developer

Felix has built a Component that implements a clever set of algorithms that are useful to many applications. He wants to encourage the widest possible integration, and so he sets both the CATALOG and IMPORTED royalty to zero. Later, he creates an additional algorithm that is very useful to a limited subset of the developers using it. He deploys a new revision of the Component that adds a modest fee. Because Radix is an immutable ledger, developers continue to freely instantiate and import the earlier free version, but if they wish to have access to the newer more featureful version, they must also accept the new royalty with the update.

As described above, it is important to note here again that once a Component is deployed to the ledger, its developer cannot silently foist a change of royalty on users or other developers. Only through the creation of a new version of a Component can the set of royalties be updated, with the old version remaining available.

4. Scale dApps Without Breaking DeFi Composability

One clear challenge faced by DLT platforms today is scalability. The rapid expansion of DeFi apps on Ethereum has pushed the platform to its limits. While Ethereum pursues its 2.0 upgrade to help alleviate the bottlenecks by 2022, other DLT technologies have entered the picture proposing new techniques to reach greater throughput of transactions.

Posting high throughput numbers alone, however, fails to encompass the full scalability requirements of DeFi. If a DeFi DLT platform is to provide a functionally viable alternative to a global network of traditional financial systems, it must be able to support a tremendous number of DeFi apps *simultaneously*, while running each app at high throughput, and without compromising decentralization. But even this isn't enough.

Perhaps the most important feature of DeFi is the interoperability of apps and assets, often called "composability". The ability to "compose" a single transaction, making use of multiple autonomous smart contracts, is energizing much of the DeFi innovation and excitement on Ethereum today. With the ability to freely compose across any set of DeFi apps, it becomes possible to build a service that, for example, instantly provides the best exchange rate for a trade across multiple automated market makers, or allows the leveraging of a crowdsourced liquidity pool to take instant advantage of an arbitrage opportunity. Crucially, these complex operations across apps must all happen in a single "atomic" step, meaning that either the entire transaction across all apps is valid and resolved all at once, or the entire transaction safely fails. This is incredibly powerful and is the basis for how DeFi dissolves the inefficiencies of traditional financial systems, replaced by fast, customizable, and interoperable DeFi financial apps.

Despite the crucial importance of composability, **most DLT solutions seeking to increase scalability do so by significantly reducing composability**. Typical approaches for scale separate apps and transactions across "shards" where they can run faster but do not have direct, atomic access to each other. Here more sharding means less interoperability, putting scalability and composability in direct conflict. While this may be an acceptable tradeoff for simple token transfers or applications that do not need to be freely and atomically composable, it makes true DeFi at scale essentially impossible.

Radix takes a different approach, designing its full technology stack for unlimited scalability without compromising security, decentralization, or composability.

Typical Sharding for Scalability

Typical scalability solutions involve some type of sharding. Whether sharding is implemented using a hub-and-sidechain architecture (like Cosmos or Polkadot), or by breaking a block into pieces for independent processing by different nodes (like Near), the idea is the same: different apps and transactions are localized to some number of separate shards where they can be run through consensus in parallel.

This parallelism achieves greater throughput, but the compromise is that communication between shards is made difficult. Different shards can be thought of as separate blockchains (in fact sometimes that is literally what they are), but where there is some method to send messages between them. **But if each shard conducts consensus independently, it is impossible to process a transaction across multiple shards atomically**. One way or another, cross-shard coordination must be done across multiple blocks on the different shards, often

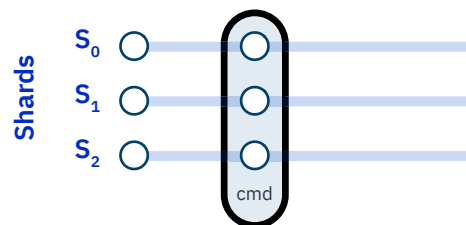
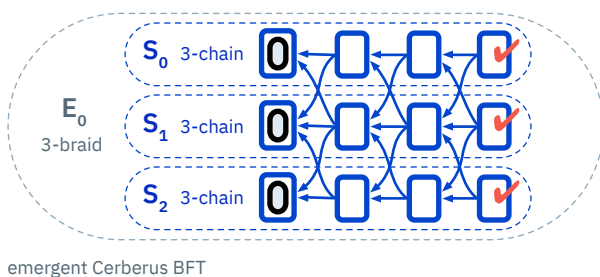
involving “receipts” or other ways of providing conditional cryptographic commitments between independent consensus processes. This makes these transactions slow, error-prone, and difficult to implement in smart contract code. Making matters worse, assignment of apps and assets to certain shards is usually static (like Ethereum 2.0), or requires significant network overhead to adjust.

We realized early on that Radix needed to start from first principles to resolve this tension between scalability and composability. First, rather than using a static set of shards, we needed to support a practically unlimited number of shards to achieve as much parallelism as possible for a global-scale DeFi platform. Second, we needed a consensus protocol able to dynamically conduct consensus, on atomic transactions (including smart contract operations), synchronously across only the relevant shards without causing the rest of the network to stall. And third, we needed an application layer able to efficiently make use of this unlimited “shard space” and multi-shard consensus.

Cerberus Consensus

A core piece of the Radix solution is our unique Consensus algorithm, **Cerberus**. Cerberus is designed around a concept we call “pre-sharding” where, rather than trying to add sharding to a monolithic ledger, we start by splitting the ledger into a “shard space” of a number of shards so large as to be practically unlimited⁵. We can use these shards to represent anything we like, and Cerberus can “braid” secure consensus across an arbitrary number of shards as required. Our [Cerberus whitepaper](#) covers this algorithm in depth. But in short, Cerberus combines three core insights:

First, we move from the typical concept of global ordering to that of partial ordering. Virtually all DLTs assume global ordering wherein all transactions must be placed on the same timeline. Some forms of sharding essentially create multiple globally ordered timelines, but keep fixed global ordering within each. Cerberus takes this concept even further, presuming that each transaction can specify precisely which shards are relevant (and thus must be ordered) *for a specific transaction*. This requires a specialized application layer that can specify how shards are used and relate to each other, which we will get to later.

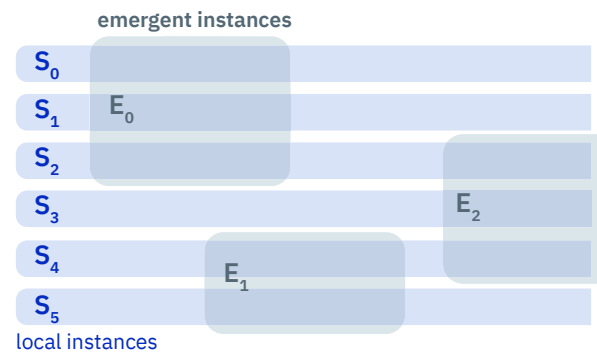


Second, now that we know which shards must be included for a transaction, we design a new form of BFT-style consensus called “braiding”. Typical BFT-style consensus uses 2 or 3 phases of signed commitments between nodes in order to confidently finalize a transaction. Cerberus’ braided consensus runs a single 3 phase BFT instance (called a “3-chain”) within *each* shard, but braids these instances together with commitments provided by the leaders of the other related shards. The result is

⁵ In concept this is similar to a typical extremely large space of public keys such that even free, random usage uses a practically insignificant amount of the available space.

an “emergent” 3-braid consensus that ensures all relevant shards can atomically commit to the multi-shard transaction.

Third, we design the protocol so that dynamic 3-braid consensus processes may run in parallel. Each shard, with its local BFT instance, can run completely independently, as can any emergent multi-shard instance (as needed for a given transaction) that isn’t related to any other at the time.



We combine these insights to create Cerberus, a new consensus algorithm designed specifically for large, diverse networks of simultaneous applications and transactions. Cerberus provides linear scalability through parallelism; more network demand can be served by ever greater numbers of economically-incentivized node-runners. As the Cerberus-based network grows, atomic composability is never compromised because direct consensus between shards happens seamlessly in response to each transaction.

Radix Engine Application Layer

The application layer of a DLT provides the interface between a developer’s smart contract code and the underlying ledger and so Cerberus requires a specialized application layer. For Radix this is the Radix Engine, and the important part for scalability is the Engine’s bottom-most layer: Radix VM.

Radix VM provides the partial ordering that Cerberus requires to braid consensus on a per-transaction basis – including transactions driven by Component Action logic. It does this by defining transactions as a related set of changes to state that are encoded across shards. For example, the ownership of two different tokens (two pieces of state) may be located on two different shards, and so an atomic transaction that transfers the ownership of both of those tokens must correctly “partially order” that transaction relative to other transactions on those two specific shards, at that moment. That is, we must ensure that changing the owner of either token doesn’t conflict with another request to change the owner of those tokens. All other requests to the network at that time don’t matter.

Radix VM also must solve another substantial problem for DeFi scalability: **concurrency**. Even if Cerberus gives us excellent ability to parallelize transactions of unrelated things, many DeFi (and other) applications are intended to handle many concurrent transactions for *highly related* things. DeFi “pools” of token reserves (such as those that power Uniswap) are an excellent example of this, where every transaction is calculated using the state of the token reserves that are themselves changed by each transaction. Many requests competing to use the pool simply cannot be parallelized.

This demonstrates that there are two fundamentally different modes of achieving high throughput of transactions on our network – one where parallelization is possible (many unrelated transactions) and one where serialization is required (many highly related transactions). These different modes are well studied, and we require two different tools to solve them: optimistic requests, and pessimistic requests.

Optimistic requests specify the full details of a state change up-front based on what the current state is believed to be. As long as nothing conflicting changes about the current state by the time the network processes the transaction, the transaction sails through. But there is always the chance that our optimism is punished and our transaction rejected if any of our assumptions are no longer valid, forcing us to retry. For most simple transactions on a DLT network this is quite unlikely and Cerberus

can naturally parallelize extremely effectively, but for something like a popular DeFi pool, transactions will constantly conflict, fail, and need to be retried.

Pessimistic requests means that a transaction locks a required part of the current state up front so that we can be confident that any well-formed request will not fail due to an unforeseen change of state. This means that requests that update a given piece of state are entirely serialized. When many requests are trying to update the same state (as with a DeFi pool), grouping all of that state together and applying serialization is the optimal solution.

So what we would really like is a hybrid solution: Simple transactions should use optimistic requests and be parallelized across many Cerberus shards as much as possible to maximize throughput. Complex transactions involving lots of related state (such as a high-volume DeFi pool) should use *pessimistic requests* allowing them to be serialized to make updates as quickly as possible to state that's collected on few (or one) shards. Radix Engine makes this hybrid solution possible through two elements: **Action-based requests** and **dynamic localization**.

As we've described, all Radix Engine transactions take the form of requests to on-ledger Actions of Components. Actions define how a given request creates an output (some *changes* to state, on their relevant shards) based on an input (some current state, on their relevant shards). Action-based requests allow both optimistic and pessimistic usage.

Let's take the example of our Token Swap Component from above, where a user can send TokenA and receive TokenB at a current exchange rate. Our swap Action was this:

Swap Action: "Any receipt of TokenA will create a *send* of TokenB by the equation $\text{TokenA} * [\text{currently defined B/A rate}]$ "

Say I own 10 TokenA and I want to send 5 of them to the Token Swap Component to receive the correct number of TokenB in return. We can use the swap Action in two ways to do this:

Optimistic request: Using the specified "swap" Action, I will send these specific 5 TokenA from my account to the reserve account; I see the current exchange rate set within the Token Swap Component ($1A \rightarrow 2.7B$), and so these specific 13.5 TokenB from the reserve account will be sent back to me.

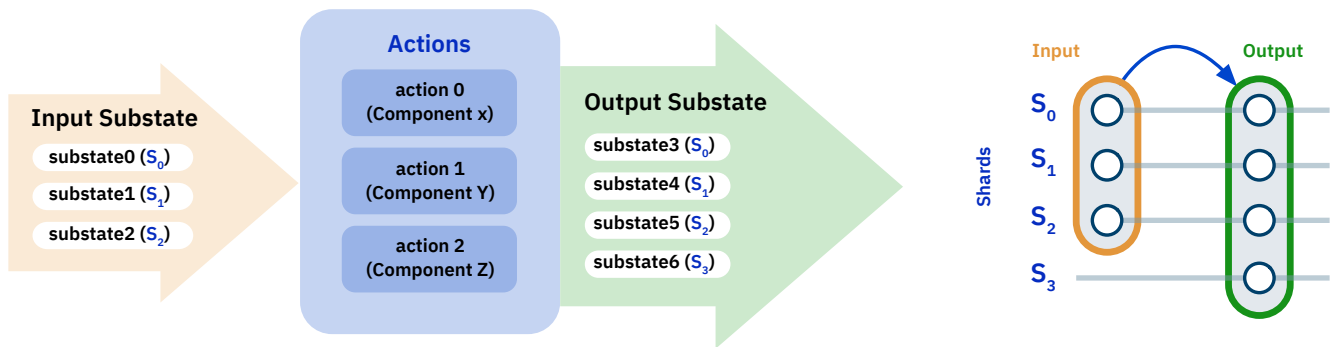
In this case, my request specifies the full transaction, including both the input state (the specific tokens) and the output state change (two transfers). The transaction is valid as long as 1) I still have those specific 5 Token B, 2) I've correctly calculated the number of TokenB, and 3) the reserve still has those specific 13.5 Token B. Here the Action essentially is only *validating* that I've specified everything correctly. If not, it fails.

Pessimistic request: Using the specified "swap" Action, I will send any 5 TokenA from my account to the reserve account.

In this case, the request only provides the minimum the Action needs. I can see that the Token Swap Component's swap Action must be told where to find a sufficient quantity of TokenA, the desired quantity to send, and that this should lead to a corresponding quantity of Token B coming back to me. But I leave it to the Action to construct (and *validate*) the final transaction and outputs. The transaction is valid as long as 1) I still have *any* 5 TokenA, and 2) the reserve still has any 13.5 TokenB.

This may seem complicated, but in both optimistic and pessimistic cases, client software can automatically construct these requests based on the user simply saying “I want to send 5 of my TokenA to the Token Swap Component’s swap Action”. The rest is completely determined by the Action’s definition (and in the optimistic case, the current ledger state known to the client). Radix will provide such client software as part of its libraries for easy integration.

A request to multiple Actions may be composed together in a single request (optimistic or pessimistic), collecting the summed inputs and producing a summed output. Cerberus only needs to know the shards on which to find the input, where to put the output, and the Actions to be used.



At left, three Actions are specified by a client along with required input substate across three shards – and the resulting output substate including those shards plus a new one identified by one of the Actions. At right, the resulting transaction on the relevant shards.

With this Action-based model giving us the ability to create both optimistic and pessimistic requests as needed, our complete scalability solution only needs Radix VM to make some intelligent decisions about how state is stored across shards. We wish to separate unrelated state (for optimistic, parallelized use), but group up highly related state (for pessimistic, serialized use). How to map different pieces of state to shards?

We could use a very simple model where we map Bitcoin UTXO-like states to individual shards. Here, each shard would have a fixed lifecycle, becoming active with a UTXO input, and then forever dormant as a UTXO output (as the input to some other UTXO shard). This approach would map very nicely to optimistic usage, making shards highly granular and unrelated for simple things like token transfers.

However this approach breaks down in situations where we generally want pessimistic transactions. A complex DeFi app could potentially touch an extremely large number of token-UTXO shards, forcing all of those shards to lock during each round of consensus. We would prefer to group such highly-related state into one shard (or a minimum number of shards) to simplify consensus and maximize throughput of serialized operation.

Fortunately the fact that we have a practically unlimited shard space, and consensus that is able to work across that shard space dynamically, allows Radix VM to perform dynamic localization. Localization means grouping a complex set of states into a single shard. Radix VM is able to dynamically shift the expression of state from more distributed, granular, parallelizable form to localized, collective, serializable form as needed. This means that there is no fixed mapping from any particular piece of state (such as the ownership of a token, or a current exchange rate) to particular shards. Radix VM is able to arbitrarily create, and change, that mapping as required to maximize throughput.

In total, Radix Engine's massive available parallelism, combined with its ability to dynamically optimize the performance of individual transactions, allow it to finally deliver the kind of multi-mode scalability needed for DeFi. More than simply delivering a high throughput number in special circumstances, Radix is designed for real-world, large-scale application usage. **Rather than fixing scalability by breaking composability, Radix delivers both without compromises making it the only layer-1 protocol designed to scale a highly interconnected, high-demand DeFi ecosystem.**



APPENDIX 3 – CONSENSUS WHITE PAPER

Cerberus

A Parallelized BFT Consensus Protocol for Radix

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v1.0

Abstract

We present Cerberus, a sharded Byzantine fault-tolerant (BFT) solution to the state machine replication (SMR) problem. While classical SMR protocols ensure global ordering of commands, Cerberus introduces a partial ordering regime that enables a novel state sharding approach. With this approach, the consensus process for commands may be massively and safely parallelized across many nodes while retaining optimistic responsiveness and linear message complexity. Cerberus is designed as a suitable core for Radix, a distributed ledger technology (DLT) platform intended for a variety of deployments including global-scale decentralized public networks.

1 Introduction

Consensus protocols are necessary to conduct state machine replication (SMR) across a network of unreliable machines. While broadly applicable and with a long history, consensus protocols designed for SMR within distributed ledger technology (DLT) networks are of intense interest. These networks range from global-scale public decentralized platforms to critical private business back-ends and often have very high transactional throughput demands. Extending and adapting existing consensus protocols to meet this demand for scalability has proved challenging.

We present Cerberus as a new scalable consensus protocol of particular utility in these high-demand DLT networks – including the Radix DLT network.

Scale Through Parallelism

Most consensus protocols today come to an agreement about the ordering of all commands (e.g. transactions or events) along a *single* timeline, stored in agreed chronological order in a blockchain or similar. This is also known as “global ordering”. Global ordering is generally required for any DLT offering traditional, imperative, “Turing complete” smart contract virtual machines where correct smart contract execution state may depend unpredictably on *any* other state within the system. Coordinating the consensus process for large numbers of commands generally incurs high communication overhead and consequently limits achievable throughput.

State sharding is a technique that can be used to parallelize consensus agreement to increase throughput. However, this parallelization is only possible if the method of sharding used enables separation of commands that are unrelated and thus may be safely processed in parallel. Sharding in this way is in direct opposition to the single timeline implied by a global ordering requirement.

Cerberus addresses this shortcoming by using *multiple concurrent timelines* that only enforce ordering of related events. To achieve this, Cerberus takes a combined approach to the consensus and application layers that does not require global ordering even for smart contract-like functionality. This opens up the possibility of safe parallelism of consensus through state sharding that substantially reduces throughput limitations.

Application Layer Interaction

Eliminating the need for global ordering in the application layer requires an alternative to the typical “smart contract” development method. We believe that a different approach is not only possible, but that it also has benefits (covered outside this paper) that go beyond removing the global ordering burden from consensus. This alternative approach breaks useful DLT application functionality into components that are functionally independent.

An exemplar of an application layer that works in this way is the Radix Engine, the application layer for Radix. The Radix Engine allows a developer to specify transactional functionality in an “asset-oriented” fashion. Useful elements of everyday transactions such as identities, money, products, or property are modeled not within “unbounded” smart contract code, but as functional, separate components that may be updated independently. The Radix Engine is able to translate these components into discrete finite state machines, which then easily translate to a shardable language Cerberus understands. The Radix Engine is, therefore, able to express a transaction as a command to Cerberus consisting of a collected set of state machine updates with explicit dependencies.

Cerberus is designed to take advantage of the highly granular partial ordering made possible by this type of application layer, enabling massive parallelism of consensus.

Related Work

The emergence of Bitcoin, distributed ledger technologies (DLT), and smart contract platforms have fueled a new era of lively research in consensus protocols. Recent developments in Bitcoin-style “Nakamoto” consensus [1][2], PBFT-like consensus [3][4], randomized consensus [5][6], and others represent advancements in fundamental consensus approaches that have diverse tradeoffs in the guarantees they can provide to users.

Cerberus’ advancement in parallelization of consensus is based on a leader-based Byzantine fault-tolerant (BFT) consensus approach. BFT protocols for state machine replication (SMR) have the advantage of strong consistency (as opposed to availability), with a long history reaching back to the work of Lamport, Shostak, et al [7]. A definitive milestone in practical BFT protocols was reached with PBFT [8].

PBFT uses a two-phase consensus process of *prepare* and *commit*. Nodes (or participants in consensus, e.g., computers in a decentralized network) select from their ranks a “leader”. A leader proposes a command by broadcasting it to the other nodes. When $2f+1$ nodes (where f is the maximum tolerated number of Byzantine faulty nodes) acknowledge the command’s validity by responding with their signature, the signatures are aggregated into a “quorum certificate” (QC) and broadcasted to the nodes as proof. The $2f+1$ requirement ensures that even double-voting faulty nodes cannot win a majority. Once $2f+1$ nodes acknowledge and sign the proof, the command is considered “committed” and final.

PBFT provided the first practical implementation of a partially synchronous BFT protocol, but it did not grapple with today’s demand for scalability in open public networks or global cloud deployments. Of particular concern for these large network deployments is the high message complexity required for each “view change”, or leader replacement, due to PBFT’s requirement for $2f+1$ QCs as a proof of safety before making progress.

More recent BFT work such as Tendermint [3] extended the state of the art in the context of large blockchain networks by reducing the messaging overhead required for view changes. Rather than requiring $2f+1$ QCs, a new leader simply “extends” its highest QC (using it as the basis for a new proposal). While eliminating substantial messaging overhead, this opens the possibility that there is another node with a higher QC. Therefore a leader must wait a defined maximum network delay time before it can be sure that it has the highest QC, preventing deadlocks. This reduces responsiveness of the network by introducing a fixed latency and therefore limits throughput by the fixed time window rather than the actual speed of the network.

More recently, the HotStuff protocol [4] presented an important extension to previous BFT approaches. HotStuff maintains optimistic responsiveness while keeping low messaging complexity by adding a third phase between prepare and commit: *pre-commit*. The pre-commit phase allows leaders to “change their mind” after voting “in the phase”, meaning that consensus can safely proceed on new proposals and the network no longer has to wait a defined network delay time. A commit occurs as soon as QCs for pre-commits have been

collected successfully. Combined with threshold signatures, this provides “optimistic responsiveness” and linear messaging complexity even for view changes. We leverage many of HotStuff’s extensions, and corresponding advantages, as we construct the sharded Cerberus BFT protocol.

Scope of this Paper

We begin by describing the Cerberus model as a 3-phase BFT consensus protocol with a sharding technique for partially-ordered commands. We then briefly analyze the model to show that it achieves linear messaging complexity while balancing the requirements for node capabilities, parallelization and security. A detailed specification of the Cerberus protocol, as well as formal proofs of safety and liveness, will follow in a later revision of this paper.

We follow the Cerberus model with topics relating to the implementation of Cerberus in practical DLT networks, focusing on the Radix public network. We describe methods to provide protocol functions outside of Cerberus consensus that are required in order to create and deploy a useful Cerberus-based network.

This paper is a working document for the Radix team and still under active research; as a result some areas will require further explanation or revision. As always, we welcome comments, collaboration and feedback.

2 Model

Overview

Cerberus solves the state machine replication (SMR) problem [15]: the fault-tolerant distribution of state across many nodes. Cerberus builds on top of BFT-style consensus, retaining two main properties of BFTs:

1. Safety under asynchrony, liveness under synchrony
2. Favoring consistency over availability (i.e. safety over liveness in periods of asynchrony)

We also make the following common BFT assumptions in our basic model. While we make these assumptions for the purposes of describing a simple form of the Cerberus model, some of these restrictions may be lifted in practical implementation (notably a static node set). We discuss some of these in the *Implementation* section of this paper.

- A static, globally-known set of nodes
- Partial synchrony (i.e. after some global stabilization time there is an unknown bound within which all messages arrive)
- $2f+1$ nodes follow protocol, where f is the number of Byzantine nodes (per shard in the case of Cerberus)
- Adversaries that are computationally incapable of breaking standard cryptographic primitives
- The availability of a practical threshold signature method to aggregate votes from nodes
- Communication between correct nodes is peer-to-peer, authenticated, and reliable

In classical BFT, a log of client commands is globally ordered so that correct nodes can agree on the final state. Performance improvements have traditionally relied on “vertical” scaling or “pushing more commands down a single pipe”. In practice, applications generally do not require global ordering of all commands; they only require ordering of commands that are interdependent.

Cerberus requires an application layer that specifies dependencies so that it may implement *partial ordering*, only ordering commands which are related. Cerberus’s “shards” represent these dependency relationships between commands. Commands that do not share a partial ordering relationship also do not share shards and can be safely executed in parallel.

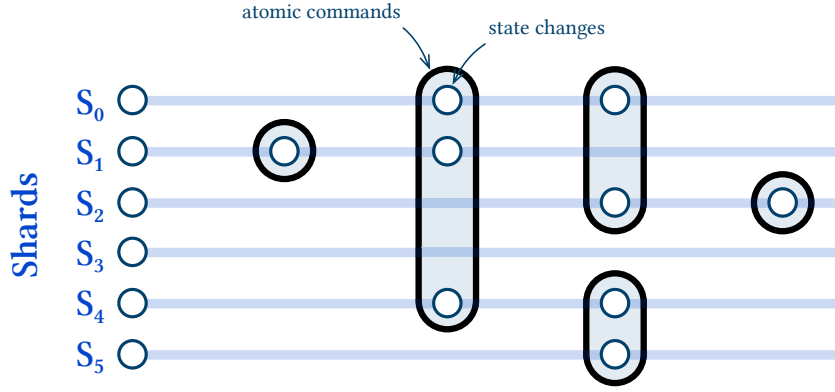
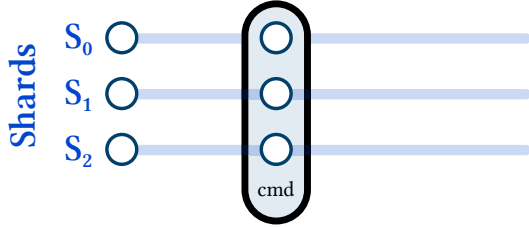
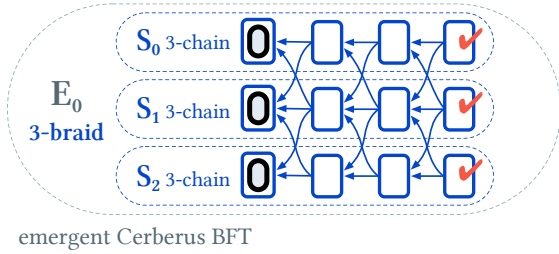


Figure 1: Partial ordering created by commands (thick edged pill shapes) grouping related shard state changes (thin edged circles) across shards s_0 - s_5

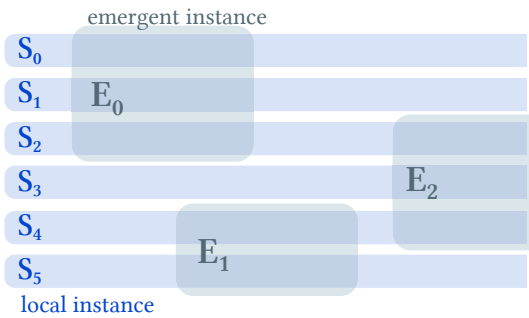
With commands sharded in this way, Cerberus conducts consensus by treating each shard as a separate “local Cerberus” BFT instance, with atomic commits of commands across arbitrary sets of shards provided by an “emergent Cerberus” BFT. We construct our description of the Cerberus protocol in three major steps:



A) Partition the application layer by formalising the partial ordering requirements of each command in the application layer.



B) Combine multiple “local Cerberus” BFT instances to form an “emergent Cerberus” instance across shards with analogous properties to a simple traditional BFT.



C) Parallelize Emergent Cerberus instances across shards, enabling massive and safe parallelization of the consensus process across the network.

A) Partition the Application Layer

Cerberus requires that its application layer specify the partial ordering requirements of its commands so that consensus may be effectively parallelized. Specifically, the implementer must define the following *pure* functions for the application layer:

1. Specify the initial state of each shard

```
INIT() -> map of shard -> initialShardState
```

2. Partition the command into its set of shards

```
PARTITION(command) -> shards
```

3. Map a command to its per-shard intermediate results

```
MAP(command, shard, shardState) -> localResult
```

4. Reduce the local intermediate per-shard results across all shards

```
REDUCE(command, shard, shardState, localResults) -> nextShardState
```

INIT defines the initial state of the application. **PARTITION** yields the shards a command must be synchronized with to maintain correct partial ordering. When a command is to be executed, it is first mapped to per-shard intermediate results via **MAP**. The intermediate results are then reduced to a global result and applied across all shards of the command via **REDUCE**. This approach is similar to that of contemporary large-scale distributed computing models [9]

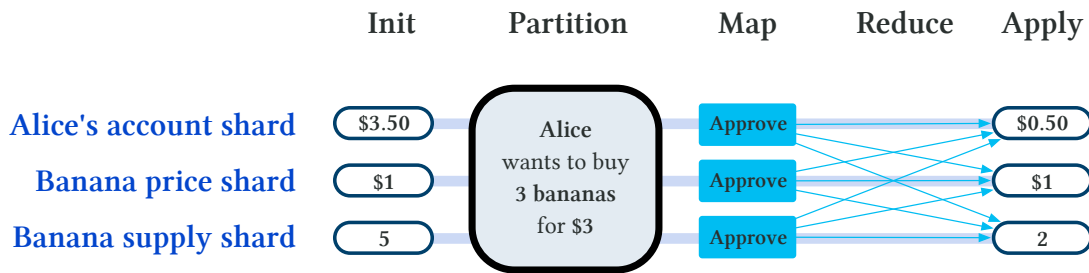


Figure 2: Example of using the application layer to partition consensus

For example, consider a simple application layer consuming a command saying “Alice wants to buy 3 bananas for \$3”. This command could map to shards representing the available banana supply, current banana price as well as Alice’s current account balance¹. Upon receiving the command, the shards will map the command of it to get their individual intermediate result. Cerberus then distributes these local results to all shards and applies them in the reduction step to yield the next shard states.

B) Local Cerberus Instances into Emergent Cerberus

We now describe how Cerberus can execute a command atomically across all shards it is partitioned to, starting with the operation of consensus within a single shard and moving on to operation of consensus across multiple shards.

Local Cerberus

We start with a single BFT instance patterned after the BFT design of HotStuff [4]. We call this a local Cerberus instance on a single shard that, operated in isolation, provides the same optimistic responsiveness and linear messaging complexity of HotStuff (as applied to a single blockchain). These advantages will serve as a useful foundation as we construct Cerberus as a multi-shard consensus protocol.

¹ Note that Radix’s mapping and usage of shards are different than the example shown in figure 2. See “Radix Engine” in the Implementation section of this paper for more information.

Specifically, we start with a “3-chain” BFT composed of prepare, pre-commit, and commit phases. A quorum certificate (QC) – an aggregation of signatures from validating nodes – is required for each phase.

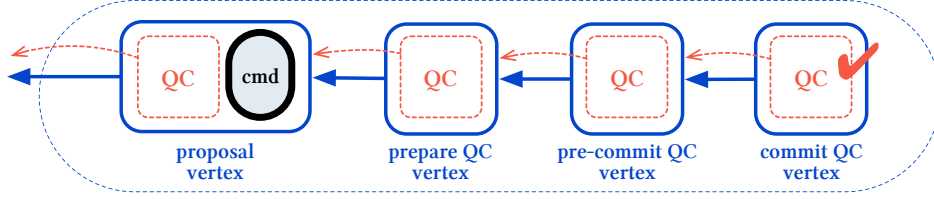


Figure 3: A local Cerberus 3-chain

Progression through each phase of consensus is marked by the creation of a “vertex” (shown as the enclosing rounded squares in figure 3) composed of that phase’s QC and a reference to the Vertex’s parent, similar to blocks in a blockchain. In a local Cerberus instance, the vertex representing a new proposal (the beginning of a new 3-chain) includes the command.

Multiple Local Cerberus Instances

We now introduce multiple instances to the model with n independent local Cerberus instances operating in parallel. For simplicity, we assume that each of these instances are served by a distinct set of nodes (i.e. each node is assigned to exactly one instance). This bi-map between nodes and instances are assumed to be static and well known to every node².

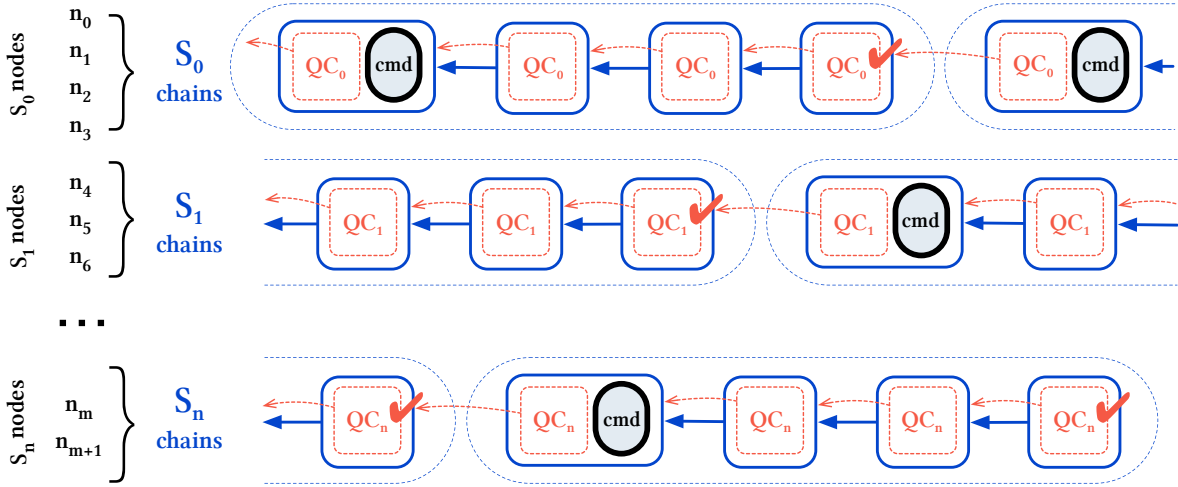


Figure 4: Nodes distributed over n parallel instances of local Cerberus (denoted S)

These parallel instances are not very useful yet however as there is no way to synchronize them.

Emergent Cerberus Instance

We now synchronize multiple local Cerberus instances. To do this, we add synchronization primitives between multiple local Cerberus instances and model the synchronized combination as a single emergent Cerberus instance which behaves like a single cross-shard BFT instance.

² We discuss practical mapping of nodes to shards and management of dynamic node sets in the Implementation section of this paper.

We implement this emergent level by mapping analogous concepts from Traditional BFT (i.e. local Cerberus instances) to multi-shard concepts in emergent Cerberus:

Traditional BFT (Local Cerberus)	Emergent Cerberus
Leader	Leader Set
Proposal	Merged Proposal
QC	QC Set
3-chain	3-braid

Emergent Leader: Leader Set

A local Cerberus instance designates a leader to drive progress in each new view via *new-view* messages from that shard's nodes. At the emergent instance level, the set of leaders elected by a set of shards is considered, in aggregate, to be the emergent leader called a "leader set".

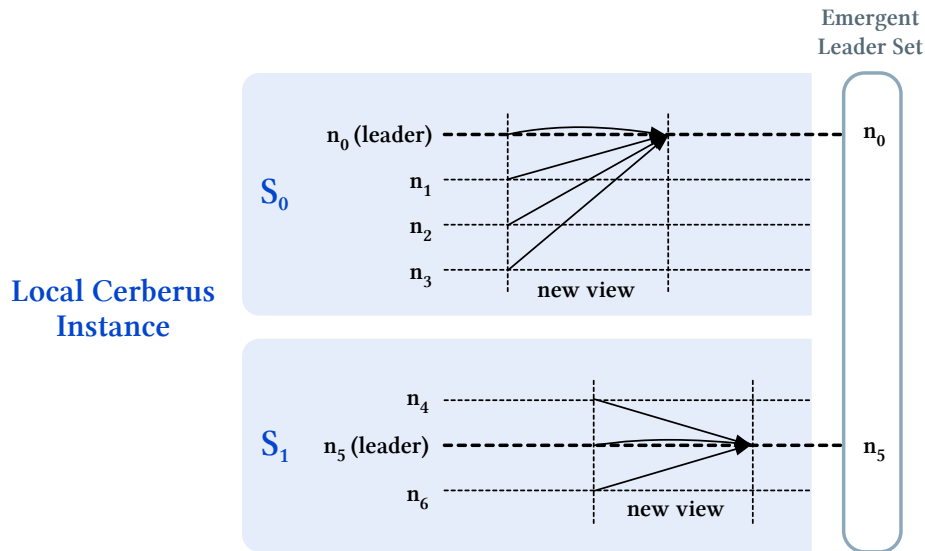


Figure 5: Leader set implicitly formed as a set of local leaders

In the example shown in Figure 5, two Cerberus shards independently elect local leaders (n_0 and n_5 , respectively) via *new-view* messages. The set $\{n_0, n_5\}$ is then considered the leader set across the two instances. Note that at this stage no node is aware of the leader set, it is only a theoretical construct for the purposes of describing and analysing the protocol.

Emergent Proposal: Merged Proposal

We now add the mechanism by which a leader set can propose a new command, or a "merged proposal".

To explain by example, each local leader in a leader set, $\{n_0, n_5\}$ in our example, broadcasts a local proposal to all nodes in the emergent Cerberus instance (the shards of which are specified by the application layer's [PARTITION](#) function). Each node then takes each proposal from each local leader in the leader set and merges them to form a merged proposal.

As commands are sharded, each local proposal also includes the local intermediate result provided by the application layer's [MAP](#) function.

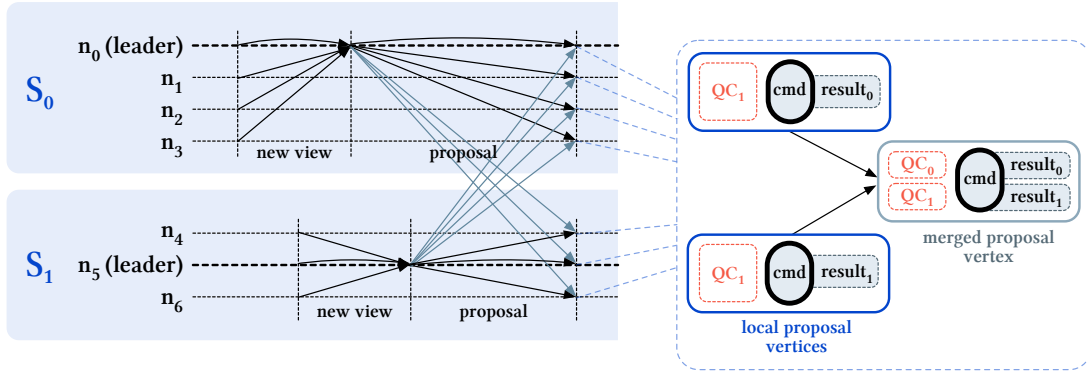


Figure 6: An emergent merged proposal is formed from a set of local proposals with equivalent commands

This merged proposal then acts as a vertex at the emergent level which begins the 3-braid between shards S_0 and S_1 .

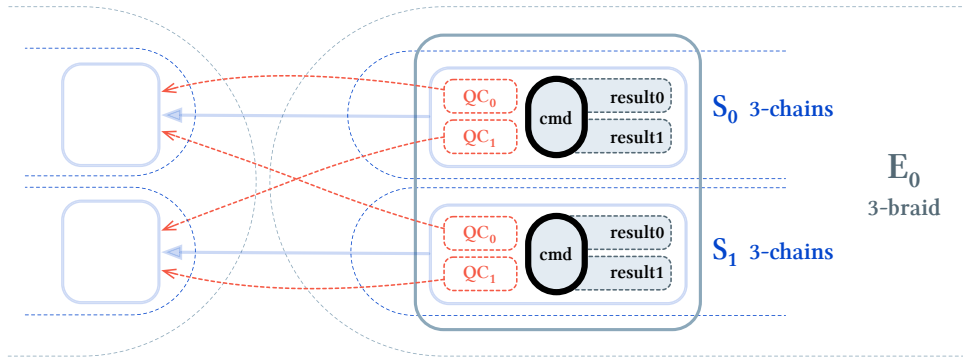


Figure 7: The merged proposal is added to the end of both chains

Emergent QC: QC Set

We now add a mechanism for the leader set to collect votes for the vertex. When a node receives an merged proposal, it sends its vote to its local leader. When the local leader has $2f+1$ votes, it creates a QC. From the emergent view, the leader set creates an emergent QC, or “QC set”, made of one QC from each shard, when each of its local leaders creates its local QC.

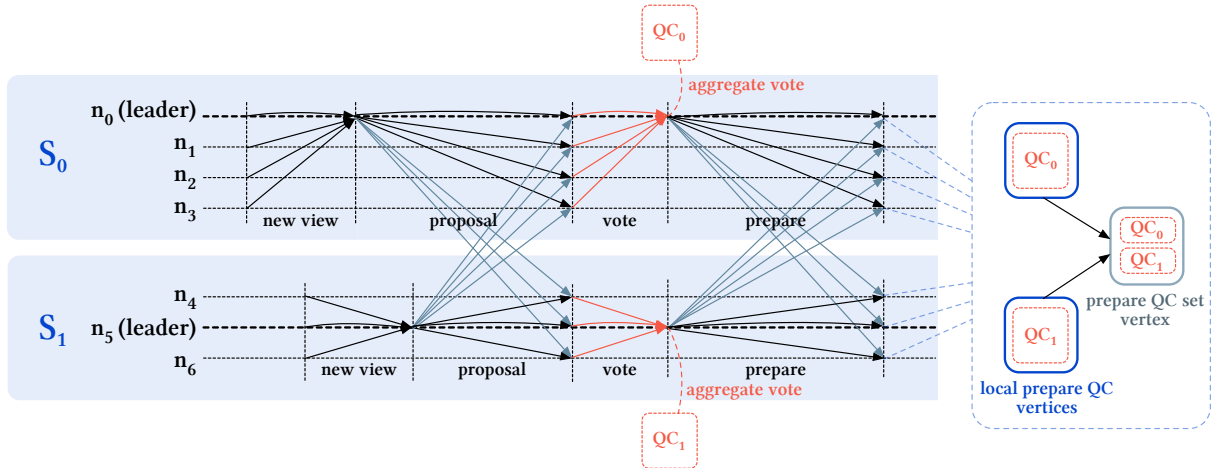


Figure 8: Emergent QC set for cmdA collected from QCs from each shard, for a single phase

Emergent 3-chain: 3-braid

The QC set is then passed on to the leader set in the next view where the process starts again repeating the process until a 3-braid is formed for cmdA.

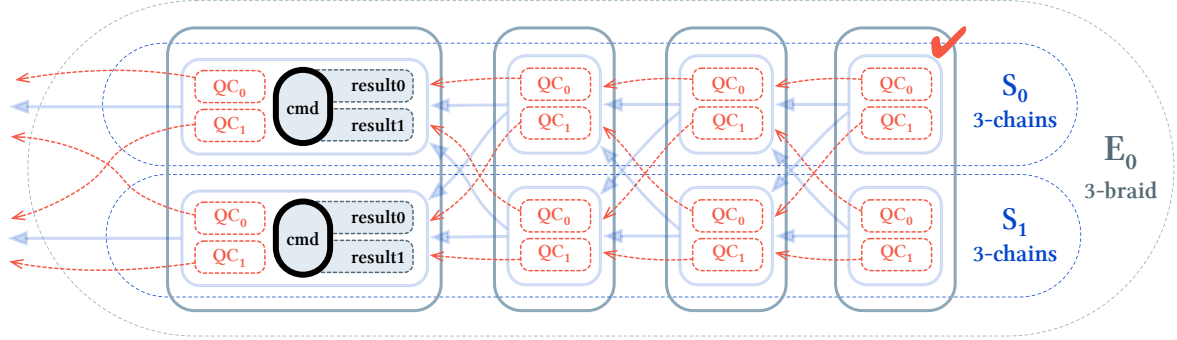


Figure 9: 3-braid in an emergent instance across two local instances

Once a 3-braid is created for a command, the command is considered “committed”.

Emergent Failure Mitigation

We now consider how the emergent Cerberus instance handles leader failures, forks, and merges.

Emergent Leader Failures

In a local Cerberus instance if a vertex fails to get created at some view, a blank vertex is added in its place so that the parent/justification equivalence check is false. In an emergent Cerberus instance, every parent of each shard must be checked.

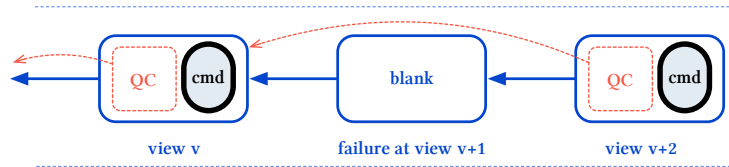


Figure 10: Local Cerberus instance failure, the parent of v+2 vertex (vertex at v+1) is not equivalent to the justification of v+2 (vertex at v).

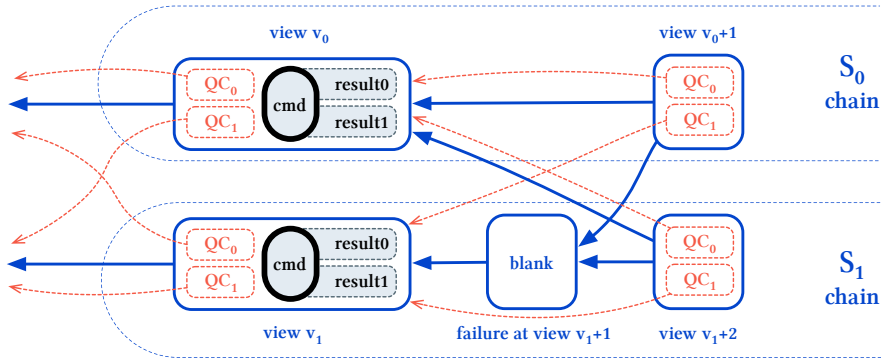


Figure 11: Cerberus failure, parent0 is not equivalent to the rest of the parents or justifications.

Emergent Forks

In a single local Cerberus instance, safety and liveness are preserved in the face of forks with checks on locked vertices and/or QC view numbers. With an emergent Cerberus instance, there is additional complication since a fork may occur outside of a local instance's view.

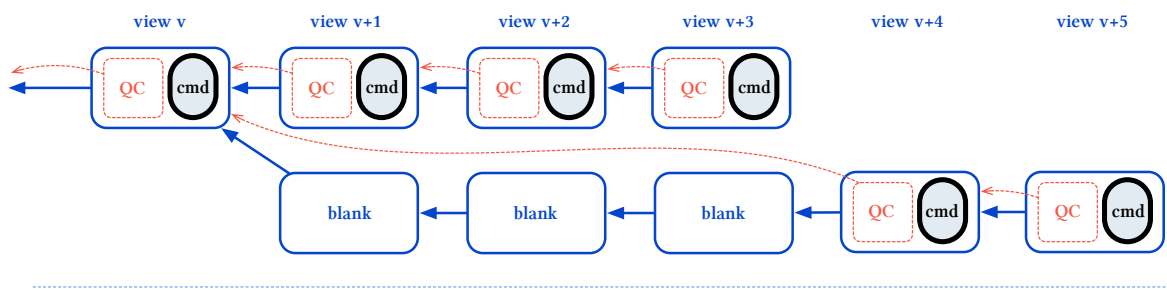


Figure 12: Local Cerberus instance fork which requires an “unlock” of the vertex with cmd at view $v+1$

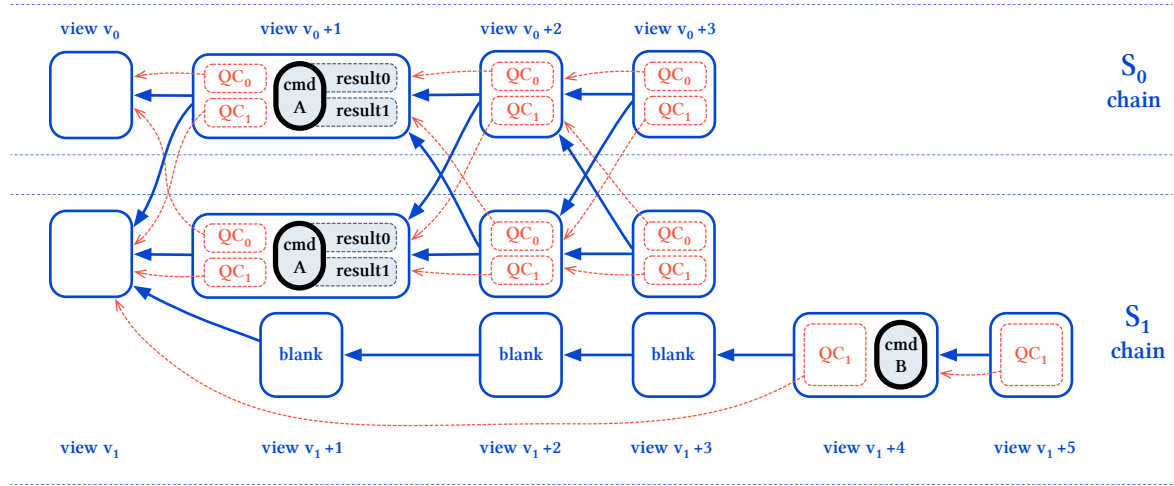


Figure 13: Emergent Cerberus fork which requires shard S_1 to “unlock” shard S_0 from the vertex with cmdA at view v_0+1

We extend HotStuff’s check in the emergent instance by including the forked instance’s higher QC in the emergent QC. This is detected when a forked instance receives a proposal that has an older, conflicting QC. The forked instance then sends the forked QC as a message to the other shards.

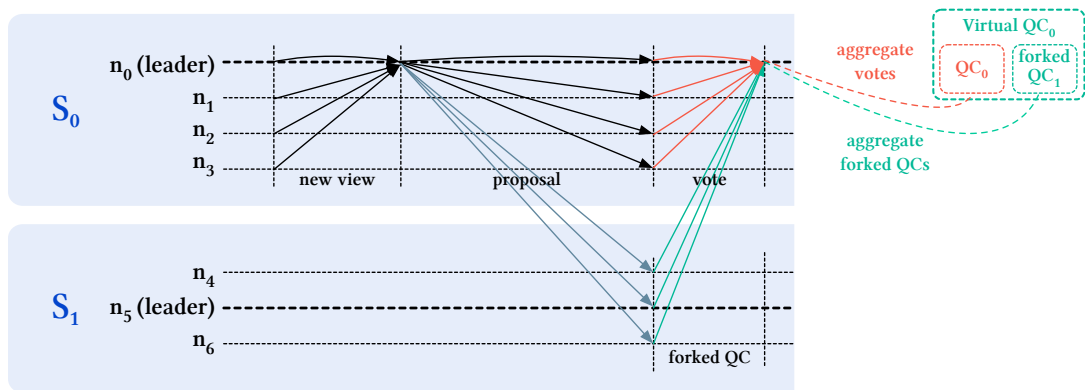


Figure 14: Recovery of a cross-instance fork in an emergent instance through forked QC

This forked local QC is then included in the emergent QC and is proof of a fork. A “virtual QC” is then created from this proof which “virtually” has a higher `viewNumber` than the current QC and extends back to the point where the forked QC is no longer forked.

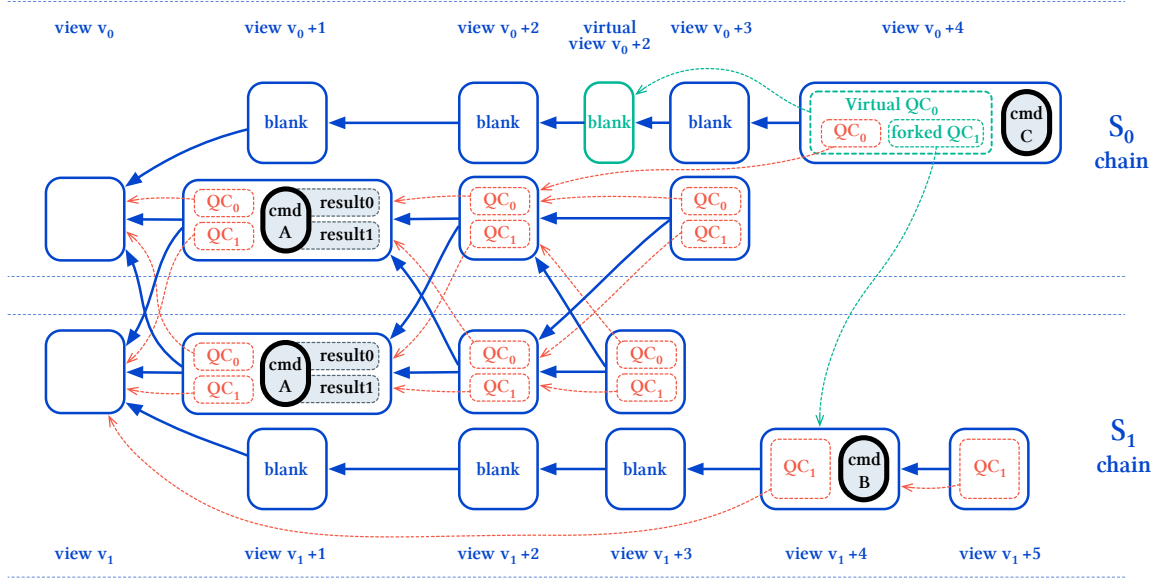


Figure 15: A Virtual QC with a virtual higher `viewNumber` is used to “unlock” a forked Emergent Cerberus instance

Virtual QCs thus act as a level of indirection. We now refer to QCs which actually contain signatures as *physical* QCs. A *virtual* QC is composed of a primary *physical* QC and an optional *forked physical* QC. A *virtual* QC composed of just the primary QC without the forked QC acts like the primary. For the purposes of simplicity, for virtual QCs with no fork, we will represent this as a single blue box. And for virtual QCs with a fork, we will represent this as a blue box with a justify arrow pointing to a virtual node. For example, Figure 15 can be reduced to the following diagram:

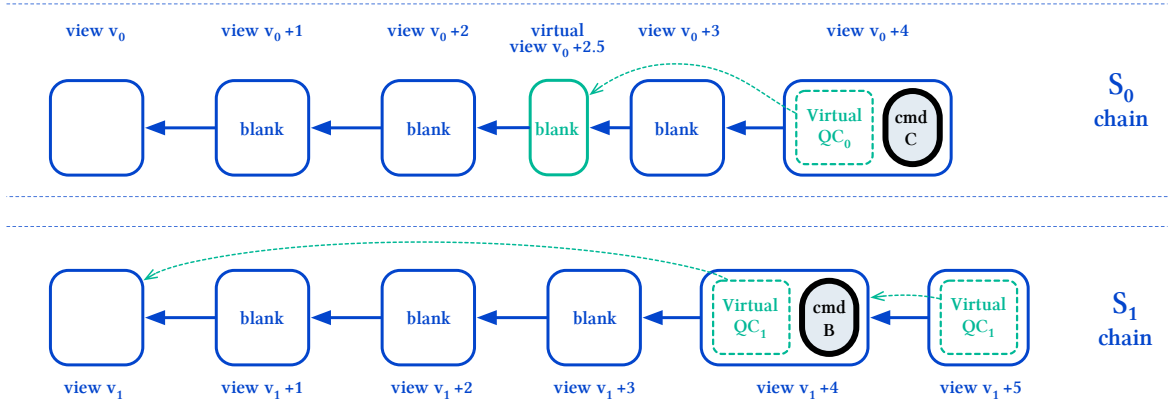


Figure 16: Virtual QCs operate as a layer on top of physical QCs and substantially reduce the complexity of fork management

Emergent Merges

As long as the leader set is correct and proposes a command, progress should always be made in order to ensure liveness. A complication arises in the case of forked local proposals, which are possible when dealing with separate independent Cerberus local instances.

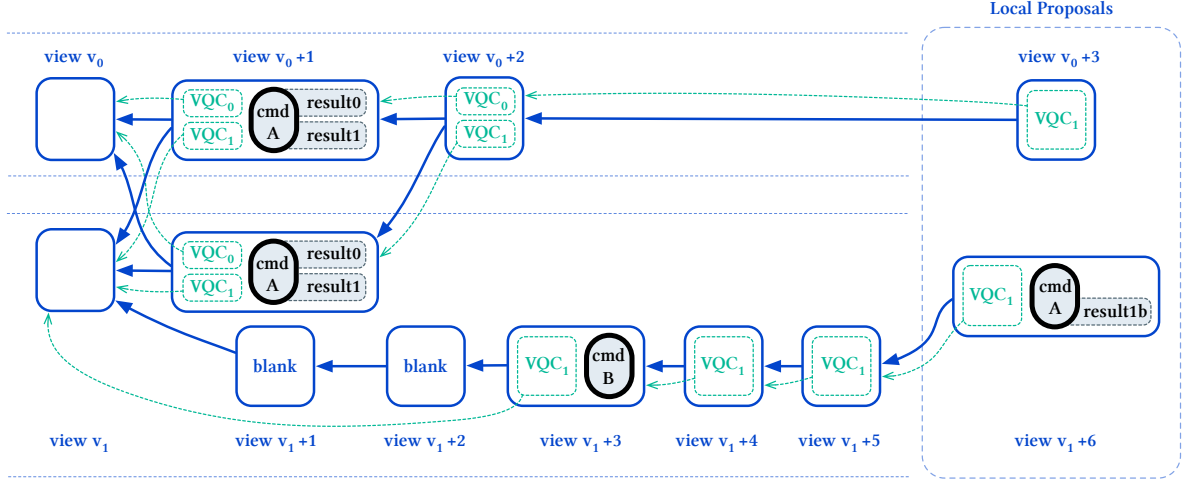


Figure 17: Local Proposals may be forked - in this example there are two vertices connected at view v_1+1 on S_i and thus is forked

To resolve forked proposals, we modify the merging of local proposals so that if there exists a fork amongst the local proposals, we reuse the Virtual QC method to effectively unlock the locked shard.

As an example, in Figure 17 VQC0 and VQC1 in views v_0+3 and v_1+6 respectively fork. To resolve this, the physicalQC of VQC1 is added as a forkedQC to VQC0. This effectively removes the fork.

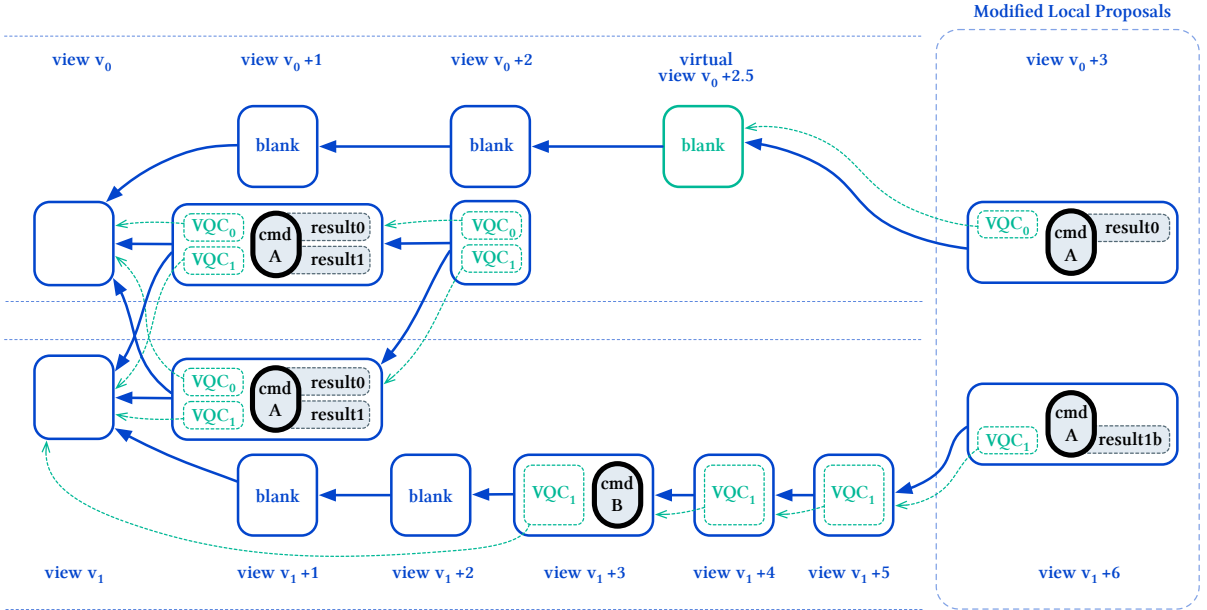


Figure 18: As the original local proposed virtual QCs forked, we modify the proposals by including forked QCs until the fork disappears. Normal merging of proposals can now continue.

The modified local proposals are then able to merge as normal without forks. The result can be seen in Figure 19.

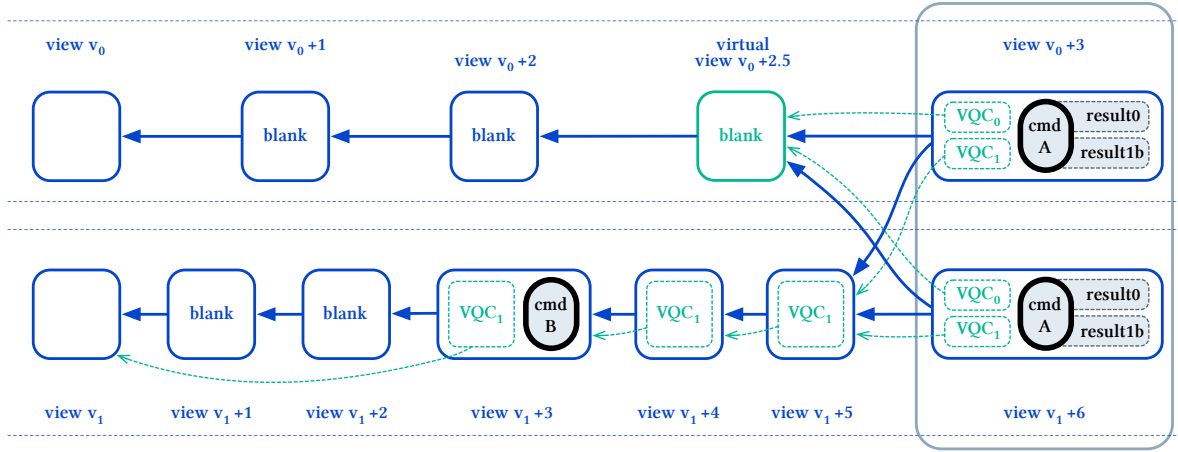


Figure 19: The merged proposal post-modification cleanly attaches to both chains

C) Parallelize Emergent Instances

Now that we have described a mechanism that combines multiple local Cerberus instances into an emergent Cerberus instance, we can now parallelize emergent instances across a fixed set of shards.

Given a set of shards $S = \{S_0, S_1, \dots, S_n\}$ we can create an emergent instance across any subset from S . Furthermore, we can create a set of emergent instances in parallel (a set of subsets of shards from S) as long as each instance is disjoint (no element in common) from others. For example, $\{\{S_0, S_1\}, \{S_2\}, \{S_3, S_4, S_5\}\}$.

The creation of emergent instances is dictated by the set of commands to be processed and the result given by the application layer's **PARTITION** function across these commands.

In the example below, **cmd0-4** defines a set of emergent instances. The partitions of **cmd0** and **cmd1** are disjoint shard sets thus they are able to create two separate emergent instances in parallel. On the other hand, **cmd1** and **cmd2** intersect at shard S_3 thus one command must be executed before the other (**cmd1** in this case).

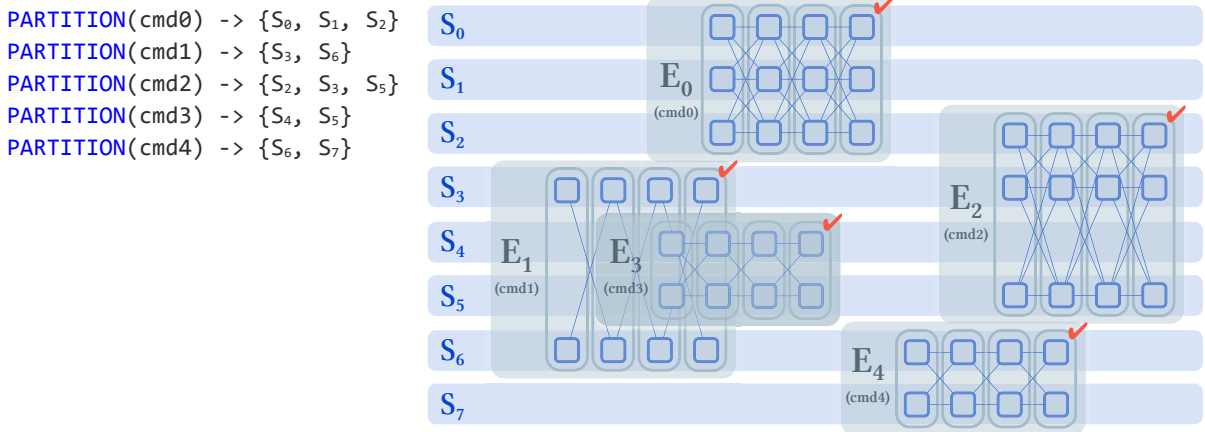


Figure 20: Example parallelization of Cerberus for commands cmd0-4 across shards S_0-S_7

The set of shards can be arbitrarily sized, enabling massive parallelization of the consensus process. Notably, this form of parallelization introduces a tunable tradeoff between security and scalability: reducing the number of shards each node covers increases parallelism and scale, but decreases the validator set size of a given shard. This capability allows the system designer to select a minimum threshold of per-shard node coverage that achieves high practical security, while maximizing parallelism and scale by allowing a large number of independent nodes. This is unique to Cerberus and discussed further in the analysis section.

Analysis

As with HotStuff, the messaging complexity of Cerberus is measured in authenticator complexity, the number of authenticators received by any given node in order to reach a consensus decision. In the nominal case using threshold signatures, HotStuff achieves authenticator complexity of $O(n)$ – linear complexity with the number of nodes n^3 .

In Cerberus, we must consider sharding. Given an evenly distributed node-to-shard mapping and single-shard commands, Cerberus behaves like HotStuff and the authenticator complexity given a correct leader is $O(n / T)$ where T is the number of shards. As T approaches 1, or a single shard consensus instance, authenticator complexity approaches $O(n)$, the same as HotStuff.

With emergent Cerberus instances, cross-shard consensus requires a separate QC per shard. Thus the authenticator complexity with correct leaders is $O(s * n / T)$ where s is the number of shards in a given command (and s must be $1 \leq s \leq T$).

This definition lets us consider the tradeoffs of deploying and using a Cerberus-based network. In effect, s / T is the burden placed on the network by the application layer, while n / T defines a tradeoff between security (more nodes per shard) and parallelism (fewer nodes per shard). For example:

- In a practical HotStuff network, a certain set number of nodes might be selected to provide the amount of per-shard node overlap needed to provide high practical security, without creating too much messaging overhead. If we apply that same set number to the number of nodes per shard in a Cerberus network, the throughput of the Cerberus network can be scaled linearly with the number of nodes, without compromising per-shard security.
- If we assume an application layer creating commands with an average low number of dependencies (relatively simple transactions), s burdens messaging very little. One can imagine implementing a public network in which greater fees are paid only for unusually complex transactions.

3 Radix DLT Network Implementation

While Cerberus is a consensus protocol that may be applied to a variety of applications, an application area of particular interest is that of scalable DLT networks such as Radix. This section describes a set of conservative and reasonably extensible potential solutions to implementation issues in deploying Cerberus in Radix, and potentially other DLT networks. While we attempt to cover some of the most important aspects of implementation, the solutions provided are not intended to be fully formed, nor is the listing of issues exhaustive.

Radix is intended for a range of network deployment types, the most challenging of which is a public permissionless network. The solutions proposed in this section are generally intended to be applicable across Radix network deployment types, but with a particular focus on a very large network with open participation.

Radix Engine Application Layer

To provide its parallelization benefits, Cerberus requires an application layer that specifies the dependencies between commands. Here we present the key interactions between one such application layer, the Radix Engine, with Cerberus consensus.

A Radix node combines a Cerberus-based Radix Ledger with an application layer called the Radix Engine. A collection of Radix nodes make up a Radix network called a “universe”, and transactions created by the Radix Engine are called “atoms”. Atoms contain a set of discrete updates to finite input/output state machines called “particles”. This is similar to the common blockchain UTXO model but extended for greater flexibility. As might be expected, atoms must be committed or rejected by Cerberus consensus atomically; either all particles within the atom are valid and the atom is committed, or one or more particles are invalid and the atom is rejected.

Radix Engine atoms and their particles map neatly to Cerberus commands that specify related state changes. The particles themselves define the mapping of the atom to shards, with each particle mapping directly to a shard (the particle/shard addressing scheme is described below in Particle Identifiers).

³ Similar to other leader-based BFT protocols, the linear message complexity of Cerberus relies on the use of a signature aggregation scheme (e.g., using BLS signatures).

Each particle/shard is, therefore, a simple state machine that looks like figure 21.

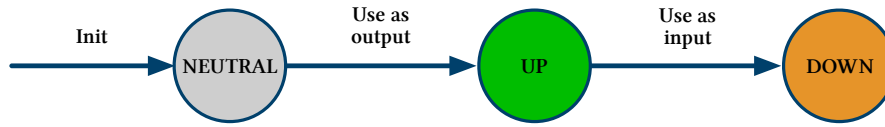


Figure 21: The defined sequential states of “particles”: neutral (default), up (active), and down (consumed)

Given the UTXO-based logic provided by the Radix engine, which dictates which input/output pairs are valid, we use Cerberus to enforce the atomic commit of updating the state of multiple particles (either up or down). An example ledger is shown in figure 22.

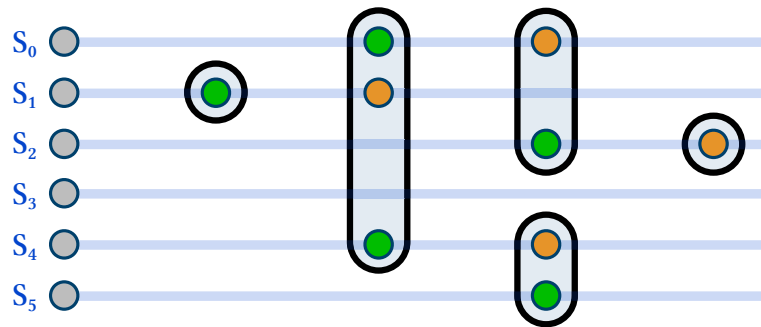


Figure 22: Thick edged pills represent commands that group related particle state changes (thin edged circles) in their respective shards.

Particle Identifiers

Particles, being the fundamental unit of the Radix Engine, must be uniquely addressable by an identifier. Inspired by Chainspace’s “Hash-DAG structure” [11], we identify a particle by the atom in which it was created to create a high-integrity data structure of transactions in which it is both hard to forge a transactions history but easy to verify it.

A particle’s identifier uses a combined hash of a containing atom’s identifier and the particle’s (deterministic) index within that atom to identify particles. A particle’s identifier is “created” when it first appears in an atom, and any subsequent changes to that particle use this same identifier.

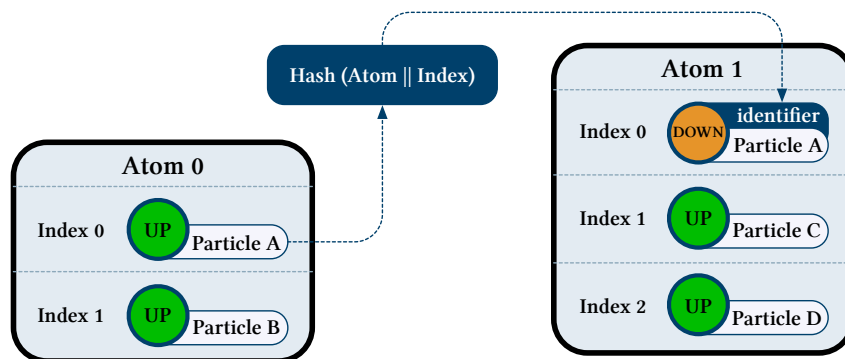


Figure 23: Using a particle’s identifier (right) as defined at the particle’s creation (left)

This particle identification scheme produces immutable, auditable chains of hashes that may be traced back to the creation of each particle. Further, since the particle identifier is based on the atom it was created in, rather than its content, there is no need for nonces to ensure uniqueness amongst particles in different atoms with identical content.

Shard Mapping

We use a one-to-one mapping of particles to shards based on particle identifiers. Distributing particles across the shard space in this manner has advantageous properties:

1. The shard space can be fixed-width, static and well-known ahead of time, removing the need for complex dynamic sharding procedures.
2. The consensus process is naturally “load-balanced” across all shards, as particle identifiers are hashes that can be expected to be nearly uniformly distributed.
3. The potential for attack or disruption for an adversary of taking over any single shard are low, as a single shard contains the state of only a single particle.

This mapping provides good consensus performance and security on transaction writes. *Transaction read* queries, however, such as requesting a complete transaction history for a given account, are more difficult since every transaction is addressed by content rather than by a single identifier.

A comprehensive solution to the contradictory requirements of consensus (e.g., shard safety, load balancing) and applications (e.g., easy account querying) requires two separate solutions, each tuned to the needs of its domain and freed from the constraints of the other. We decouple the shard mapping used by consensus from that used by applications with a proposed *query layer* which operates independently on top of Cerberus. The architecture is similar to other distributed, scalable systems such as search engines on top of the web, domains on top of the internet, or queries on top of IPFS [12] or BitTorrent [13].

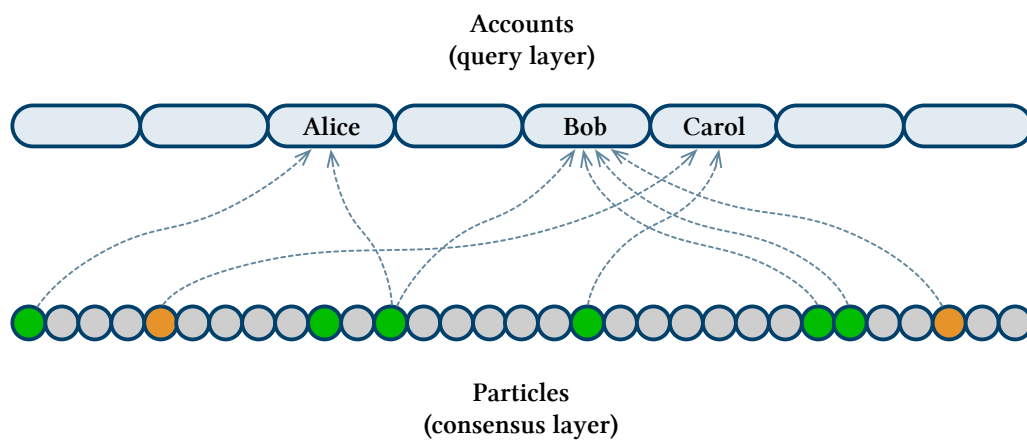


Figure 24: A query layer utilizing a secondary index on accounts

A simple query layer could be one based on a secondary index on accounts as shown in figure 24. This can be supported as long as the query layer can retrieve up-to-date particles.

Some practical solutions to retrieving the latest state are:

1. Let the query layer eavesdrop on every message in the consensus layer. The query layer can then tell which transactions have been committed to.
2. Have the query layer “crawl” through transactions. Since practically every transaction is linked to another, a crawler can easily walk the chain of transactions.

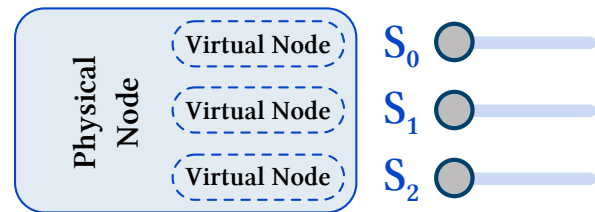
Once the latest state has been received it can be submitted to any of the many available distributed key/value databases or a decentralized DHT such as Kademlia [14].

A more specific description of the precise query layer mechanism to be used in Radix is left for future work.

Virtual Single-shard Nodes

As a theoretical model, Cerberus assigns one shard to every node. In practice, mapping nodes to *machines*, with only one shard per machine, would be an inefficient use of resources since shards represent particles that are both ephemeral and typically inactive. Most machines will have hardware capable of serving a very large number of shards simultaneously.

To retain the simplicity and correctness of Cerberus, we *virtualize* an arbitrary set of single-shard nodes in every physical node (i.e. machine). The virtual nodes behave exactly as their physical counterparts would, with the only difference being that multiple virtual nodes map to the same physical machine (with the same address and public key).



Global Shard

In Cerberus, shards allow different nodes to participate in only a subset of the network’s total consensus activity in processing general client request commands (i.e. processing transactions). However in a practical DLT network, certain functions outside of general client requests require consensus and visibility of all nodes (e.g. churn management). We therefore introduce the concept of a “global shard” that can support these functions. As opposed to regular shards that are mapped to subsets of nodes (and machines) as described above, the global shard is mapped to *all* nodes. All nodes participate in consensus on special-purpose commands there, and store the global shard’s state.

Consensus in the global shard operates identically to other shards, using the same decentralized consensus process in the form of a single local Cerberus instance. Rather than conducting consensus on general client commands, the global shard conducts consensus on the state of a set of global parameters that are required for network operation, such as the current node set and state of staking as described below. It may also be used for useful cross-network functions such as network time.

While regular shards in Radix may be considered ephemeral (i.e. related only to short sequence of up/down particle updates), the global shard can be thought of as a sort of narrow-purpose blockchain that is operated continuously. Compared to command throughput of the general network, the required rate of consensus state changes in the global shard is low and is therefore safe to be conducted globally without significant impact to performance.

Node Sets and Churn

Cerberus requires that all nodes must know the set of all nodes. However a practical DLT network requires tolerance of churn as nodes choose to enter and leave the network. To maintain this dynamic node set, we use an identity register that is updated in the global shard.

The identity register uses the notion of an *epoch* that captures the node set and shard mapping at a certain point in time. Epochs are globally known time periods (e.g., 1 hour or 1 day) during which the node set is fixed. Each node keeps track of the current highest epoch and includes it in all messages and client commands. At the end of an epoch, all nodes must reach consensus on the next node set by aggregating *leave* and *join* requests into a special command on the global shard.

Sybil Resistance

As a public network, we must protect against the possibility of an attacker gaining undue control of the node set and gaining the ability to break safety/liveness at will (the equivalent to the 51% bitcoin attack). We plan on employing Proof of Stake (PoS) in the BFT setting similar to Casper, Tendermint, and others to protect against Sybil attacks.

Rather than having equal voting power across all nodes, each node will have voting power relative to the amount of stake they have registered in order to compete for selection as a validator. This state will be maintained by the global shard using special *stake* and *unstake* commands. As all nodes must be a part of the global shard, each node will have complete information on the mapping between nodes, their shards, and their stakes. This allows every node to correctly execute consensus in Cerberus using validator sets selected using stake.

An in-depth description of the application of PoS to Cerberus within the Radix public network is left for future work.

Leader DOS Resistance

With our proposed leader election mechanism, future leaders are known well in advance and thus particularly vulnerable to denial-of-service (DOS) attacks. There are several possible mitigation strategies:

- Use sentry nodes as full proxies between validators (as in Tendermint [10])
- Use a leaderless model at the cost of quadratic rather than linear message complexity
- Use randomness to make the next leader less predictable (and thus give adversaries less time). This strategy would require the use of a verifiable random function, an area of ongoing research.

A deeper analysis of possible applications of these strategies within Cerberus for Radix public or private network deployments is left for future work.

Client Command Requests

Cerberus assumes that client command requests are known by leaders for them to select for proposal. A mechanism is required to reliably convey requests to leaders. A few possibilities exist, depending on the specifics of anticipated network usage.

The most common choice is an approach wherein client requests are broadly shared and a pool of pending requests is loosely synchronized (commonly referred to as a “mempool”).

Command Proposal Selection

Cerberus, being a leader-based consensus, requires a mechanism for reliable selection of client requests for proposal. We propose a random selection biased towards the oldest unapproved client requests that will eventually select all client commands. For liveness, we rely on the majority of nodes following this protocol to prevent attackers from taking advantage of predictable selection. However, it should be noted that the choice of proposal selection mechanism has no impact on safety, only liveness.

4 Acknowledgements

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APPENDIX 4 – ECONOMICS WHITE PAPER

Economic Model

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 Locked on October 30th, 2020



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Executive Summary

The RADIX PROTOCOL (the 'Protocol') is public open source code that may be downloaded and run without any permission by any person or company. When configured correctly this Protocol will autonomously connect to other computers running the same Protocol and create a shared public ledger. A version of this shared public ledger will be published for adoption and will become identified as the RADIX LEDGER/NETWORK.

The RADIX TOKEN (XRD) is to be a non-redeemable Token. The RADIX or RAD is intended to have two principal functions on the RADIX LEDGER: first, it may be used to pay for services for use of the Network (NETWORK FEES); second it may be used as part of the mechanism to secure the RADIX LEDGER/NETWORK against attack.

Most on-Network operations will incur a fee that is payable in RADIX TOKENS. The RADIX TOKEN is intended to facilitate the mechanism defined by the RADIX PROTOCOL whereby users pay for use of the RADIX LEDGER/RADIX NETWORK.

The RADIX LEDGER is to be secured against attack by a Proof of Stake sybil prevention mechanism. Token holders can choose to Stake a portion of the RADIX TOKEN they control (the STAKE) to engage in the Proof of Stake network. Alternatively, token holders may delegate a portion of the RADIX TOKENS they control to a Node run by someone else. Staking is designed to cause the cost of attacking the RADIX LEDGER to rise in proportion to the total value of RADIX TOKENS Staked.

The RADIX PROTOCOL¹ will allocate XRD to those who choose to delegate their STAKE to NODE RUNNERS that choose to participate in the Proof of Stake mechanism to secure RADIX LEDGER. It is intended that the RADIX PROTOCOL should exhibit a constant rate of new token emission. These new tokens are programmatically generated by the RADIX PROTOCOL and delivered automatically to all delegators of STAKE, in proportion to the STAKE they delegate.

As the RADIX LEDGER will be based on a Proof of Stake protocol it is desirable that there is a broad decentralised ownership of RADIX TOKENS before the mainnet is instantiated.

Prior to release of the first RADIX LEDGER, an ERC20 token called the E-RADIX will be generated which can be exchanged for a RADIX TOKEN when the RADIX LEDGER (mainnet) is instantiated.

It is envisaged that a maximum of 24Bn RADIX TOKENS will be generated by the RADIX PROTOCOL over the life of the Public Network. 18.3% are to be issued as E-RADIX TOKENS on the Ethereum network, 31.6% are to be pre-allocated as RADIX TOKENS at genesis of the RADIX LEDGER, and the remaining 50% released as new NETWORK EMISSION paid to STAKING delegators over a 40-year period. Token STAKING may commence once the RADIX LEDGER is instantiated.

The balance between market supply and demand of RADIX TOKENS will be improved by a mechanism which will use the market price of the RADIX TOKEN to determine how many RADIX TOKENS will be unlocked, making them available to be Staked or otherwise used. The design of the UNLOCK MECHANISM is intended to promote supply and demand that tends towards equilibrium.

The UNLOCK MECHANISM will initiate shortly after the release of the E-RADIX TOKEN and will continue on the RADIX LEDGER once instantiated.

IF THE TOKEN UNLOCKING MILESTONES ARE NOT REACHED SOME TOKENS MAY REMAIN INDEFINITELY LOCKED.

¹ A automatic, autonomous process that is a function of the correct running of the Radix Public Network by a decentralised network of Nodes. Once set in motion RADIX TOKENS (JERSEY) LIMITED (RTJ) will have no control over this process.

Token Overview

This section covers key details around the likely use of the RADIX TOKEN, some security considerations for Proof of Stake, and how public control is achieved and maintained.

System Stages

Cerberus represents a novel consensus mechanism to underpin a decentralised, public network that can support and connect billions of users.

Not all features of Cerberus will be enabled in the first version of the Radix Public Network. RADIX TOKENS (JERSEY) LIMITED (RTJ) will publish THE RADIX LEDGER in stages, starting with one called RPN-1 (Radix Public Network 1).

This will start with a basic, “unsharded” version that will establish the foundations of the subsequent networks. It is anticipated that sharding will be implemented in later phases after the basic security model in RPN-1 has been tested.

This conservative approach to launching a public network is intended to ensure a short timeline to the first release and a greater degree of staged security testing.

Transaction Fees

The RADIX PROTOCOL will require that a TRANSACTION FEE in RADS be submitted with all requests for any NETWORK OPERATION requiring a change of ledger state (e.g. transaction, token creation, messaging, conditional state creation).

Once the requested NETWORK OPERATION has been validated, the Transaction Fee is burnt. The frequency and volume of token burn is therefore directly related to frequency and volume of Network use.

Proof of Stake Considerations

Staking is the process by which a user can lock their RADIX TOKENS so that they can participate in the Proof of Stake consensus mechanism and earn NETWORK EMISSION. A user must lock their “Stake” either to a Node Runner client that they run, or delegate their stake to another Node Runner. Any Node Runner that has RADIX locked to it is called a Staking Node. The locked Stake is then used by the Staking Node to participate in validating transactions and other on-ledger operations. This work forms the core of running the RADIX LEDGER.

As Cerberus is a BFT-style consensus algorithm, it has strict bounds on liveness and safety. Controlling greater than one third of currently Staked tokens gives the controller the ability to both halt liveness (stop the RADIX LEDGER from being able to reach consensus) and break safety (cause a double-spend).

For the unsharded RPN-1, the number of Staking Nodes is targeted to be approximately 100. Later versions of RADIX LEDGER are planned to substantially increase the number of nodes that can act as full consensus nodes and move the RADIX LEDGER to a fully sharded configuration. Any balance holder that is not running a [Staking Node](#) may delegate their stake to a Staking Node to share in RADIX LEDGER [NETWORK EMISSION](#).

Staking Nodes protect the RADIX LEDGER from attack and are the backbone of its security. To ensure a truly decentralised network, RADIX LEDGER EMISSION earned by the Staking Nodes should represent enough value to cover the expenditure of running a Staking Node, with enough left over to encourage participation.

The higher or lower the market price of a RADIX TOKEN, the higher or lower the value of NETWORK EMISSION. Over time, the value of RADIX LEDGER EMISSION is likely to affect the number and professionalism of users who are willing to lock their holding of RADS and participate in running Staking Nodes. It is likely that a decrease in the value of the tokens could make the RADIX LEDGER less secure.²

The E-RADIX TOKEN

Achieving an early and a broad spread of Token distribution is essential to ensure sufficient decentralisation of the public network and that participants are incentivised as swiftly as possible from the moment of instantiation.

A RADIX ERC20 (E-RADIX) token will be released prior to the commencement of the RADIX LEDGER. The sale of the E-RADIX token is intended to promote distribution of Stake and achieve a high degree of decentralisation prior to the release of the RADIX LEDGER (mainnet).

In addition, as ERC20 tokens are one of the most commonly issued token types, generating an ERC RADIX token will also facilitate faster integration into the existing DeFi ecosystem on Ethereum, infrastructure such as exchanges, and user services such as wallets, further assisting distributed stake building.

On instantiation of the RADIX LEDGER, the E-RADIX will be exchangeable 1:1 for RADIX TOKENS.

E-RADIX TOKEN Sale

In Q4 of 2020, 642m RADIX TOKENS are made available for purchase. It is expected that the receipts will form part of the permanent endowment for RTJ, a wholly owned subsidiary of the RADIX FOUNDATION.

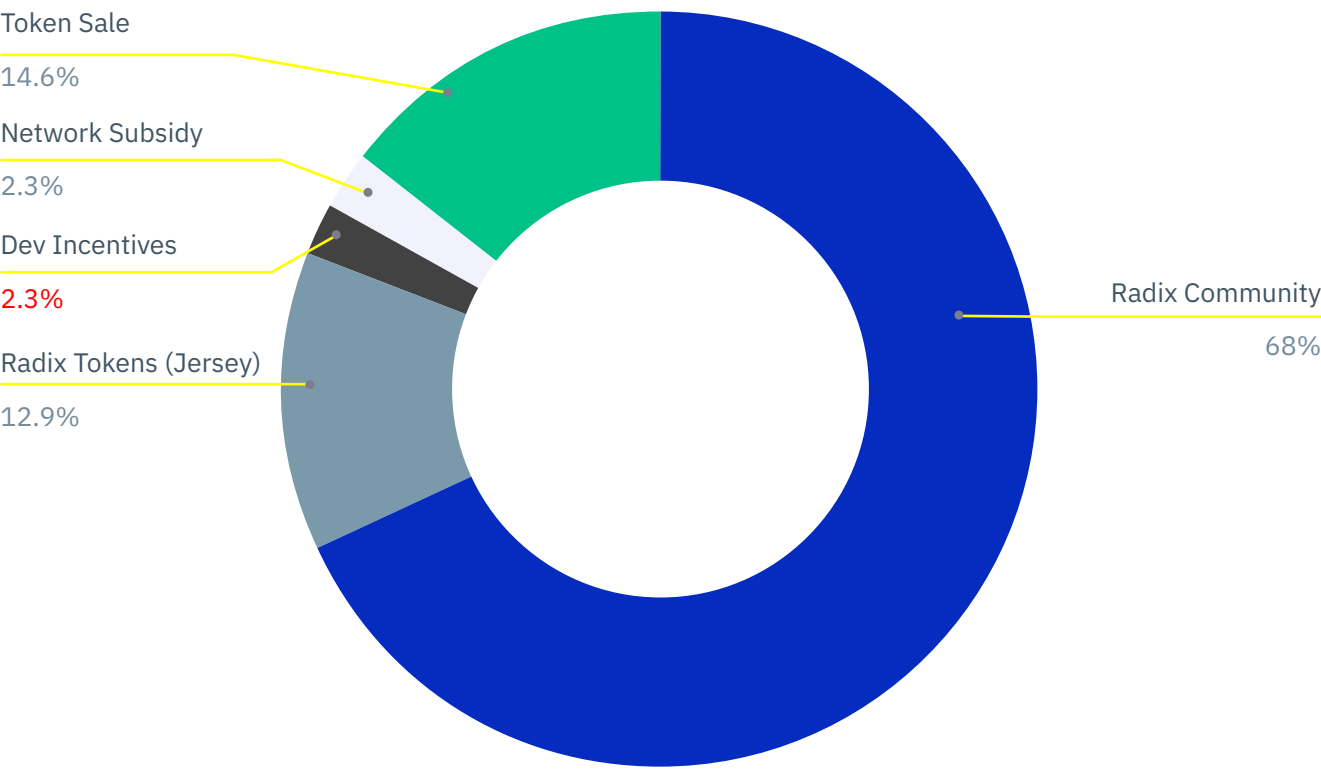
Any unsold tokens will be retained in the RTJ treasury and will follow the [E-RADIX TOKEN Unlocking Schedule](#).

To make further enquiries please visit www.radixdlt.com for more information.

² This is based on the assumption of rational market behaviour under normal market conditions

E-RADIX TOKEN Distribution

A total of 4.41Bn E-RADIX will be distributed (INITIAL SUPPLY) as both locked and unlocked tokens on the Ethereum Public Ledger. This chart shows the proposed distribution (ceteris paribus) as a % of INITIAL SUPPLY:



See the [Table of E-RADIX and RADIX Distributions](#) for further details.

E-RADIX TOKEN Unlocking Schedule

Of the 4.41Bn INITIAL SUPPLY of E-RADIX TOKENS, 4.2Bn will be distributed as 99% locked and 1% unlocked E-RADIX TOKENS. A further 200m E-RADIX TOKENS are unlocked for the Liquidity Incentives, and 10m as a marketing float. **These 210m E-RADIX TOKENS are not included in the Circulating Supply tables below³.**

E-RADIX TOKENS will be unlocked as the market price for E-RADIX TOKENS increases. Many blockchain projects use time-based unlocking which is potentially less secure as they may cause a less than optimal alignment between the incentives to acquire tokens/price with the need for security.

As each price milestone is passed, more tokens will be unlocked.

Assuming a linear relationship between supply and demand, an increase in price can be reasonably assumed to reflect a proportionate increase in demand. It is assumed that releasing tokens to feed that demand, will maintain price stability of the tokens. It is therefore assumed that demand will increase as adoption of the RADIX LEDGER increases. The algorithm deployed to release tokens has been designed to allow the market price to increase whilst keeping supply and demand in equilibrium, with upward pressure on price being exerted by functional demands for the RADIX LEDGER.

³ The 200m E-RADIX tokens are expected to be released via a long term liquidity incentivisation mechanism. The 10m marketing token float is subject to a variety of different lockups.

The 99% locked will remain locked until the 1-week Simple Moving Average (SMA) of the E-RADIX TOKEN crosses the following token price milestones:

SMA Milestones	% Total Unlocked	Circulating Supply of E-RADIX TOKENS	Total Market Value of Circulating E-RADIX TOKENS
NA	1%	42,000,000	\$1,638,00
\$0.05	5%	210,000,000	\$10,500,000
\$0.07	10%	420,000,000	\$29,400,000
\$0.09	15%	630,000,000	\$56,700,000
\$0.11	20%	840,000,000	\$92,400,000
\$0.13	25%	1,050,000,000	\$136,500,000
\$0.15	30%	1,260,000,000	\$189,000,000
\$0.17	35%	1,470,000,000	\$249,900,000
\$0.19	40%	1,680,000,000	\$319,200,000
\$0.21	45%	1,890,000,000	\$396,900,000
\$0.23	50%	2,100,000,000	\$483,000,000
\$0.25	55%	2,310,000,000	\$577,500,000
\$0.27	60%	2,520,000,000	\$680,400,000
\$0.29	65%	2,730,000,000	\$791,700,000
\$0.31	70%	2,940,000,000	\$911,400,000
\$0.33	75%	3,150,000,000	\$1,039,500,000
\$0.35	80%	3,360,000,000	\$1,176,000,000
\$0.37	85%	3,570,000,000	\$1,320,900,000
\$0.39	90%	3,780,000,000	\$1,474,200,000
\$0.41	95%	3,990,000,000	\$1,635,900,000
\$0.43	100%	4,200,000,000	\$1,806,000,000

IF THE TOKEN UNLOCKING MILESTONES ARE NOT REACHED SOME TOKENS MAY REMAIN INDEFINITELY LOCKED.

The tracking and triggering of these milestones will commence after the initial distribution of the E-RADIX TOKENS to the **E-RADIX TOKEN Sale** purchasers, and a reliable price feed can be established by RTJ for the E-RADIX TOKEN price (as reported by Coin Gecko or similar).

Market experience suggests that during periods of low market liquidity there is likely to be an increase of price sensitivity (whereby small buy/sell volumes are predicted to have a disproportionate effect on price), thereby potentially triggering the release of large numbers of token tranches in a short time frame.

To mitigate this risk of rapid tranche unlocking, each time a new tranche is unlocked the SMA measure is reset to let the newly unlocked supply settle in the market. For example, if the price moved up to \$0.43 in the first two weeks, it would only unlock a further 4% of supply, taking total unlocked supply to 5%. If it still stayed above \$0.43 for the next two weeks, it would simply unlock the next tranche of 5%, taking the total unlocked supply to 10%, and so on.

After each unlocked the SMA measurement restarts from the first hour after the new tranche has entered circulation. This mechanism is designed to limit the frequency at which price-based unlocking can occur, making sure a minimum of 7 days must elapse before a new tranche can be unlocked.

To avoid supply shocks caused by sudden or violent volatility, only one new tranche can be unlocked at any given time.

Any E-RADIX still locked when the RADIX LEDGER (mainnet) is instantiated will remain locked on the Ethereum Public Ledger until such time as all milestones have been achieved.

E-RADIX TO RADIX TOKEN BRIDGE

Following RPN-1 launch, a 1-way “token bridge” service will be provided from Ethereum to the RADIX LEDGER. This service will allow holders to exchange unlocked E-RADIX for unlocked RADIX TOKENS.

Once mainnet is launched, all E-RADIX will be 1:1 exchangeable for RADIX TOKENS held securely in a 3rd party custodian.

E-RADIX holders who wish to use the bridge will send E-RADIX TOKENS from their ERC20 wallet to an Ethereum address provided by the service and will then receive an equal number of RADIX TOKENS to a Radix wallet address that they provide.

Because each circulating E-RADIX TOKEN will be backed by a custodied RADIX TOKEN, these tokens share a single circulating supply equal to the total supply of RADIX.

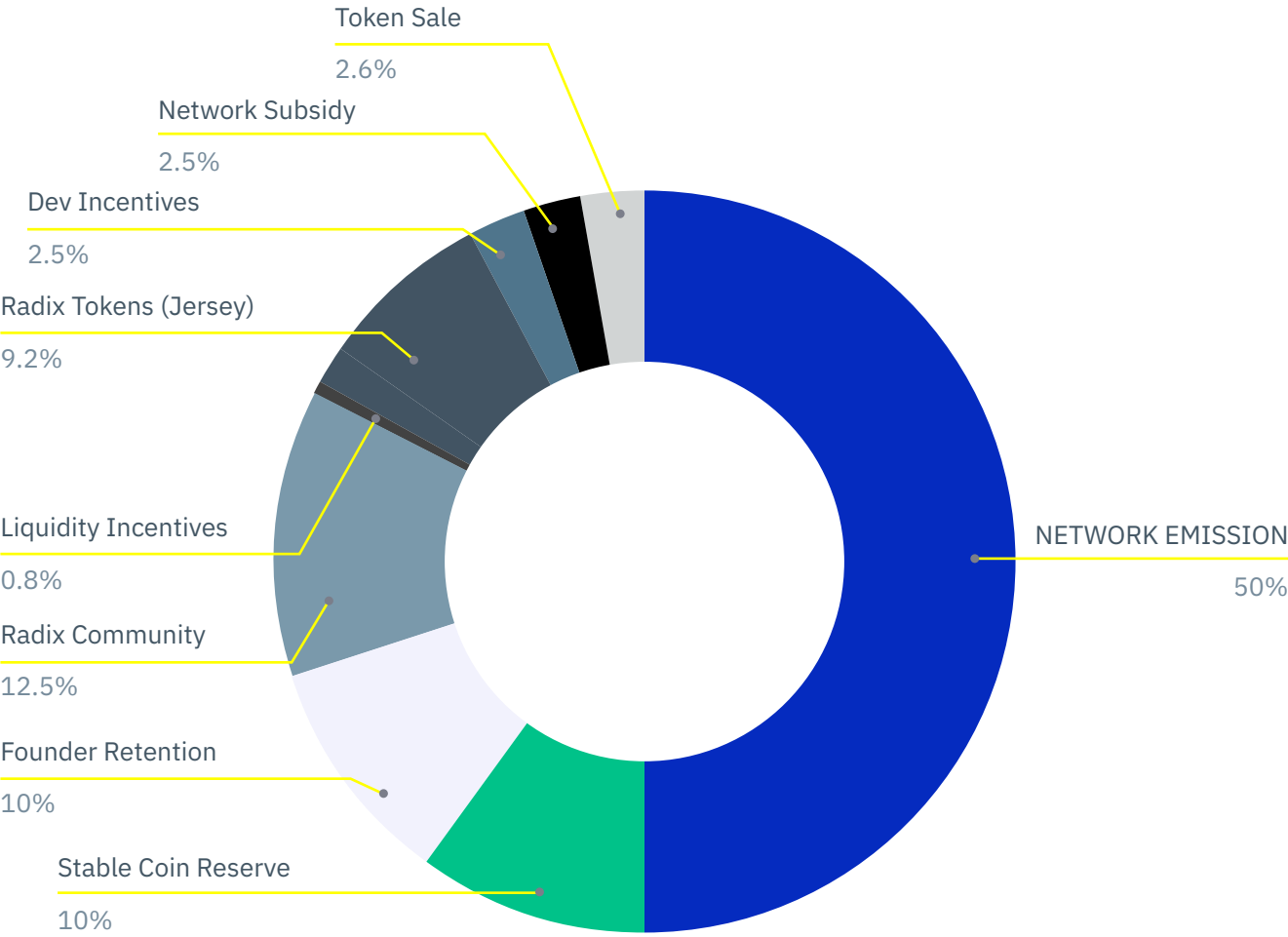
Initially, when there are only E-RADIX TOKENS, the unlocking milestones will follow the E-RADIX TOKEN market price. Following publication and adoption of the RADIX LEDGER, once the market trading volume of RADIX TOKEN exceeds that of E-RADIX TOKEN, the price data source will be switched permanently to follow the RADIX TOKEN price.

RADIX TOKEN DISTRIBUTION

The generation of NETWORK EMISSION by the RADIX LEDGER will commence on the instantiation of the RADIX LEDGER. This NETWORK emission is capped at 12Bn RADIX TOKENS over a minimum 40-year period. For more details please see the [NETWORK EMISSION](#) section.

In addition to the 4.41Bn E-RADIX TOKENS, a further 7.59Bn RADIX TOKENS will be created at instantiation of the RADIX LEDGER. These RADIX TOKENS will be instantiated as locked tokens for: the [FOUNDERS](#) (Radix DLT); the RADIX LEDGER [Subsidy](#); the [Dev Incentives](#); a mainnet endowment for [RADIX TOKENS \(JERSEY\) LIMITED](#); and an indefinitely locked allocation for the [Stable Coin Reserve](#).

This chart shows the distribution (ceteris paribus) as a % of maximum supply (24Bn RADIX TOKENS).



See the [Table of E-RADIX and RADIX Distributions](#) for further details.

RADIX TOKEN UNLOCKING SCHEDULE

The price based unlocking mechanism starting with the E-RADIX TOKEN is intended to be replicated and apply to the RADIX TOKENS on the RADIX LEDGER.

Of 12Bn tokens issued or exchangeable for RADIX TOKENS on the Radix Public Network, 2.4Bn are locked in the Stable Coin Reserve, and 4.41Bn will have already been issued as E-RADIX TOKENS, leaving 5.19Bn to be unlocked via the price based unlocking mechanism on the RADIX LEDGER (mainnet).

The 5.19Bn RADIX TOKENS must also, separately, achieve the same SMA price targets as the E-RADIX TOKENS, with the same SMA period for each tranche to unlock. This means that even if the E-RADIX TOKEN has achieved 100% unlocking by the time the Radix Public Network is instantiated, the starting supply of unlocked RADIX TOKENS will still be 1% of the 5.19Bn locked RADIX TOKENS on RADIX LEDGER (mainnet).

System Components

This section introduces the key system components of the Radix Public Network and some notes about how they interact with each other.

Node Runners & Staking Nodes

The Radix Public Network will be effectuated by the engagement of a network of Nodes (computers running RADIX LEDGER client software). Those that choose to run Nodes are called Node Runners. Anyone may become a Node Runner by downloading and running the Node Runner client. Any Node Runner that has **RADIX locked** to it is called a **Staking Node**.

Although anyone may run the Node Runner client, only Staking Nodes are able to directly participate in consensus. For the first version of the Radix Public Network (RPN-1), an initial maximum of 100 Staking Nodes will be permitted. Subsequent sharded versions of RADIX LEDGER is predicted to substantially extend this number.

Any RADIX balance holder that wishes to Stake but does not wish to run a Staking Node, or has insufficient balance to do so themselves, will be able to delegate their stake to any of these 100 Staking Nodes. Our target is to make the required balance less than 20 RADIX so that virtually anyone may participate in RADIX LEDGER's security and earn from RADIX LEDGER EMISSION should they wish to do so.

NETWORK EMISSION

The RADIX PROTOCOL will periodically create new RADIX TOKENS which will be paid to all delegators that have STAKED tokens to Staking Nodes that have been participating in consensus. These new tokens, called RADIX LEDGER **EMISSION**, are earned by delegators in proportion to the total amount of stake delegated.

NETWORK EMISSION is generated by the Radix Protocol at a rate of 300m RADIX TOKENS per annum. The total cap of 24Bn tokens would be reached 40 years from the release of the Radix mainnet. As Staking Nodes are the main beneficiaries of the transaction fee burn created by utility of RADIX LEDGER, RADIX LEDGER has 40 years to reach the stage that burn from utility sufficiently covers the cost of running a Staking Node.

This NETWORK EMISSION will be distributed (at least daily) in equal parts to all valid delegators. Each staking period a fixed amount of new RADIX TOKENS are available to be earned by the Staking Nodes.

From the perspective of a delegator, assuming the amount they stake is constant, the lower the Sum of all delegated Stakes, the larger the total earnings per period that delegator will earn. This creates higher potential earnings for delegators at the start of RADIX LEDGER's life, when total staked supply is low.

Prior Funding and Distribution Details

MAXIMUM SUPPLY = 24Bn = 4.41Bn E-RADIX TOKENS + 19.59Bn RADIX TOKENS.

Radix NETWORK EMISSION. (0% E-RADIX Supply. 50% of Maximum Supply).

12Bn Total = 0 E-RADIX TOKENS + 12Bn RADIX TOKENS.

Generated by the RADIX LEDGER Protocol directly, these are capped at a maximum of 12Bn tokens programmatically generated and allocated over a 40-year period.

Radix Community. (68% E-RADIX Supply. 12.5% of Maximum Supply).

3Bn Total = up to 3Bn E-RADIX TOKENS

Between the years 2013 and 2017 members of the early Radix community contributed approximately 3,000 BTC to Dan Hughes who initiated the theoretical development of the RADIX PROJECT.

In 2017 Dan Hughes assigned all the code and technology he had developed, which was then incomplete, to RADIX DLT LIMITED.

In recognition of the contributions made by the early contributors RADIX DLT and DAN HUGHES have agreed with these community members to procure the delivery to them of 3Bn RADIX TOKENS, creating an approximate token price of 1 BTC per 1m RADIX when the RADIX LEDGER is instantiated.

Founder Retention. (0% E-RADIX Supply. 10% of Maximum Supply).

2.4Bn Total = 0 E-RADIX TOKENS + 2.4Bn RADIX TOKENS.

FOUNDER TOKENS are retained by Radix DLT Ltd, which has funded and developed the RADIX technology since 2017 until the present day.

Radix DLT Ltd will hold the tokens it retains from the instantiation of the RADIX LEDGER as a treasury asset of the company.

RADIX DLT LIMITED will not receive E-RADIX TOKENS.

SALE OF EQUITY (RADIX DLT LIMITED). Between the years 2018 and 2019 a total of circa 10% of the equity in Radix DLT Ltd was sold for \$6m in two rounds to investors including VC LocalGlobe and the co-founder of TransferWise, Taavet Hinrikus, giving them exposure to the 2.4Bn RADIX TOKENS that Radix DLT Ltd holds in treasury.

E-RADIX Token Sale. (14.6% E-RADIX Supply. 2.7% of Maximum Supply).

642m Total = 642m E-RADIX TOKENS.

642m RADIX tokens are to be reserved for the current token sale.

RADIX TOKENS (JERSEY) LIMITED. (8.3% E-RADIX Supply. 9.8% of Maximum Supply).

2.358Bn Total = 368m E-RADIX TOKENS + 1.99Bn RADIX TOKENS.

RADIX TOKENS (JERSEY) LIMITED (RTJ) is a wholly owned subsidiary of the RADIX FOUNDATION. The RADIX FOUNDATION is a not-for-profit company limited by guarantee, registered in the United Kingdom.

The objects of the RADIX FOUNDATION are to promote the use and development of the RADIX LEDGER as well as provide education and support for the Radix community and ecosystem, and operate incentive programs, events and more. Any unsold tokens will be retained in the RTJ treasury and will follow the [E-RADIX TOKEN Unlocking Schedule](#).

Dev Incentives. (2.3% E-RADIX Supply. 2.5% of Maximum Supply).

600m Total = 100m E-RADIX TOKENS + 500m RADIX TOKENS.

A healthy DeFi ecosystem must involve, and create value for, the individual developer as well as for large projects. Indeed, it is individuals who often lay the coding groundwork that makes the larger projects possible.

These individuals are so important that we expect to add to the RADIX LEDGER and the RADIX PROTOCOL a built-in incentivisation mechanism of developer royalties. Crucially, these developer royalties would be designed to be long term and self-incentivising.

This Dev Incentive grant aims to bootstrap the self incentivising developer ecosystem on the RADIX LEDGER. To learn more, please see our [DeFi White Paper](#).

Network Subsidy. (2.3% E-RADIX Supply. 2.5% of Maximum Supply).

600m Total = 100m E-RADIX TOKENS + 500m RADIX TOKENS.

The Network Subsidy is intended to help improve the performance of the RADIX LEDGER and the RADIX NETWORK. This is a discretionary pool to fund grants that are expected to be administered via RTJ. These grants are expected to be used to incentivise both DeFi-related and performance-related activities on the RADIX NETWORK/RADIX LEDGER, depending on what is most needed by the ecosystem.

Simple examples of this would be test net incentives prior to the launch of the RADIX LEDGER (mainnet), or an early performance subsidy for any mainnet Staking Nodes that consistently meets or exceeds a set of minimum specifications for computing power, bandwidth, and responsiveness.

Stable Coin Reserve. (0% E-RADIX Supply. 10% of Maximum Supply - locked).

2.4Bn Total = Indefinitely locked to the mainnet.

Some components of an ecosystem are intrinsically important to its success. Decentralised stable coins provide a mechanism for converting native tokens into stable exchange tokens and back again, function as a helpful leverage mechanism, create large amounts of lockup, and provide a useful decentralised stable store of value for other applications to use. Intrinsically, decentralised stores of value also introduce a degree of systemic risk when pegged to an external store of value (such as the USD).

To help bootstrap important projects of this type, the Stable Coin Reserve will be formed. This Reserve holds 2.4Bn RADIX TOKENS that are indefinitely locked and entirely removed from circulating supply. The Foundation has a 10-year period in which to review, disburse this reserve to support a systemic stabilization protocol, which may or may not require the use of the Stable Coin Reserve. Should the Stable Coin Reserve not be needed, it will remain locked and be destroyed after the 10-year period.

The Stable Coin Reserve will not be issued as E-RADIX TOKENS.

Table of E-RADIX and RADIX Distributions

E-RADIX Tokens (ERC20)

Allocation	eXRD (bn)	(%)	Unlocking Schedule?	Description
Radix Community	3.000	68.03%	Yes	Genesis Allocation for 3000 BTC
Radix Tokens (Jersey)	0.358	8.12%	Yes	Awareness programs and OpEx
Token Sale	0.642	14.56%	Yes	New Token Buyers
Liquidity Incentives	0.200	4.54%	Yes*	Liquidity Mining Programs
Dev Incentives	0.100	2.27%	Yes	New Developer Incentives
Network Subsidy	0.100	2.27%	Yes	Security & Performance Incentives
Marketing	0.010	0.23%	No	Marketing Related Expenses
Total E-RADIX	4.410	100%		

**the liquidity incentives program is intended to be distributed over a 6 month period. Other unlocking schedules are price unlocking.*

RADIX Tokens (Mainnet)

Allocation	eXRD (bn)	(%)	Unlocking Schedule?	Description
Radix Tokens (Jersey)	1.790	34.49%	Yes	Awareness programs and OpEx
Founder Retention	2.400	46.24%	Yes	Team Allocation
Network Subsidy	0.500	9.63%	Yes	Security & Performance Incentives
Dev Incentives	0.500	9.63%	Yes	New Developer Incentives
Total E-RADIX	5.190	100%		

NETWORK EMISSION (Mainnet)

Allocation	XRD (bn)	(%)	Unlocking Schedule?	Description
NETWORK EMISSION	12.000	NA	NA	40 year programmed release for network security

Stable Coin Reserve (Mainnet) Non-Circulating

Allocation	XRD (bn)	(%)	Unlocking Schedule?	Description
Stable Coin Reserve	2.400	NA	NA	Future stable coin project supply

